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8. Controller software CNC

This machine, software operation is divided into:

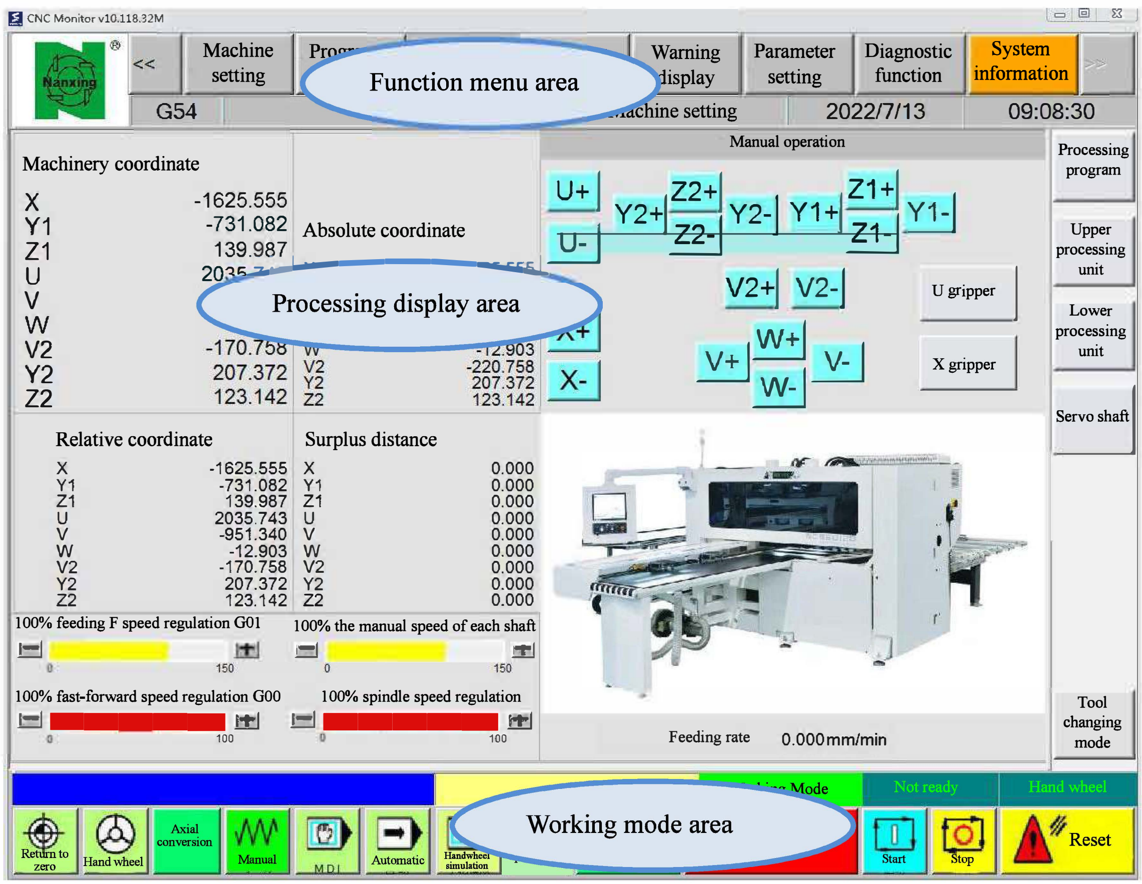
1. CNC controller software, see Chapter 8;
2. PCDrillCAD processing drawing software, see Chapter 9.

Main interface

Find and click the "

" shortcut icon on the computer cover to enter the main interface of the CNC controller, as shown in the figure below.

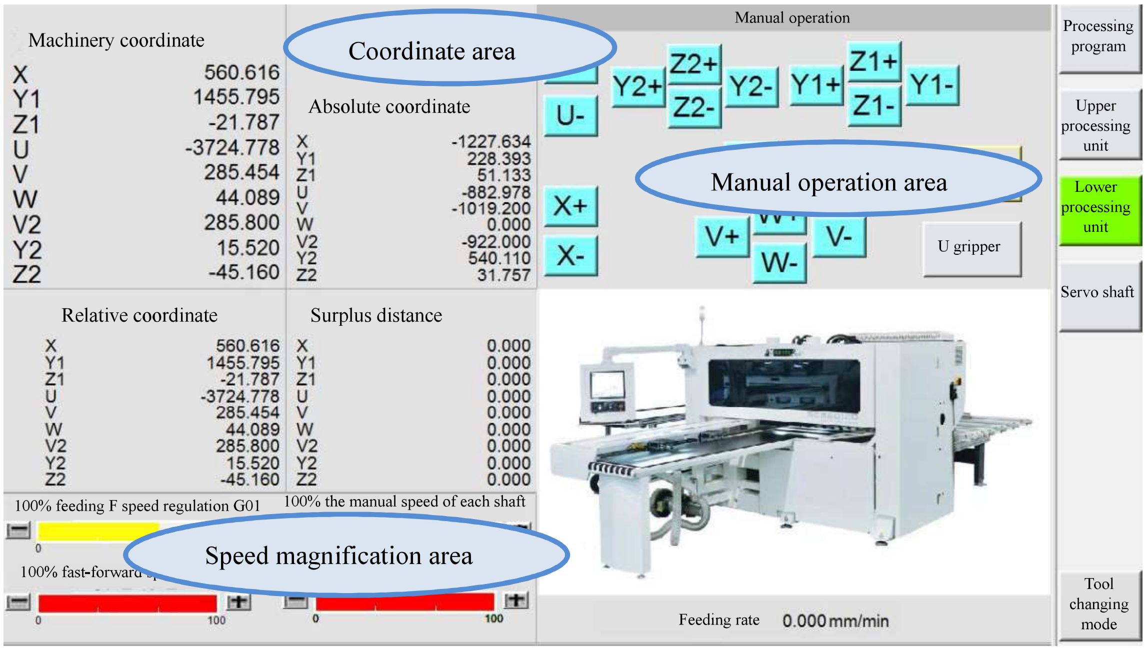
The controller interface can be divided into three areas: processing display area, function menu bar area, and working mode area.



Software upgrades, some icons may have minor changes.

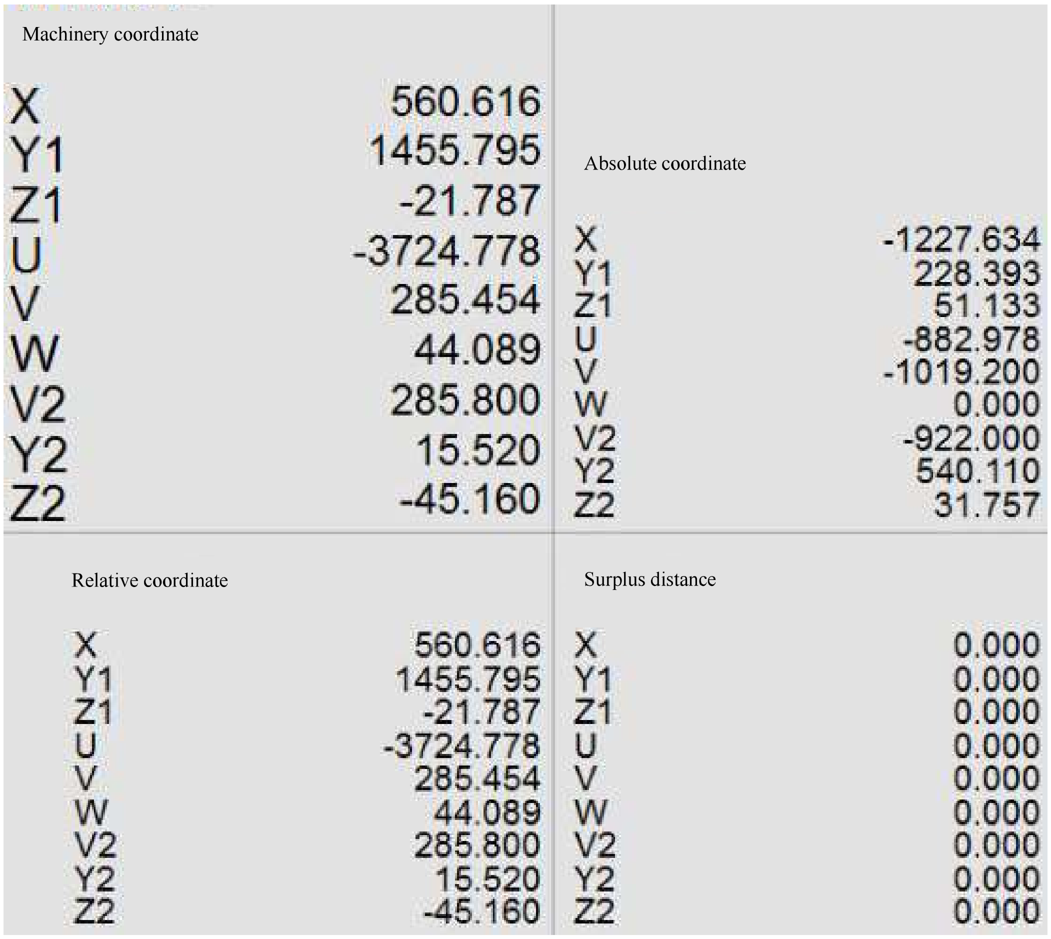
Processing display area

Processing display area of the main interface, which can also be divided into a coordinate area, a speed magnification area, and a manual operation area.



8.2.1 Coordinate area

It is used to display the position of each axis of the machine in real time, including "machine coordinate", "absolute coordinate", "relative coordinate" and "remaining distance".

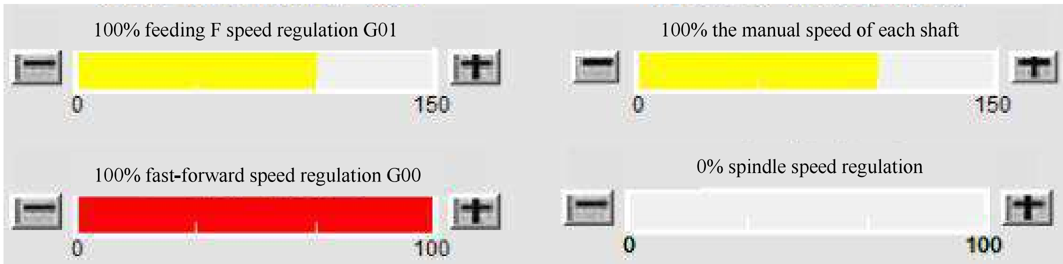


1. **The machine coordinate** refers to the position of each axis in the mechanical coordinate system G53.
2. **The absolute coordinate** refers to the position of each axis in the workpiece coordinate system G54.
3. **The relative coordinate** refers to the position of each axis relative to the previous fixed point; you can move each axis to a certain point and then clear the relative coordinate, and then move the axis, you can know the displacement of the current position relative to the previous point.
4. **The remaining distance** refers to the real-time display of the distance between the current position and the target position in the program, is only valid when the program is running.

Skill	Click on the main interface **"machine setting"**, and then click on the **"coordinate switching"** function, you can switch between different coordinate systems.

8.2.2 Speed magnification area

Displays the current spindle speed and G01 feed rate, and adjusts the machine's working magnification.



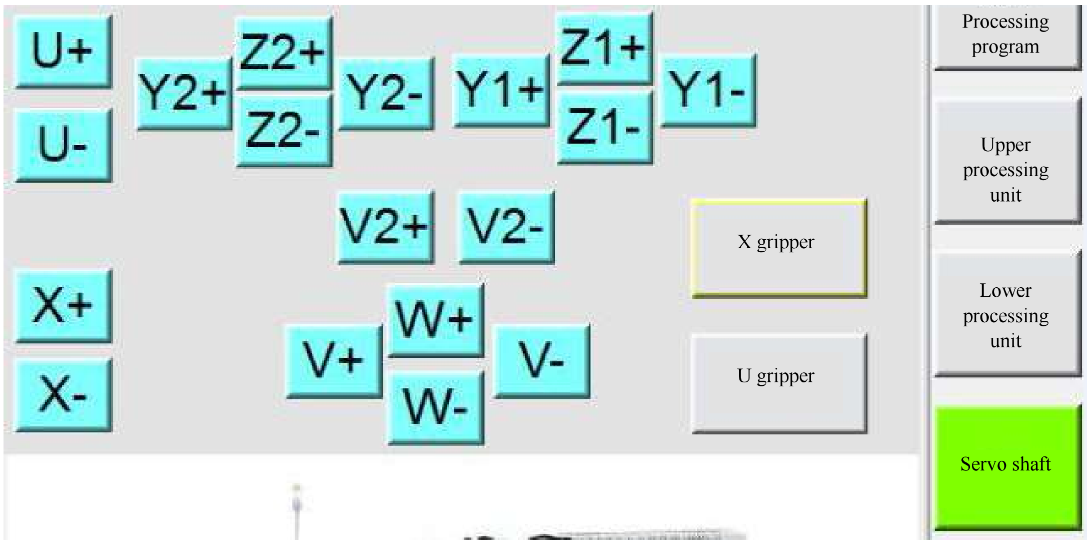
1. **Fast-forward speed regulation G01:** Adjust the feeding speed of each axis during the program running process, and the feeding speed set by the program when it is 100%.
2. **Fast-forward speed regulation G00:** Adjust the G00 speed of each axis during the program running process, and the G00 speed set by the controller when it is 100%.
3. **Manual speed of each axis:** Adjust the moving speed of each axis in manual mode, and the manual speed set by the controller when it is 100%.
4. **Spindle speed regulation:** Adjust the percentage of spindle speed, which is the spindle speed while 100 percent.
5. **Feeding rate:** Display the moving speed of X1 in real time.

Prompt	The speed adjustment will affect the processing quality and tool life of the sheet, so do not change it arbitrarily.

8.2.3 Manual operation area

In manual or handwheel mode, operate the corresponding actions of the machine.

1. **Servo axis:** Control each axis to move in positive and negative directions.



2. **Upper processing unit:** Control the operation of each part of the upper processing unit.

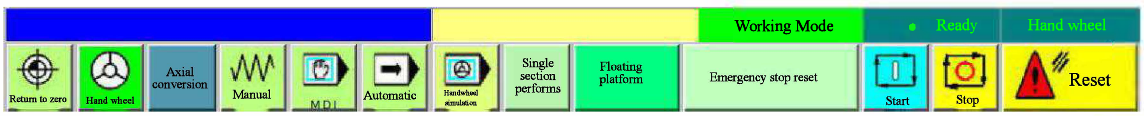
3. **Lower processing unit:** Control the operation of each part of the lower processing unit.

Working mode area

The upper right corner displays the current processing status and working mode of the machine.

By setting "parameter 3418" to 1, enter "engineering mode".

(In the engineering mode, many machine error prompts will be blocked, and the machine will lose its individual protection function, so that each shaft of the machine can move, which may cause the danger of collision.)

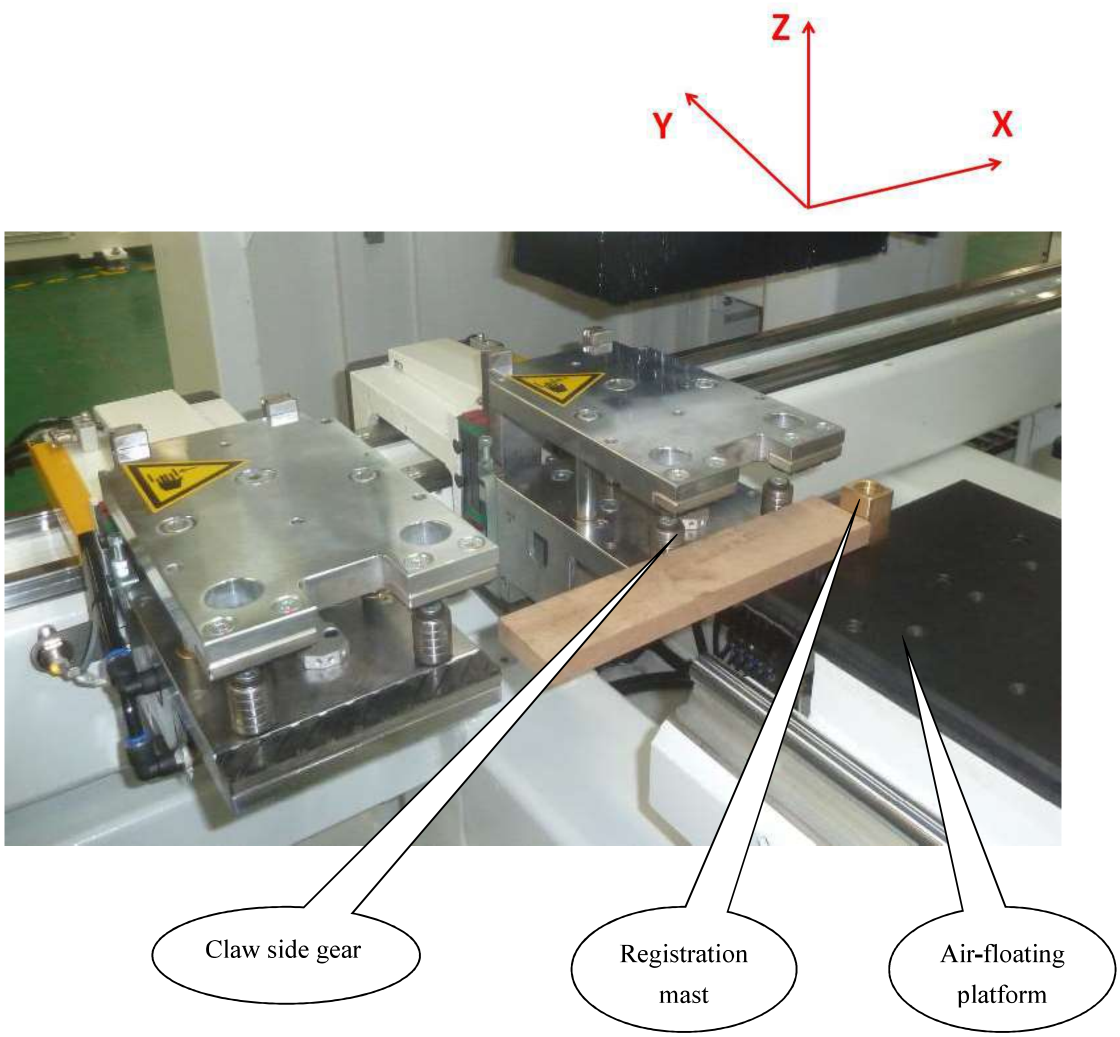


◆ Setting of workpiece origin G54

<u>Processing origin of the machine</u>

The machine uses a Cartesian coordinate system. The processing origin of the machine is:

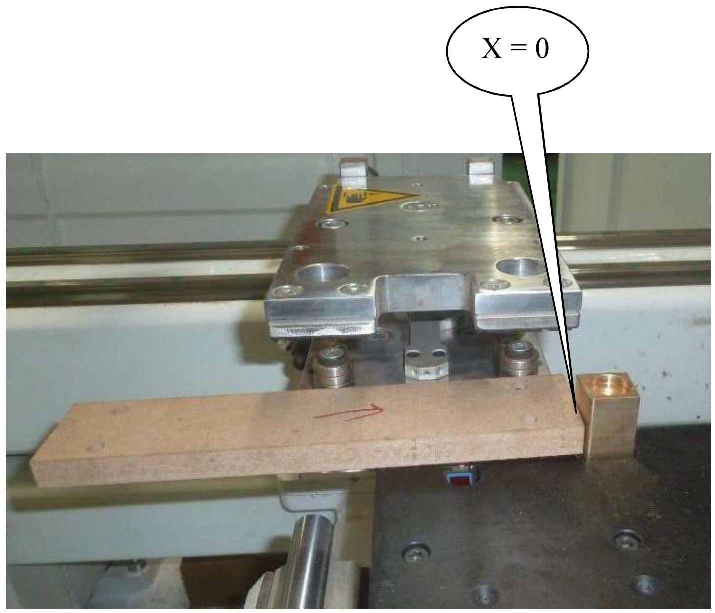
1. X axis: the front side of the positioning post;
2. Y axis: right side of the claw side crosspiece
3. Z axis: the surface of the air-floating platform.
4. U3 axis: loading table surface
5. V3 axis: loading table surface



The origin of the claw in the X axis direction

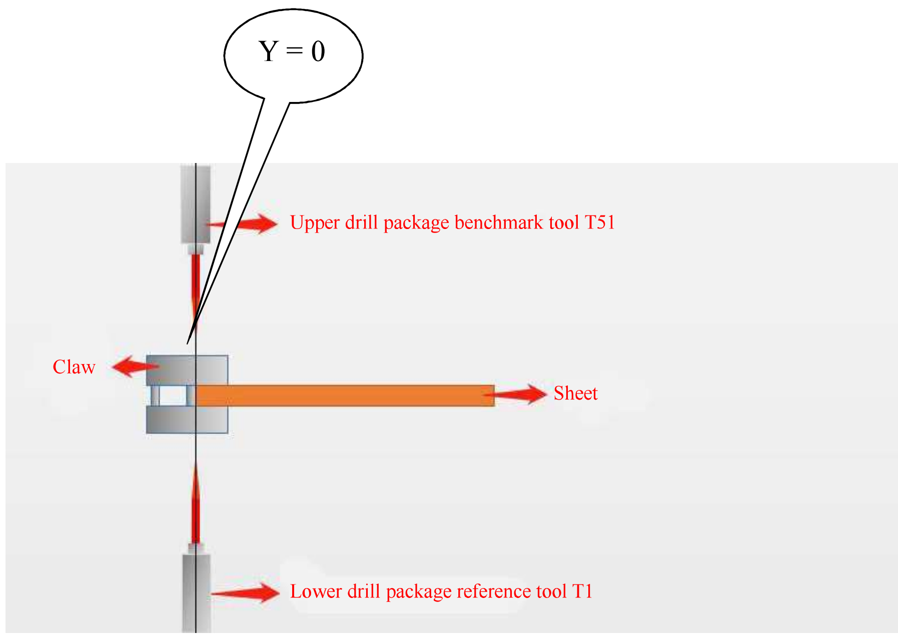
Align the front side of the claws with the front side of the sheet and hold the sheet against the positioning posts. The contact surface between the front side of the claws, the front side of the sheet and the positioning post is the origin (zero point) of the claws.

The X axis of the front claw and the U axis of the rear claw have the same origin.



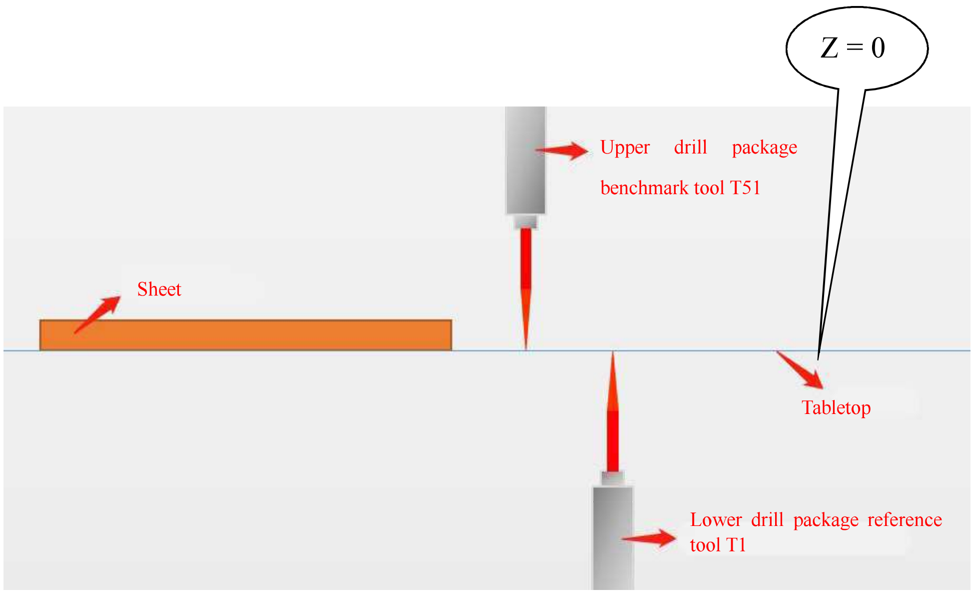
The origin of the drill box in the Y axis direction

1. Origin coordinate setting of the Y1 axis on the upper drill box: Take out the reference tool T51 of the upper rotary box 1, and move the upper drill box Y1 axis so that the tool nose is just aligned with the contact surface between the side of the claw sheet and plate and the side gear of the claw.
2. Origin coordinate setting of the Y2 axis on the upper drill box: Take out the reference tool T151 of the upper rotary box 2 and move the upper drill box Y2 axis so that the tool nose is just aligned with the contact surface between the side of the claw sheet and plate and the side gear of the claw.
3. Origin coordinate setting of the W axis on the lower drill box: Take in the reference tool T1 of the lower rotary box, move the W axis on the lower drill box so that the tool nose is just aligned with the contact surface between the side of the claw sheet and plate and the side gear of the claw.



The origin of the drill box in the Z axis direction

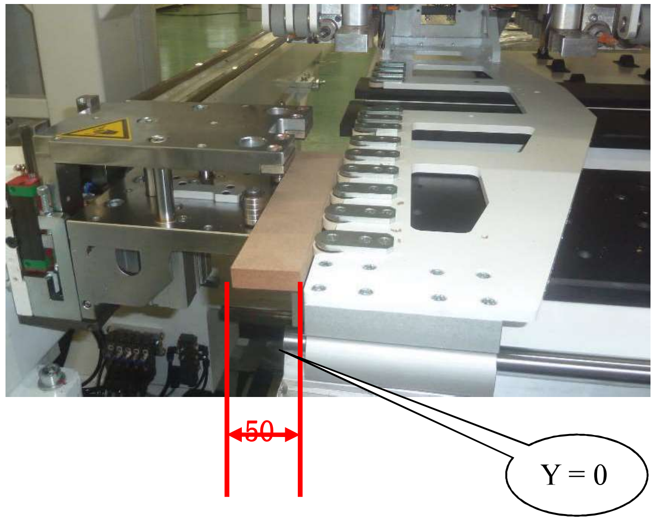
1. Coordinate setting of the Z1 axis on the upper drill box: Take out the reference tool T51, and move the Z1 axis so that the tool nose just touches the tabletop.
2. The coordinate setting of the Z2 axis of the upper drill box: Punch down the reference tool T151 and move the Z2 axis so that the tool nose just touches the tabletop.
3. Coordinate setting of the Z axis of the lower drill box: Take out the reference tool T1, and move the Z axis so that the tool nose just touches the tabletop.



Origin of side crosspiece V axis

1. Move the side crosspiece V axis laterally to the position close to the sheet;
2. The distance between the positioning post and the V axis side crosspiece, that is, the width of the sheet;
3. If the distance (or width of the board) is 50mm:
4. In the position of V axis G54, fill in -50.

If the side crosspiece is on the other side of the sheet, fill in 50 at the position of V axis G54.


50
Y = 0

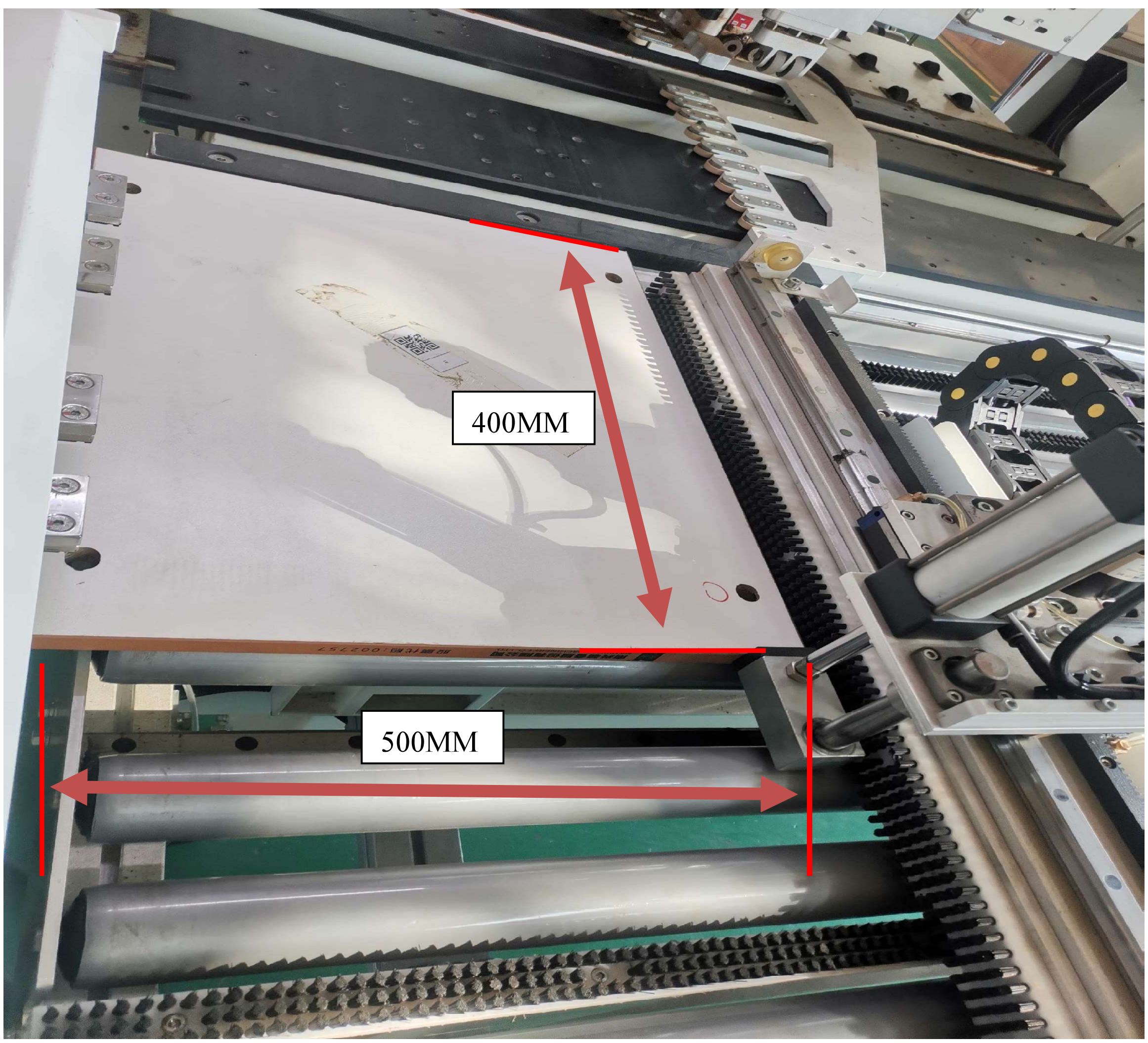
Original points of side thrust U3-axis and V3-axis

1. Move the side crosspiece V3 axis laterally to the position close to the sheet;
2. The distance between the positioning post and the V3 axis side crosspiece, that is, the width of the sheet;
3. If the distance (or width of the board) is 500mm:
4. In the position of V3 axis G54, fill in -500.

If the side crosspiece is on the other side of the sheet, fill in 500 at the position of V axis G54.

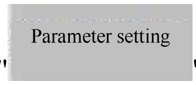
5. Move the side gear U3 axis vertically to a position close to the sheet;
6. The distance between the positioning post and the U3 axis side crosspiece, that is, the width of the sheet;
7. If the distance (or plate length) is 400mm:
8. Fill in - 400 in the G54 position of U3 axis.

If the side crosspiece is on the other side of the sheet, fill in 400 at the position of U3 axis G54.

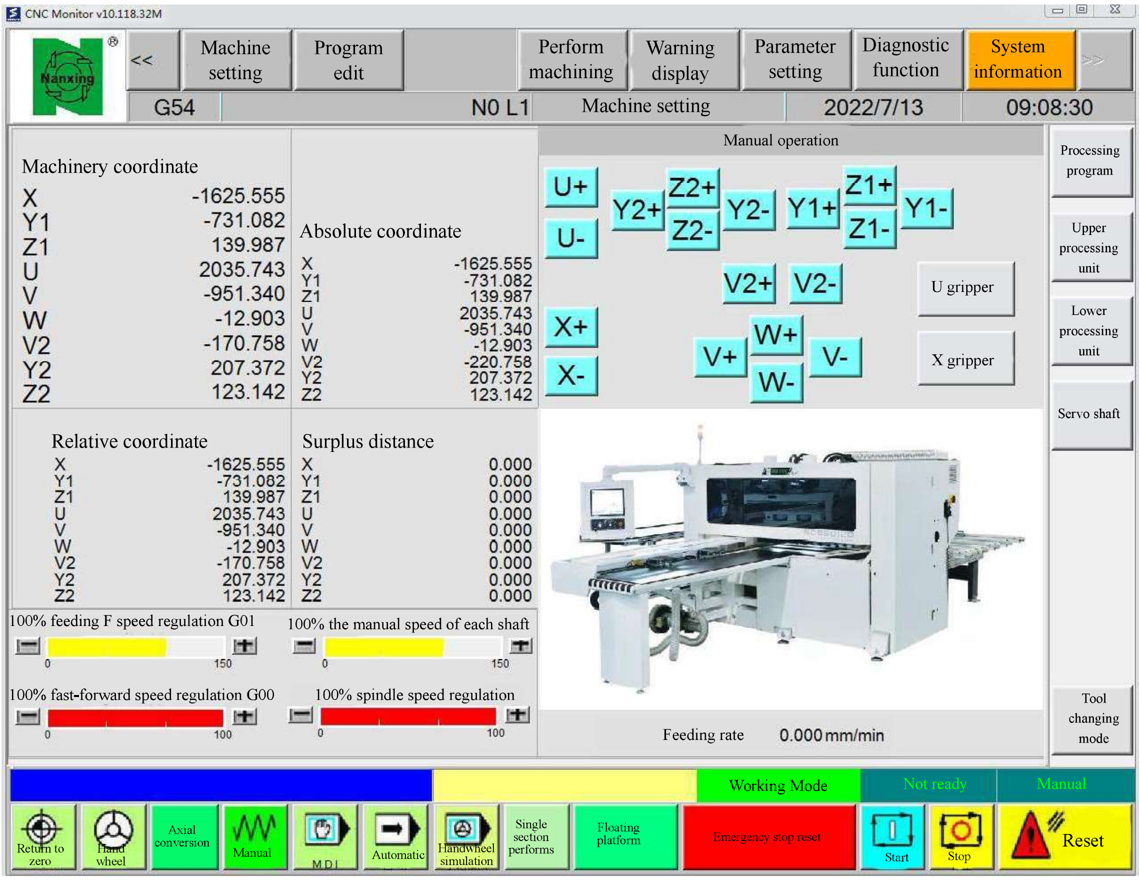


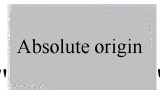
Origin point setting of each axis

For absolute coordinates, the steps for setting the origin of each axis are as follows:

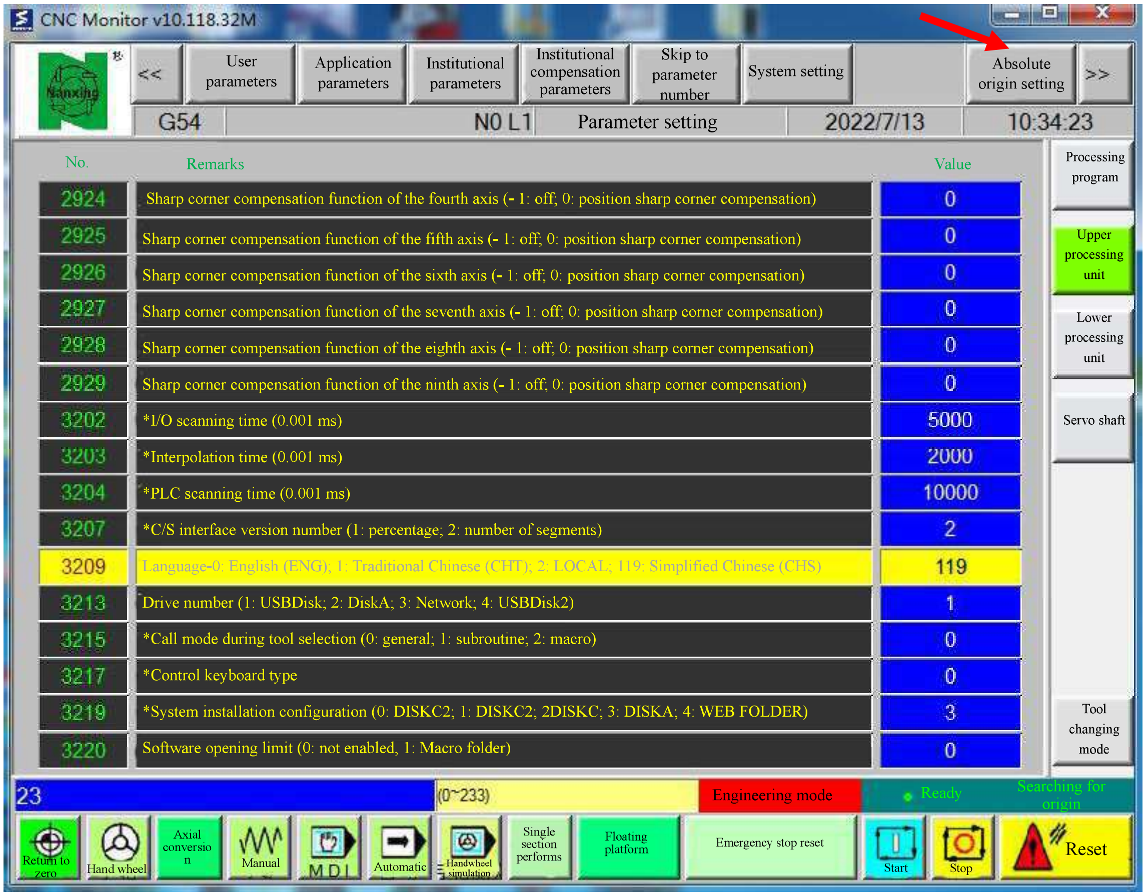
1. Manually move the axis to the target zero (origin) position.
2. Click the "

" icon on the main interface.



3. Click the "

" icon.



4. Select the specified axis and follow the steps.

◆ Handwheel mode

Manual operation of the machine is allowed, and the movement of each axis of the machine is carried out by shaking the handwheel. See Chapter 5.

◆ Manual mode

Manual operation of the machine is allowed, using the axial movement button to operate the movement of each axis.

◆ MDI mode

Enter the MDI (Manual Data Input) operation interface to edit/execute the MDI program.

◆ Auto mode

In this mode, the processing file can be selected for automatic processing, which is the standard mode of use in normal production.

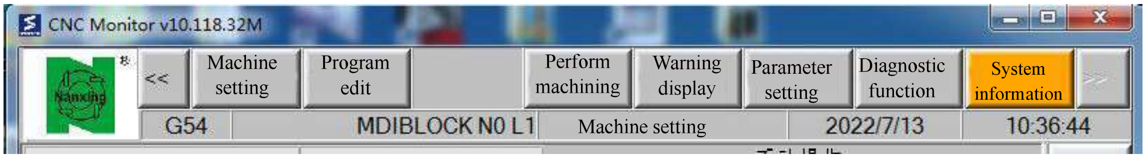
◆ Handwheel simulation

After starting the program, control the movement by shaking the handwheel.

Skill	Handwheel simulation function, often used to test the program or check the movement position.

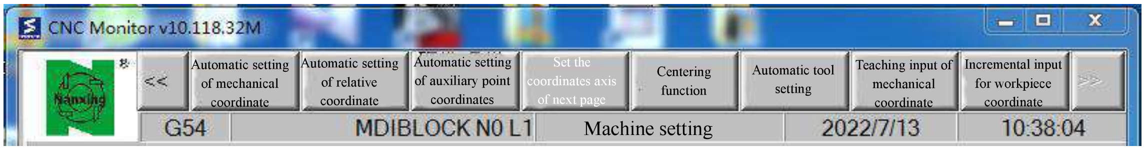
Function menu area

It is used to switch the operation pages of different items to the main interface area for display. The lower left corner of this area displays the program name of the currently selected processing program.



15.4.1 Machine setting

Machine setting is mainly used to set the workpiece coordinate system, the external offset of the coordinate system, etc.



Workpiece coordinate system setting

Access the machine setting from the main menu, and select the workpiece coordinate system to access the workpiece coordinate system operation interface. In this interface, you can set the working origin of the absolute coordinate system and change the external offset value of each axis.

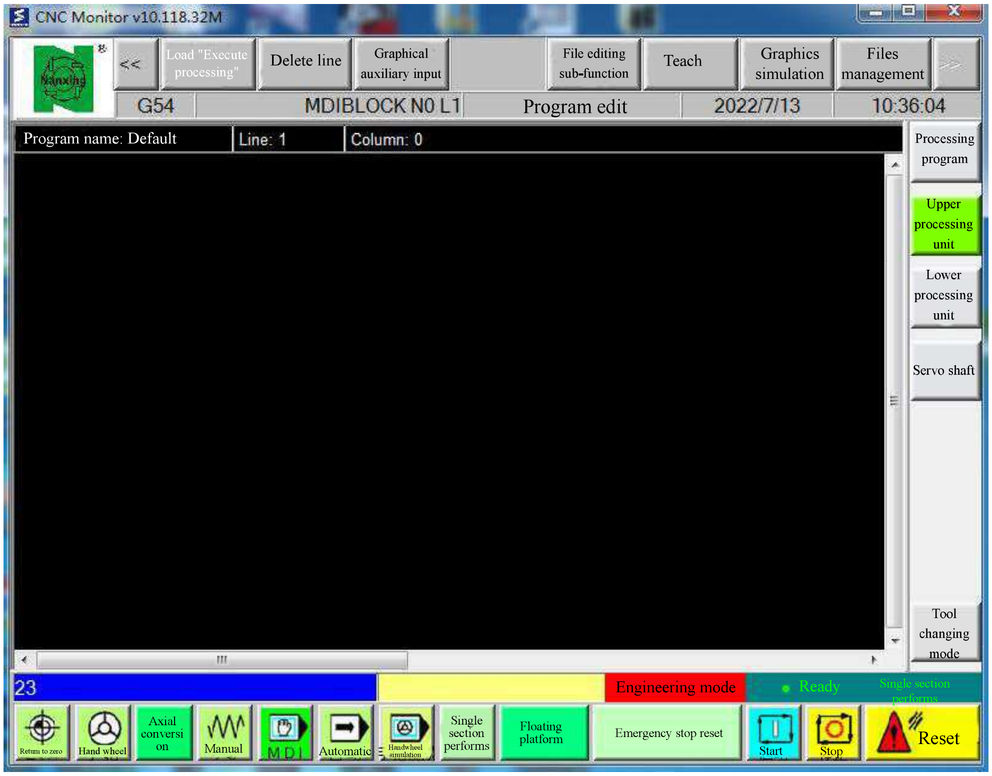
Coordinate system setting interface

External coordinate offset value: Based on the absolute coordinate system, increase the offset value by the specified amount.

Skill	External offsets are axis-based and are valid for all absolute coordinate systems.

15.4.2 Program edit

You can edit or modify the program directly here, and it will take effect immediately.


CNC Monitor v10.118.32M
<<
Load "Execute processing"
Delete line
Graphical auxiliary input
File editing sub-function
Teach
Graphics simulation
Files management
>>
G54
MDIBLOCK N0 L1
Program edit
2022/7/13
10:36:04
Program name: Default
Line: 1
Column: 0
Processing program
Upper processing unit
Lower processing unit
Servo shaft
Tool changing mode
23
Engineering mode
Ready
Single section performs
Return to zero
Hand wheel
Axial conversion
Manual
MDI
Automatic
Handwheel simulation
Single section performs
Floating platform
Emergency stop reset
Start
Stop
Reset

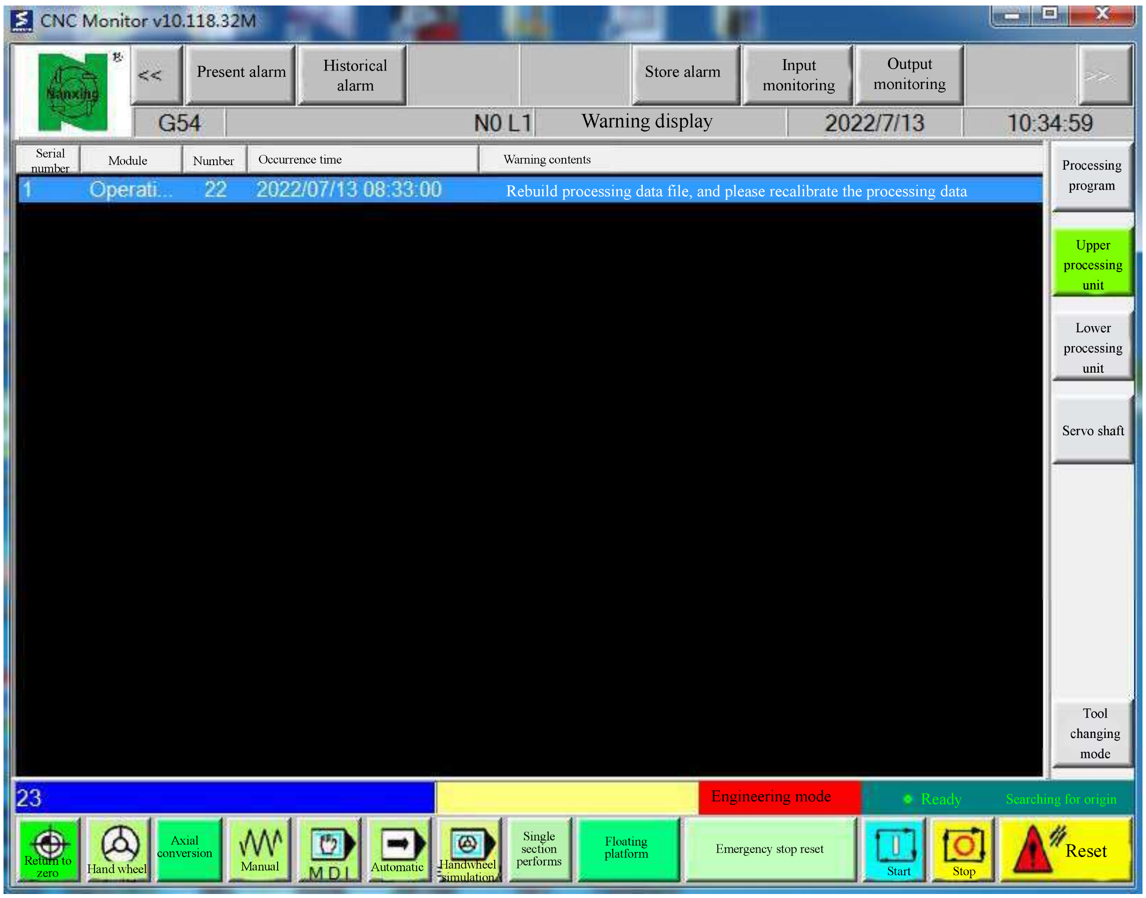
15.4.3 Perform machining

Here, you can select the processing file for processing; the processing program, coordinates, speed and magnification, the number of workpieces, the current processing code, etc. will be displayed.

15.4.4 Alarm display

Click the "

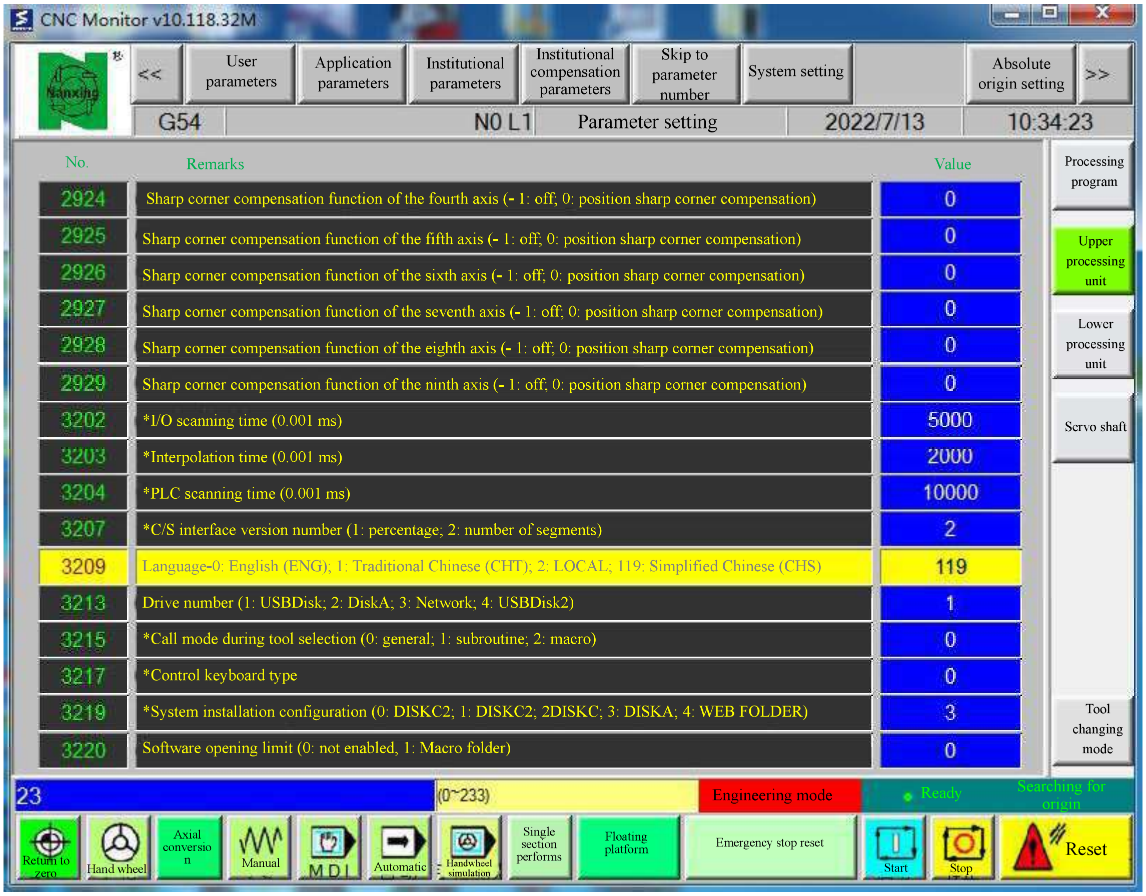
" icon with the left mouse button, and the machine equipment will enter the alarm display screen.



Alarm messages are displayed in the alarm bar. Trained and qualified personnel can deal with them according to the prompts in the alarm information. If you have any questions, please contact the customer service department.

15.4.5 Parameter setting

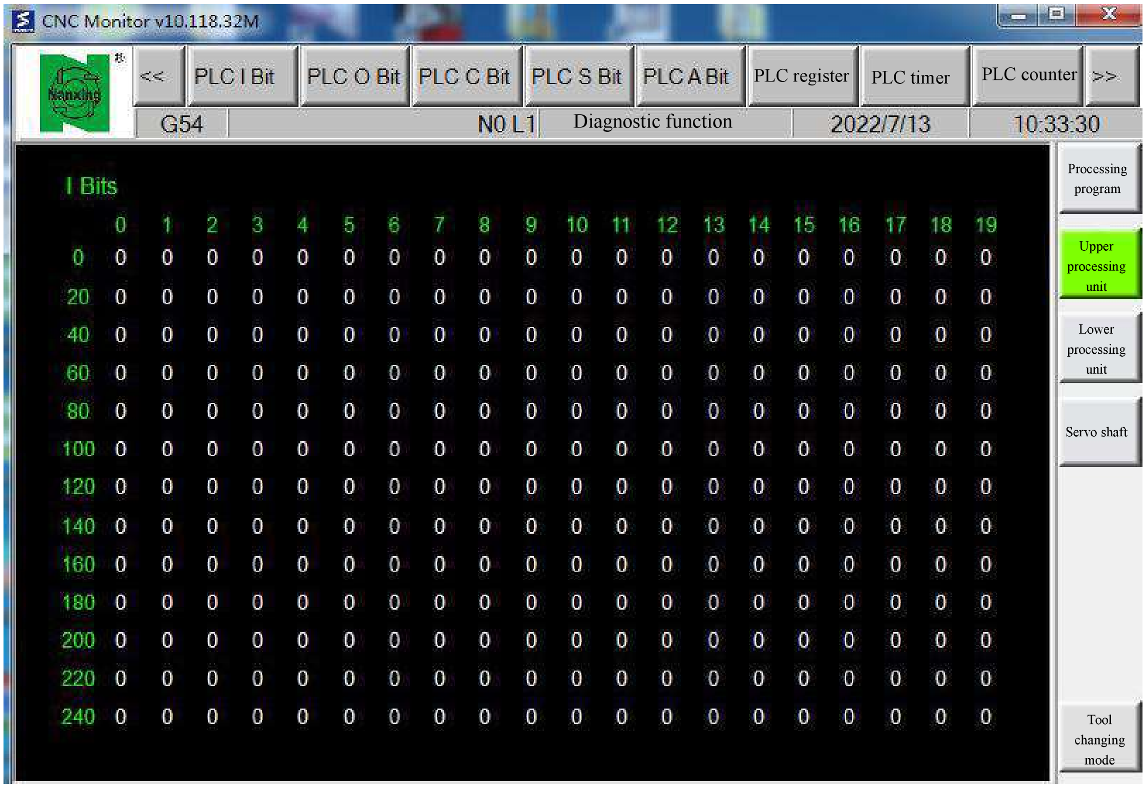
Various parameters can be set here, such as user parameters, application parameters, mechanism parameters, system setting, origin setting, etc.



⚠Attention:	The machine parameters have been set before leaving the factory, please do not modify them without authorization! Parameter modification requires a password. If it must be modified, please complete it under the guidance of professionals!

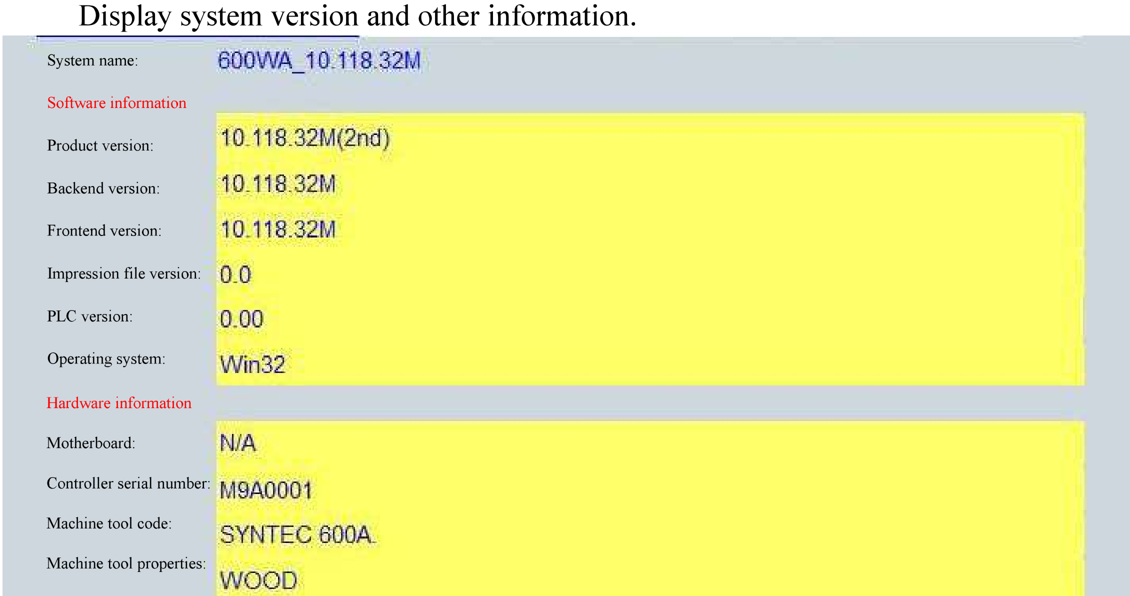
15.4.6 Diagnostic function

You can view register values, PLC ladder diagrams, etc. here to facilitate troubleshooting.



15.4.7 System information

Display system version and other information.

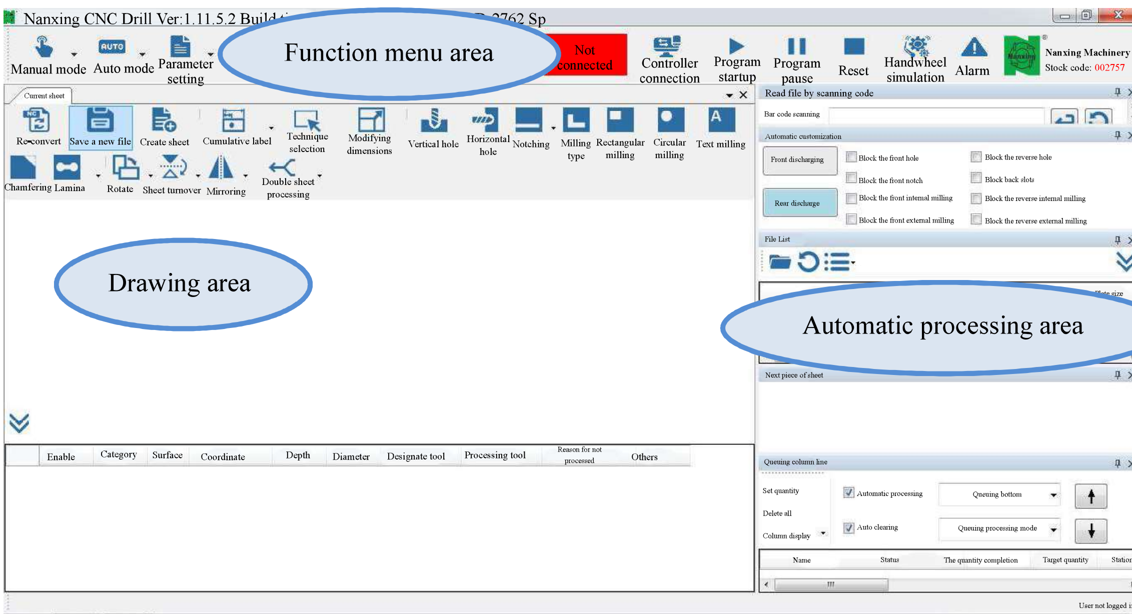


9. Processing drawing software PCDrillCAD

Main interface

Find and click the "  " shortcut icon on the desktop, and the following main interface will pop up.

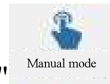
The main interface includes: function menu bar area, automatic processing area, and drawing area.



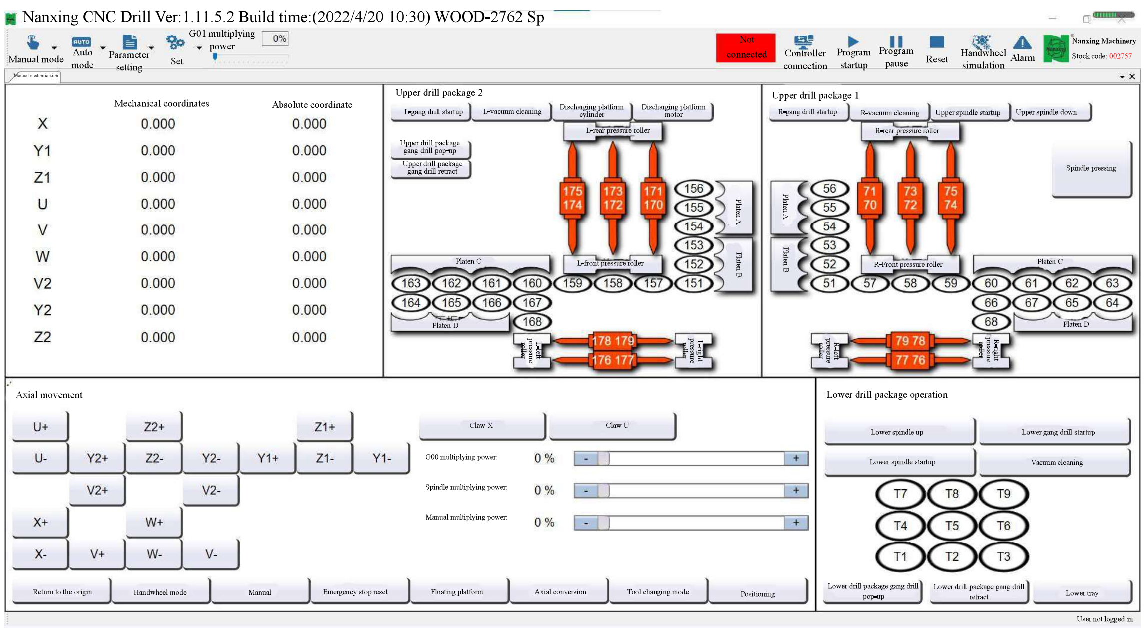
Function menu area

Function menu area, including the following functions.

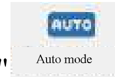
9.2.1 Manual mode

Click the "

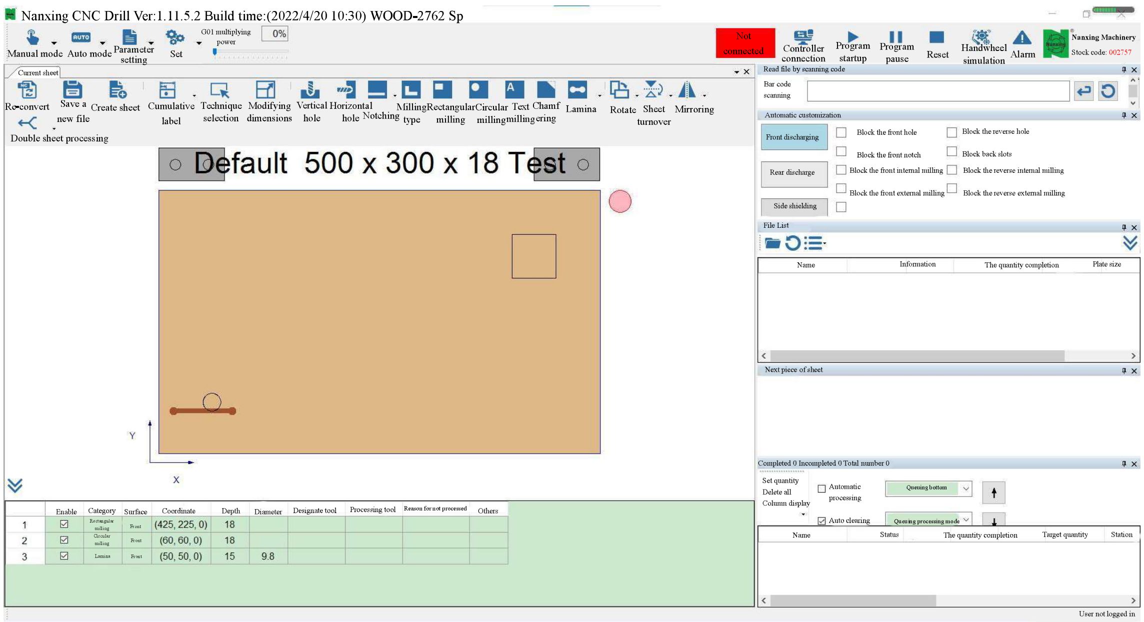
" icon on the main interface, and the following manual mode page will pop up. The controller is in manual mode. Corresponding manual operations can be performed, including axial movement, speed multiplying power adjustment, upper drill box operation, and lower drill box operation.



9.2.2 Auto mode

Click the "

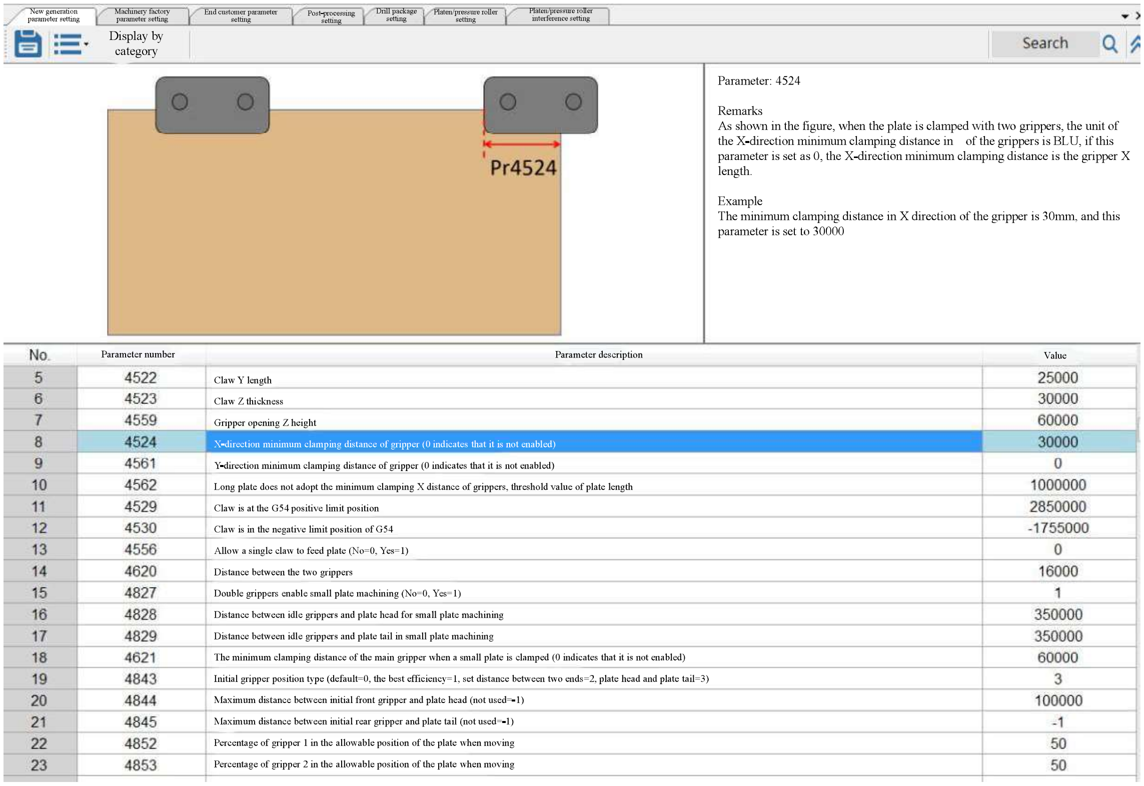
" icon on the main interface, and the following automatic mode interface will pop up. The controller is in automatic mode.



9.2.3 Parameter setting

Click the "

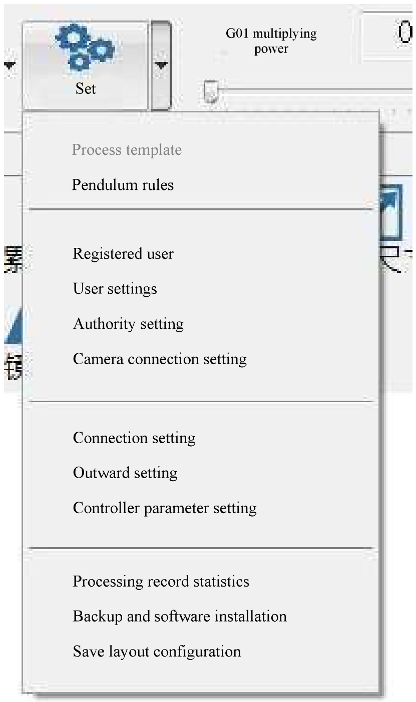
" icon to access the parameter setting mode, including various parameter setting, drill package setting, and platen setting.



No.	Parameter number	Parameter description	Value
5	4522	Claw Y length	25000
6	4523	Claw Z thickness	30000
7	4559	Gripper opening Z height	60000
8	4524	X-direction minimum clamping distance of gripper (0 indicates that it is not enabled)	30000
9	4561	Y-direction minimum clamping distance of gripper (0 indicates that it is not enabled)	0
10	4562	Long plate does not adopt the minimum clamping X distance of grippers, threshold value of plate length	1000000
11	4529	Claw is at the G54 positive limit position	2850000
12	4530	Claw is in the negative limit position of G54	-1755000
13	4556	Allow a single claw to feed plate (No=0, Yes=1)	0
14	4620	Distance between the two grippers	16000
15	4827	Double grippers enable small plate machining (No=0, Yes=1)	1
16	4828	Distance between idle grippers and plate head for small plate machining	350000
17	4829	Distance between idle grippers and plate tail in small plate machining	350000
18	4621	The minimum clamping distance of the main gripper when a small plate is clamped (0 indicates that it is not enabled)	60000
19	4843	Initial gripper position type (default=0, the best efficiency=1, set distance between two ends=2, plate head and plate tail=3)	3
20	4844	Maximum distance between initial front gripper and plate head (not used=-1)	100000
21	4845	Maximum distance between initial rear gripper and plate tail (not used=-1)	-1
22	4852	Percentage of gripper 1 in the allowable position of the plate when moving	50
23	4853	Percentage of gripper 2 in the allowable position of the plate when moving	50

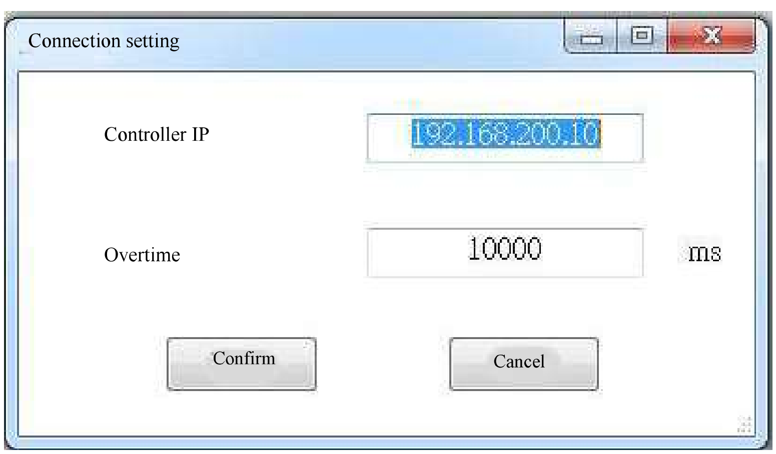
Drill package setting: Set the parameters of the drill box. See Chapter 9.5 "Drill Package Setting".

9.2.4 Setting mode



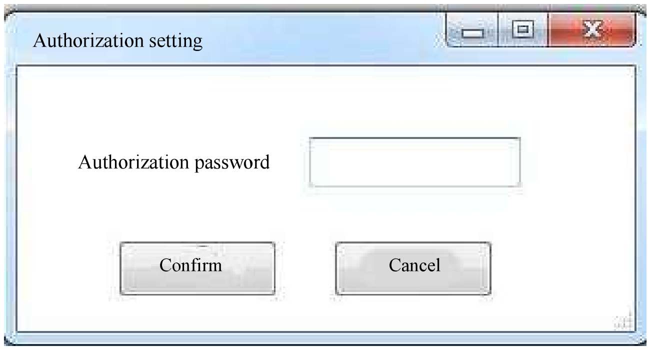
Connection setting

After setting the IP address of the background controller and the timeout time, click OK, and then click the controller connection to complete the connection between the software and the background.



Authority setting

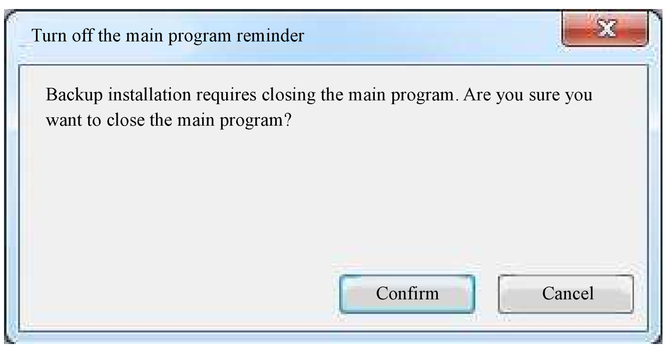
Before modifying the parameters, you need to enter the password to enable the permission.



Save layout configuration

After dragging and modifying the layout of the main interface, click on Save the Layout Configuration to complete the layout save.

Backup restore



Backup or restore current CAM software
Backup and restore is only for CAM, it cannot back up or restore the controller.

9.2.5 Processing multiplying power G01

Slide axis: Set the processing magnification when the controller in process. Take 10% as the unit, range: 0%~200%.

9.2.6 Controller status display

The controller includes the following status displays:

- Not connected: The controller is not connected
- Not ready: The software is not well-matched

- Ready: waiting for processing
- In process
- temporary suspended
- Single block stop
- In execution

Manual mode button is not enabled when the controller mode is in process, pause, single block stop or in execution.

When the controller and the program are online, the controller connection button cannot be enabled.

9.2.7 Controller connection

Click the " **Controller Connection**" icon to connect the processing drawing software with the controller software.

9.2.8 Program start

Click the " **Program Start**" icon to start the processing program currently specified by the controller.

9.2.9 Program pause

Click the " **program pause**" icon to pause the program action of the controller.

9.2.10 Reset

Click the " **Reset**" icon to reset the processing state of the controller.

9.2.11 Handwheel simulation

Click the " **Handwheel Simulation**" icon to perform handwheel simulation processing on the machine; see Chapter 5 "Axis and Handwheel".

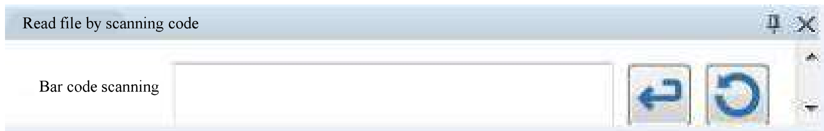
9.2.12 Alarm display

Displays the controller's current alarms. When an alarm occurs, red and blue flash alternately.

Automatic processing area

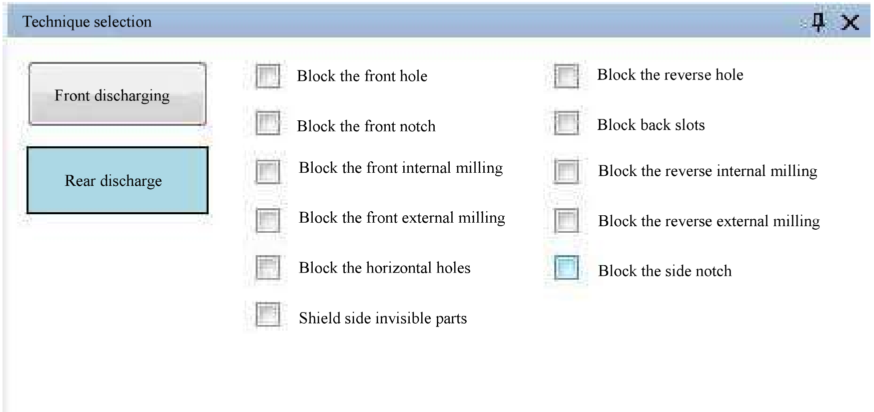
The automatic processing area includes the following functions.

9.5.1 Read file by scanning code



Barcode scanning, read processing program files.
Code scanning gun: Scan the bar code of the sheet, convert the processing information into NC file, and access the queue.
Input box: Enter the sheet and plate ID by yourself, press the Enter button, and convert the processing information to NC file, and enter the queue.
Enter the wrong barcode: If the barcode cannot be found after scanning, it will automatically turn red, and it will show that the XXX barcode cannot find the file.

9.5.2 Technique selection



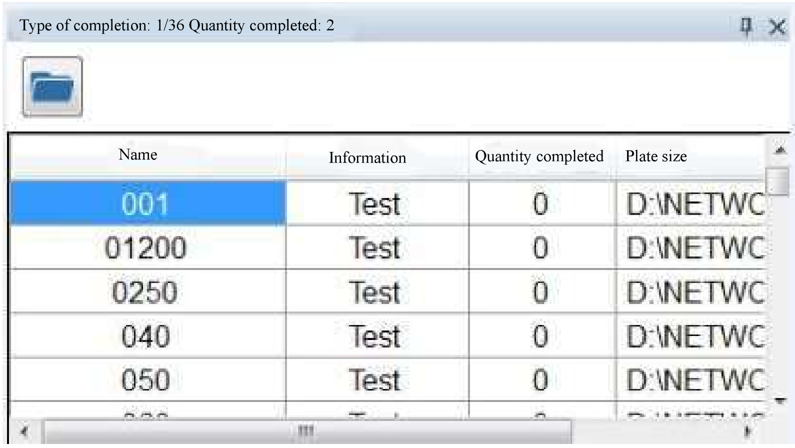
Discharging mode

Front discharging: After processing, the material is unloaded from the front end
Rear discharging: After processing, the material is unloaded from the rear end

Technique selection

Block front holes: When processing front sheets, ignore front holes.
Block back holes: When processing back sheets, ignore back holes.
Block horizontal holes: When processing side sheets, ignore side holes.
Block front slots: When processing front sheets, ignore front slots.
Block back slots: When processing back sheets, ignore back slots.
Block front internal milling: When processing front sheets, ignore front internal milling.
Block back internal milling: When processing back sheets, ignore internal milling.
Block back external milling: When processing back sheets, ignore external milling.
Block back external milling: When processing back sheets, ignore back external milling.

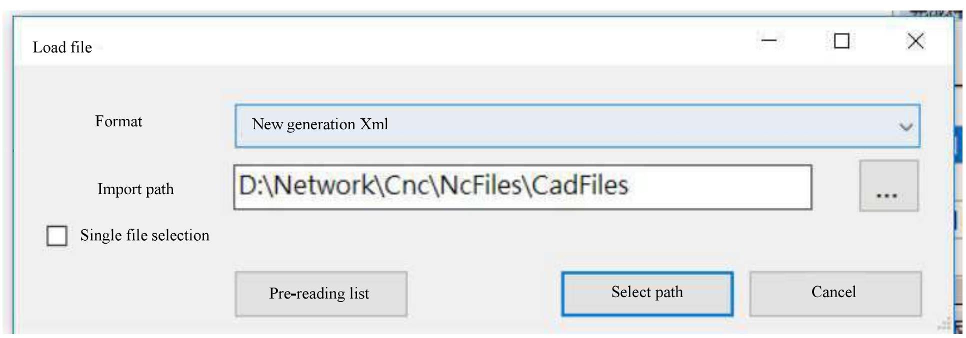
9.5.3 File List



Display information: Sheet name, information, completed quantity, path.

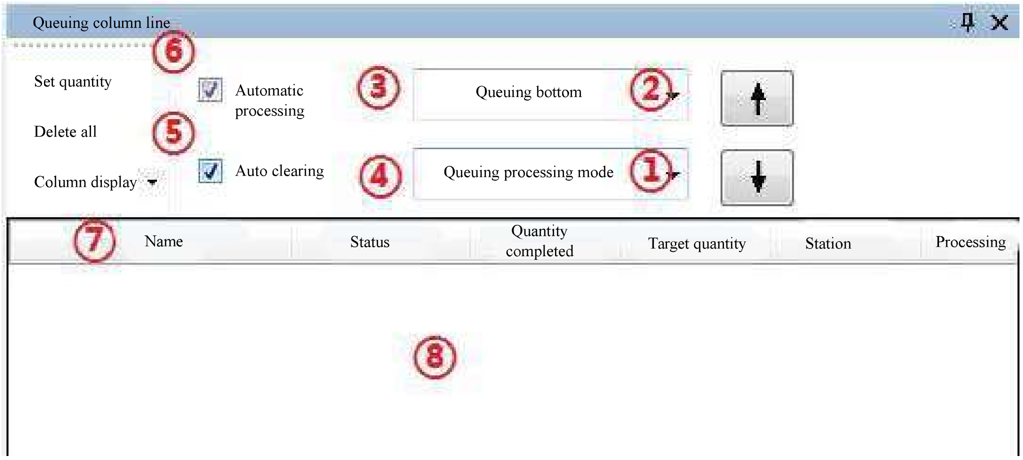
Type of completion: Remind that there are several types of sheets that have been processed, and count the types of sheets whose "Completed Quantity" is not 0.
Completed quantity: All types of sheets and plates, the total completed quantity of processing.

Click the "" icon to enter "Load File", as shown below:



Import steps: Select the format of the imported file → select the folder to be imported → press OK.
Single file selection: After checking it, you can select a single file to load.
Supported formats: Ender, Bpp, Xml, Xml, MPR Ban, Pytha DXF, Xml, PDX

9.5.4 Processing mode



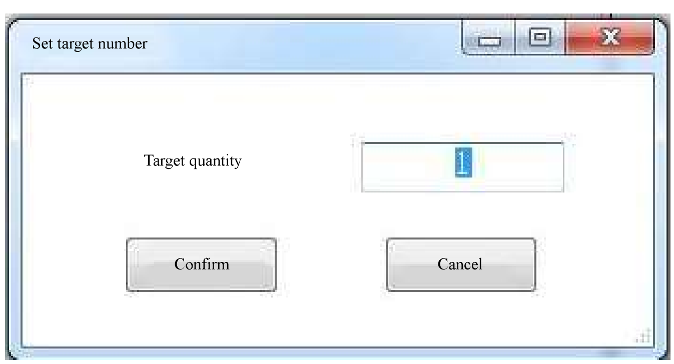
1) **Operation mode:**

- Queue processing mode: Keep processing the same sheet, and will not process the next sheet until the target quantity is reached.
- Cycle processing mode: After completing one sheet, skip to the next sheet processing and continue the cycle.

2) **Queue mode:**

- Line up to the top: Put new items to the top.
- Line up to the bottom: Put new items to the bottom.

3) **Automatic processing:** After the sheet and plate is processed, the next sheet and plate is processed immediately.
4) **Automatic clearing:** When the completed quantity of sheets reaches the target number, it will be automatically cleared from the standing row.
5) **Delete all:** Delete all items in the queue.
6) **Set Quantity:** Set the number of sheets in the row



7) **Stationed list:** Contains name, status, completed quantity, number of goals, information, path.

Status description:

- Completion: The processing of this project is completed.
- NC has been generated: The processing program of this project has been generated.
- Assigned processing: This item has been assigned to the main processing file.
- In progress: This item is being processed.
- File switchover failed: The file switchover of this item failed. You can right-click to display an error message to understand the reason.

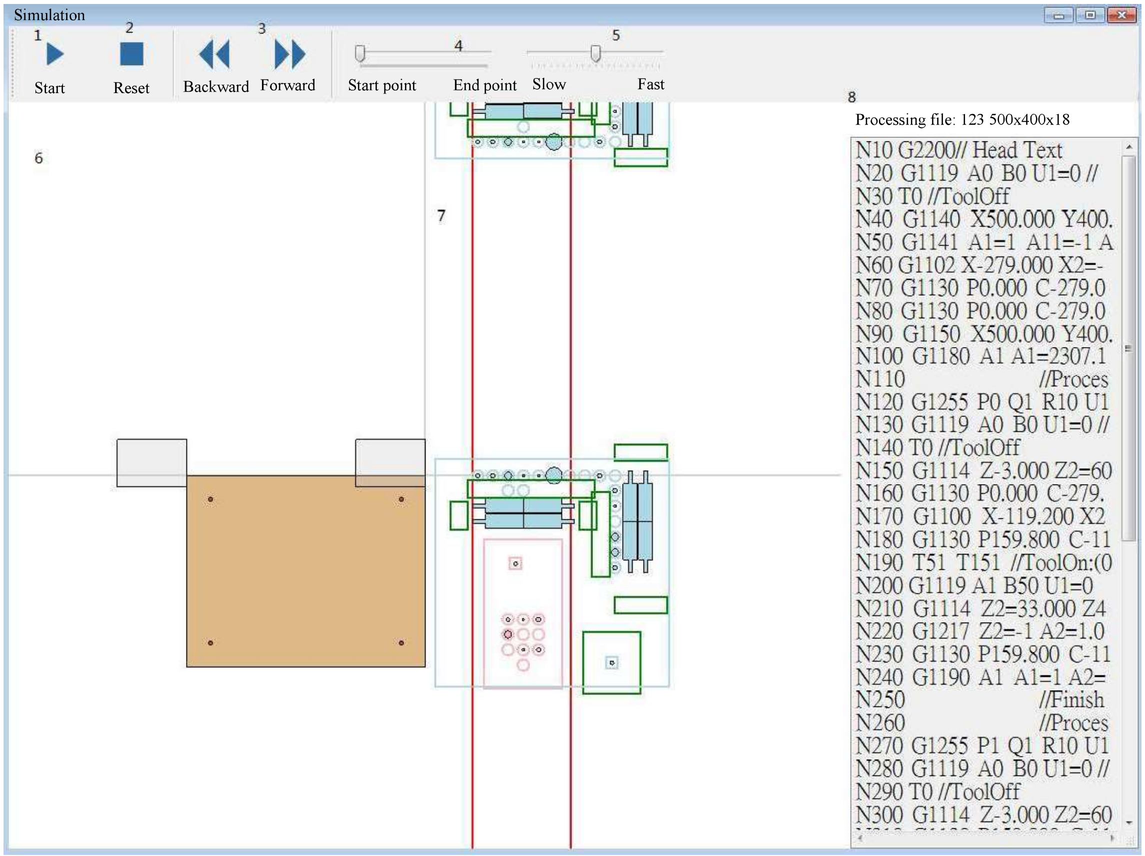
8) Right-click function:

Select a resident item and right-click it to pop up the following functional options:

- Display an error message: View the reason for the failure of the file switchover
- Set to top: Move this item to top of the row
- Reset completed quantity: Reset the completed quantity of this project to zero
- Specified processing: Specify the processing of this item
- Simulate: Simulate this project
- Delete: Delete this item
- Delete all: Delete all items in the resident column

9.5.5 Graphical simulation

Select a resident item, right-click, and select Simulate.



1) **Start:** Start the simulation, it will become the pause button when pressed

Pause: Pause the simulation, when pressed it becomes the start key

2) **Reset:** Bring the simulation back to the starting point

3) **Forward:** One time unit forward

Next: One time unit in the future

4) **Adjust the simulation progress:** Adjust each momentary picture from the start point to the end of the simulation

5) **Adjust the simulation speed:** The farther to the right, the faster

6) **Axial position information:** Display the position of each axis

7) **Simulation screen**

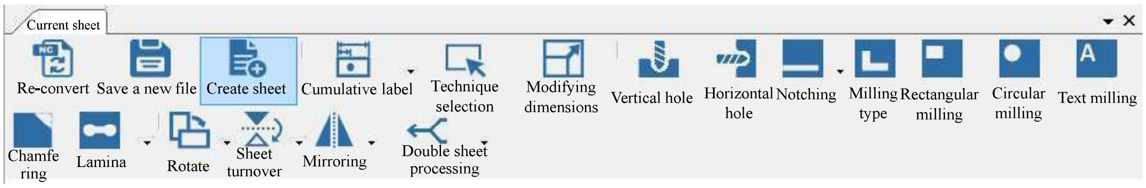
- Display graphic simulation
- Click and hold the mouse button to pan
- Scroll the mouse wheel to zoom

8) **NC**

- Display processing file information
- Click on any line of the simulation screen to start the simulated action of the line

Drawing area

The drawing area has the following functions:



1. Re-convert
2. Create sheet
3. Save a new file
4. Cumulative indication
5. Technique selection
6. Modifying dimensions
7. Vertical hole
8. Horizontal hole
9. Slotting
10. Milling type
11. Rectangular milling
12. Circular milling
13. Chamfering
14. Rotation
15. Mirror
16. Flipboard

9.4.1 Re-convert

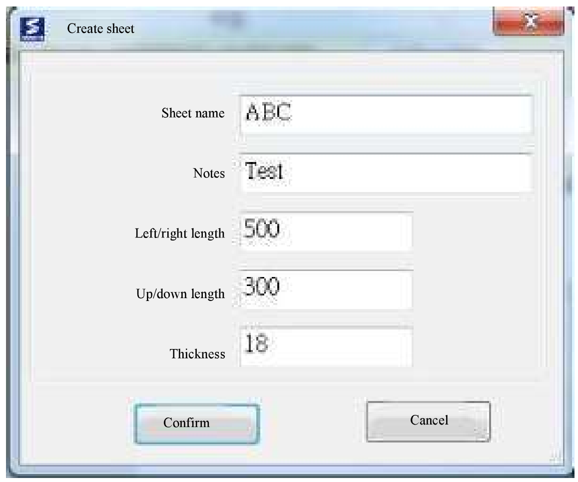
Click the " 

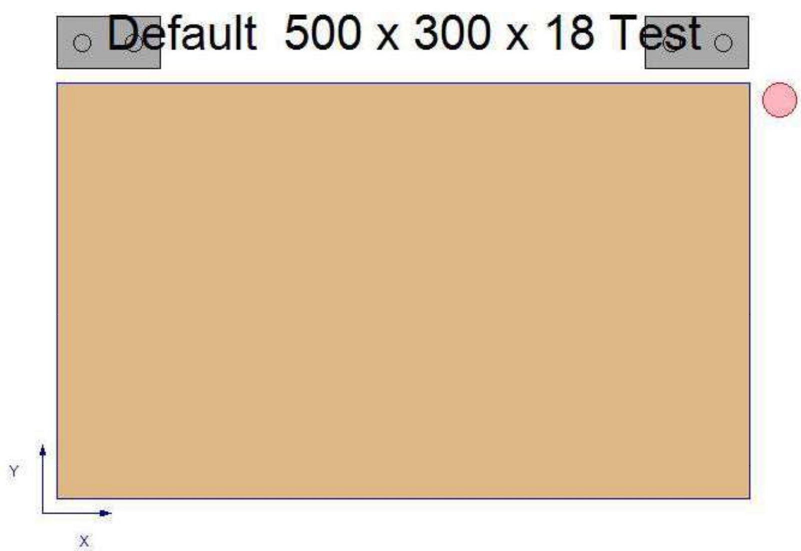
" icon to regenerate a new processing file.

9.4.2 Creating sheets

Click the " 

" icon, enter the sheet information, and create a new sheet.

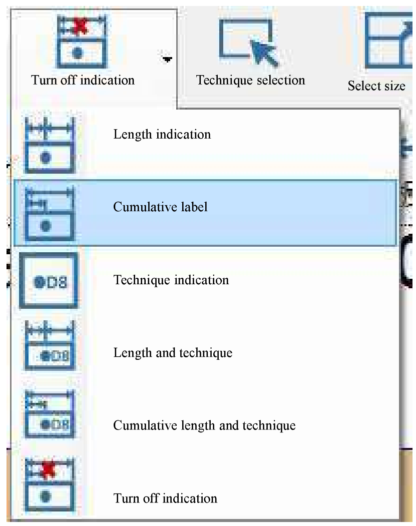




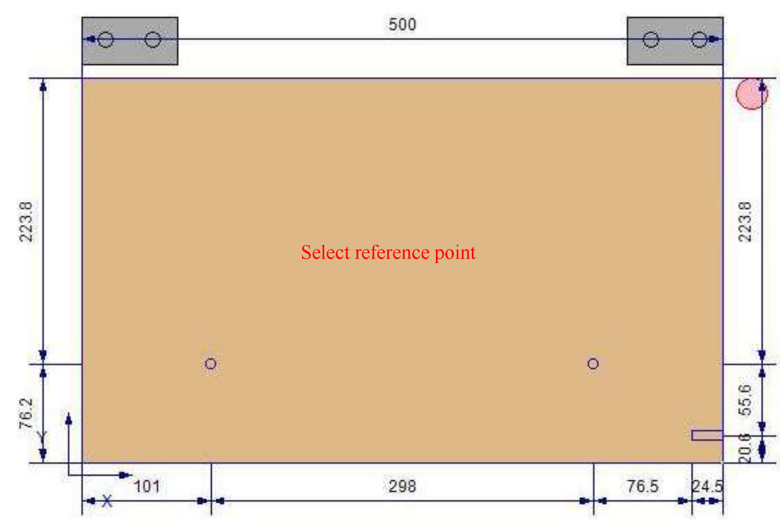
9.4.3 Save a new file

Click "Save a new file" to save the instance of the current project in the specified location in the specified file format.

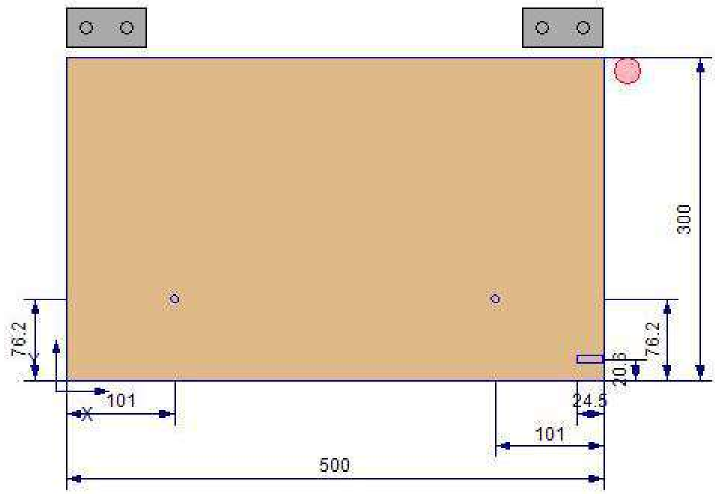
9.4.4 Indication



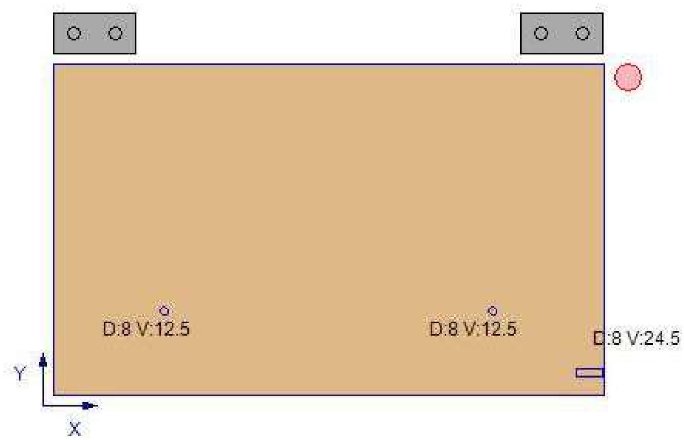
1. **Display length indication:** Indicate the distance between the craft and the edge of the sheet, and between the crafts.



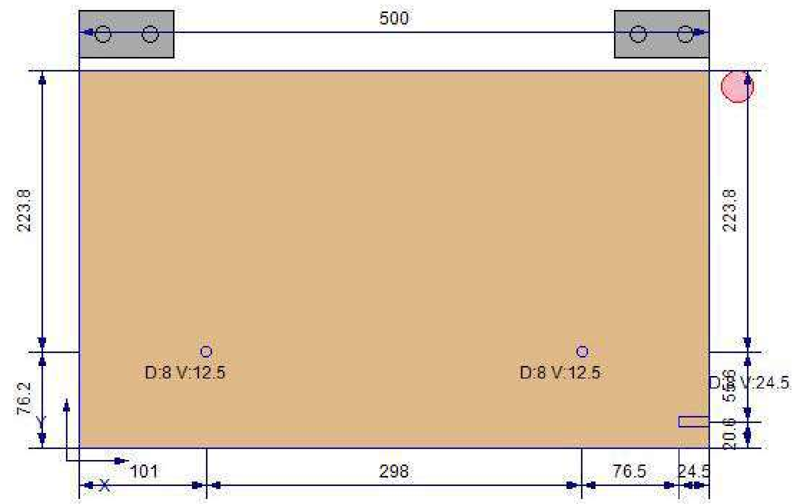
2. **Display length indication (additive):** Indicate the relative distance between the craft and the edge of the sheet.



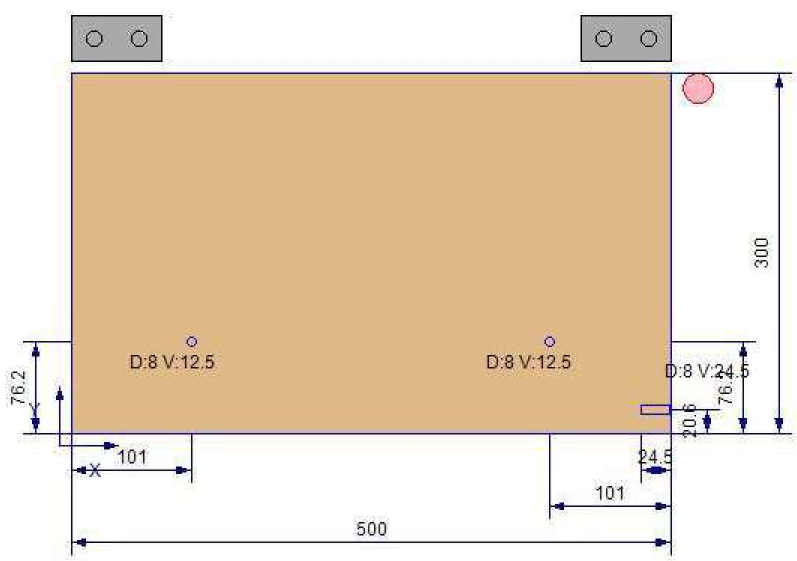
3. **Display technique indication:** Only the technique information (D: diameter, V: hole depth) is marked.



4. **Display length and technique:** Indicate the relative distance between the technique and the edge of the sheet, and indicate the technique information (D: diameter, V: hole depth).



5. **Display length and technique (cumulative):** Mark the relative distance between the technique and the edge of the sheet, and mark the technique information (D: diameter, V: hole depth).

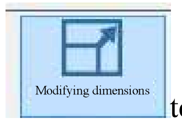


6. **Turn off the indication display: no indication is displayed.**

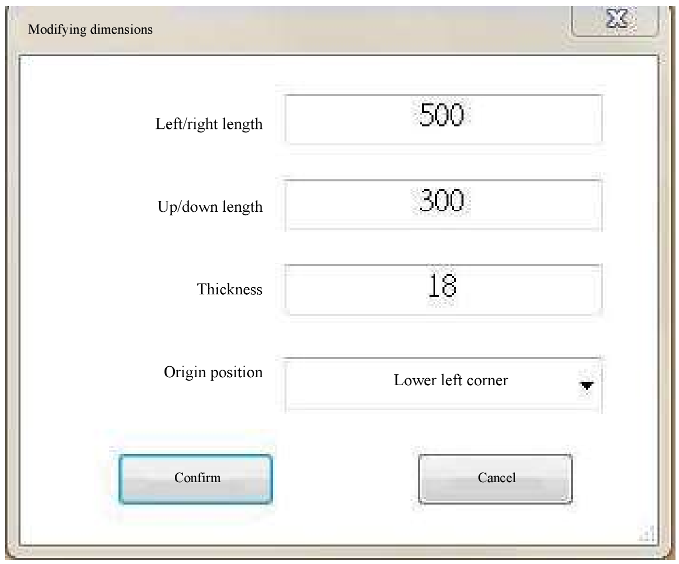
9.4.5 Select technique

Re-select any process.

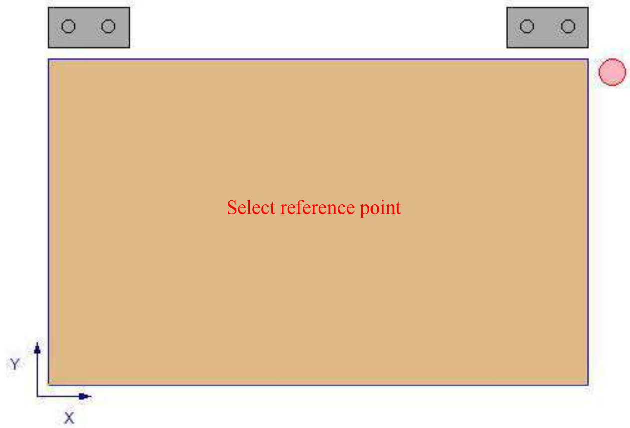
9.4.6 Modifying dimensions

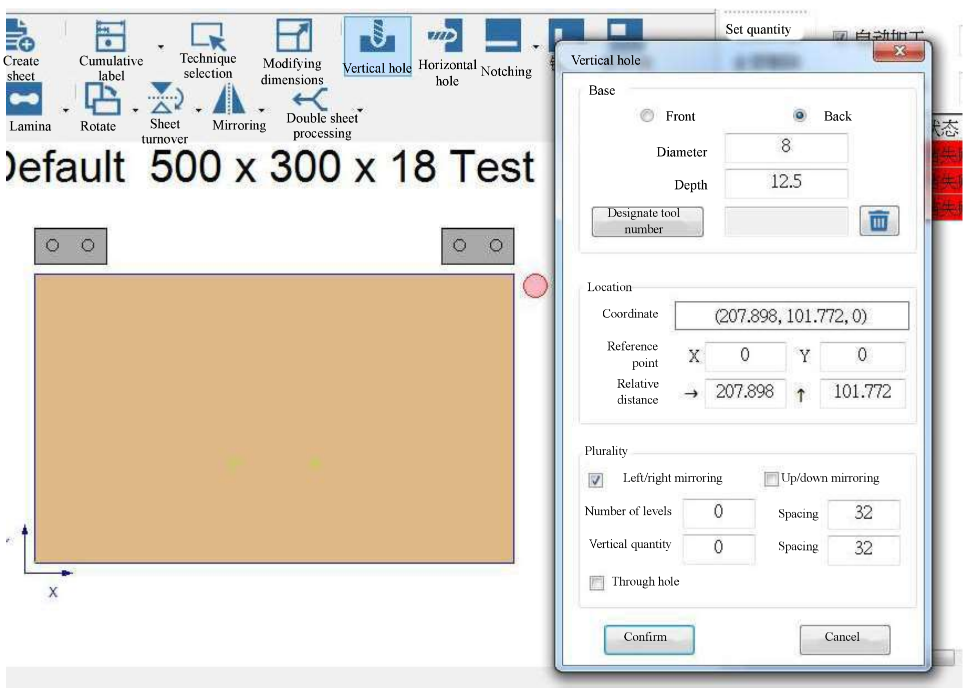
After creating or loading a sheet, if you need to modify the technique or sheet dimension, you can click 

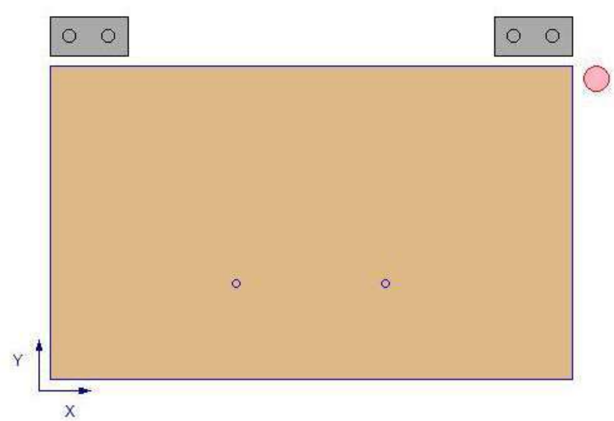
to modify it.



9.4.7 Vertical hole







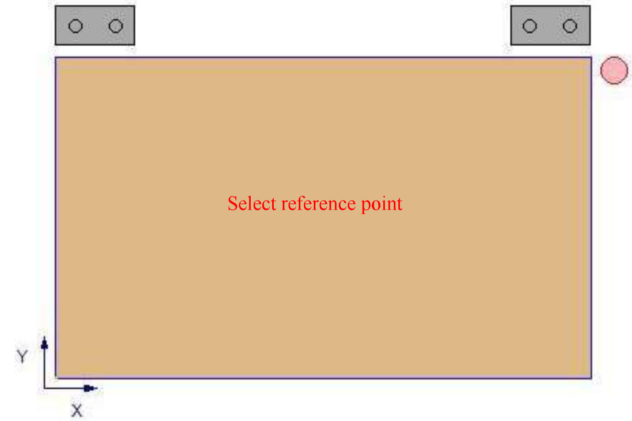
Step: Select the reference point (in any position of the sheet and plate) → select the process point → enter the related information of hole → after pressing confirm, a vertical hole has been established.

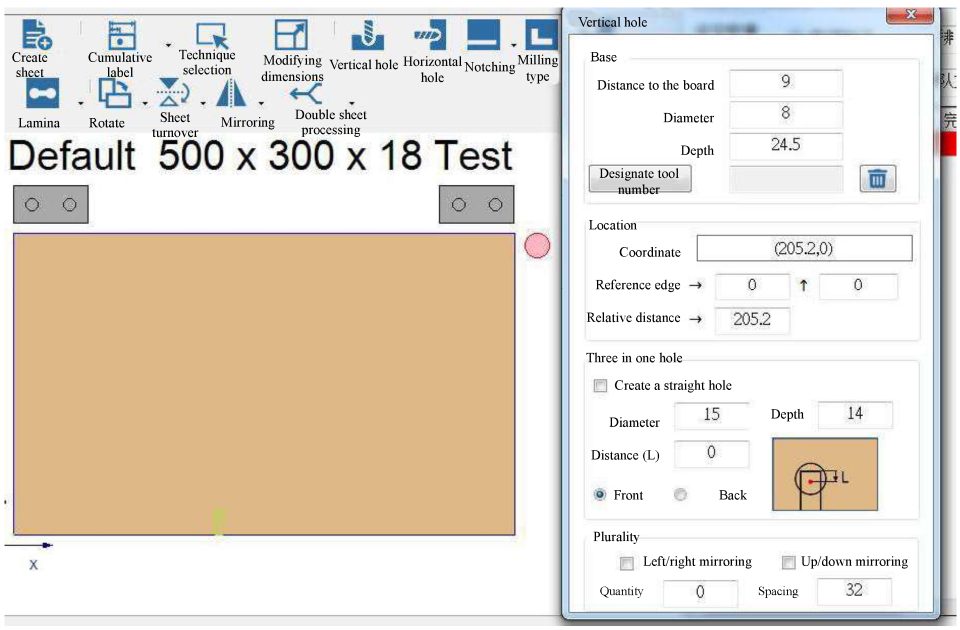
Related information of vertical hole:
Basics: Select hole location [front/back, enter hole diameter size and depth

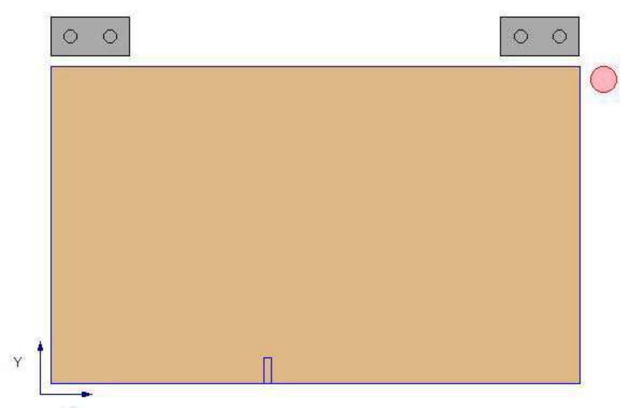
Location:
Coordinates: Display coordinates relative to the origin of the sheet (lower right corner)
Reference point: The position of the reference point
Relative distance: The relative distance between the process point and the reference edge

Plurality:
Left/right mirroring: Duplicate the same hole on the other side in the left/right direction
Up/down mirroring: Duplicate the same hole on the other side in the up/down direction
Horizontal quantity/vertical quantity/spacing: Copies the same hole with the entered amount of distance and direction
Through hole: a hole that is set to pass through

9.4.8 Horizontal holes







Step: Select the reference point (on the four sides of the sheet and plate) → select the process point → enter the related information of hole → after pressing confirm, a horizontal hole has been established

Related information of side hole:

Basics: Enter the distance from the board, the diameter and depth of the hole

Location:

Coordinates: Display the coordinates relative to the origin of the sheet (lower left corner)

Reference edge: The distance of the reference edge from the origin of the sheet (lower left corner)

Relative distance: The relative distance between the process point and the reference edge

Plurality:

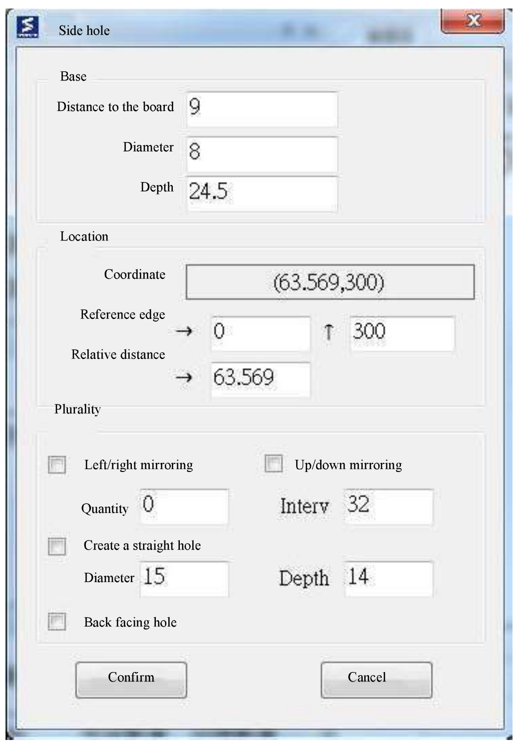
Left/right mirroring: Duplicate the same hole on the other side in the left/right direction

Up/down mirroring: Duplicate the same hole on the other side in the up/down direction

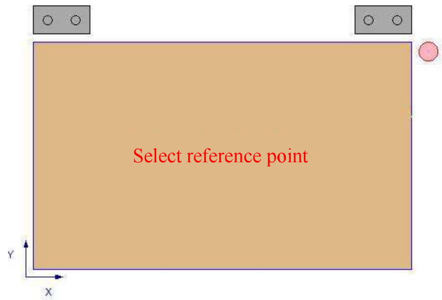
Quantity/spacing: Copy the same holes with the entered number of spacing and orientation

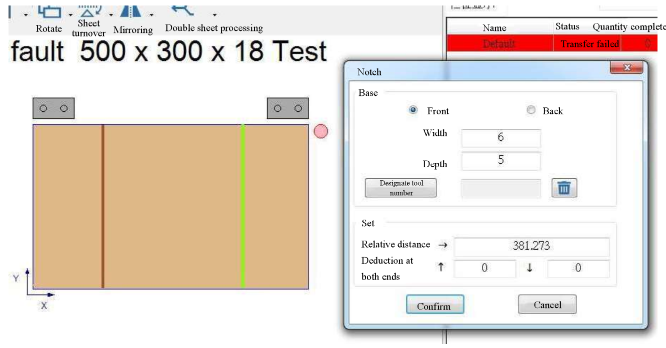
At the bottom of the horizontal hole, create a vertical hole

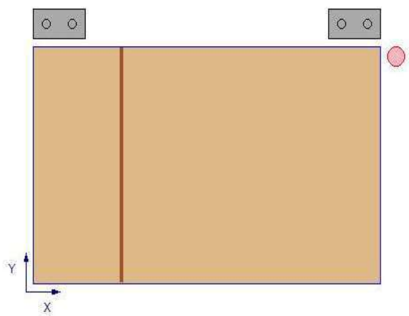
Create vertical holes on the back



9.4.9 Slotting



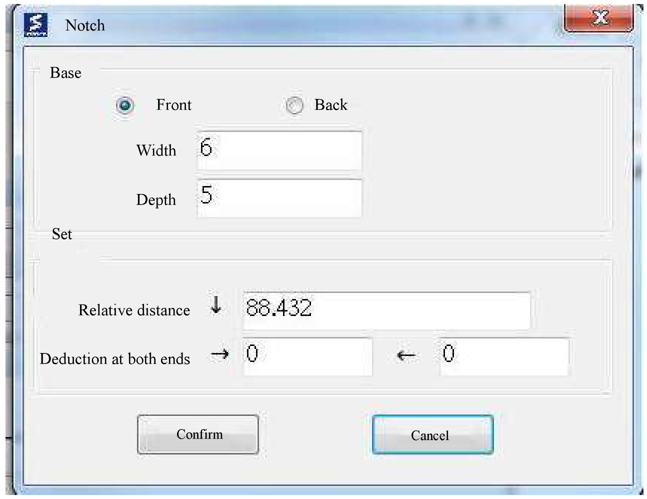




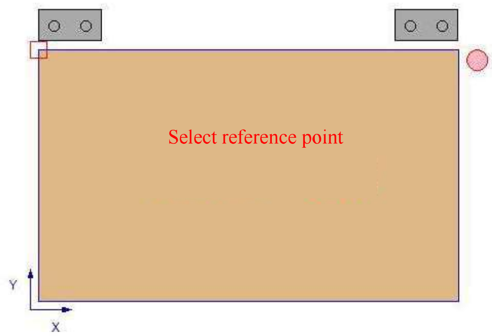
Steps: Select the reference point (on the four sides of the sheet) → select the technique point → enter the relevant information of the groove → press the confirmation, a slot has been established.

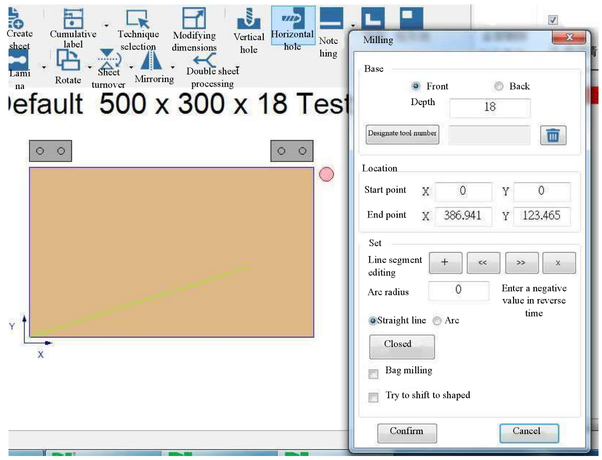
Related information of notch:
Basics: Select slot location [front/back, enter slot width and depth

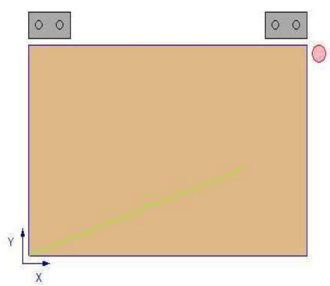
Set
Relative distance: The relative distance between the process point and the reference edge
Deduction at both ends: How far is the distance from the sides of the input notch to be inward?


Notch
Base
Front
Back
Width
6
Depth
5
Set
Relative distance
88.432
Deduction at both ends
0
0
Confirm
Cancel

9.4.10 Milling type







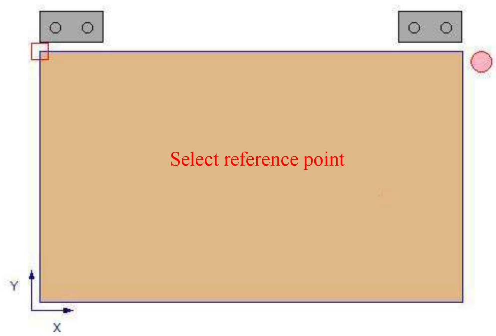
Steps: Select the start point → select the endpoint → enter the related information of milling type → after pressing confirm, a milling type has been established.

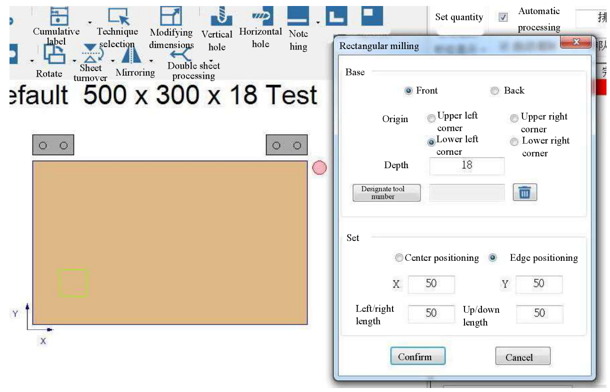
Related information of milling type:
Basics: Select milling type position [front/back, enter slot width and depth

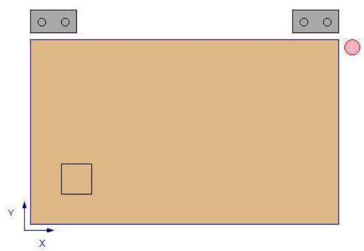
Location:
Start/End: Enter X, Y position

Set
Line segment editing: + add line, <</>> select previous or next segment, x delete line segment
Arc radius: Only applicable to the arc option, the arc radius can be adjusted (if the input value is less than the minimum arc radius, it will be changed to the minimum arc radius)
Line/arc/round: milling type line segment type
Closed: Add a new line segment to close the two line segments

9.4.11 Rectangular milling





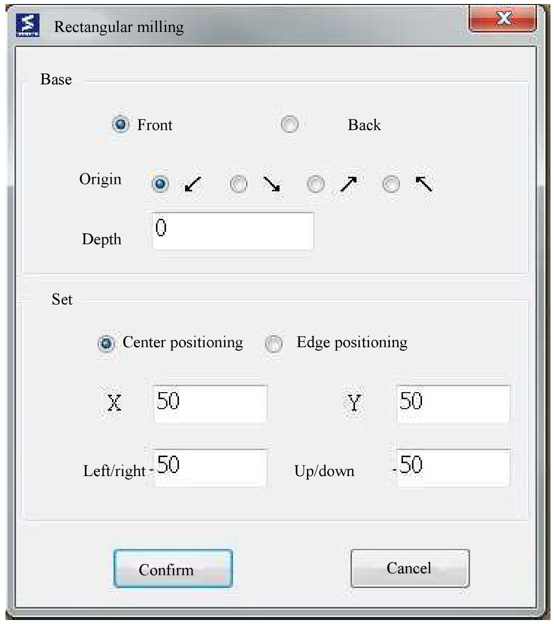


Steps: Select the reference point (at the four corners of the sheet and plate) → enter the relevant information of rectangular milling → after pressing confirm, and a rectangular milling has been established.

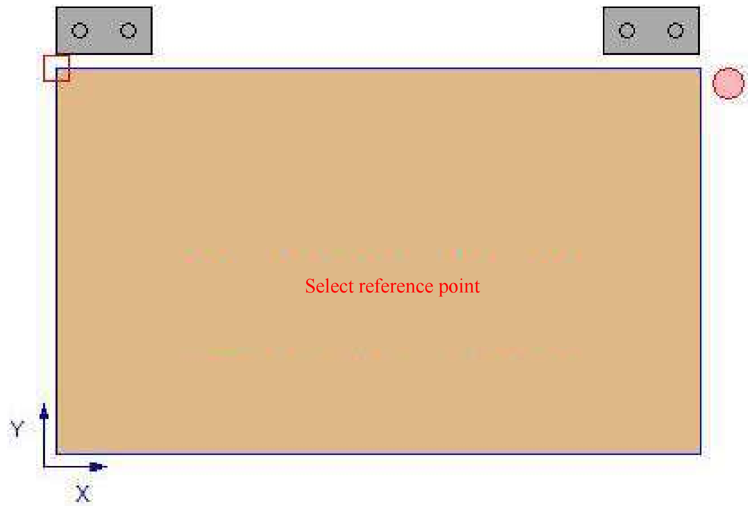
Related information of rectangular milling:
Basic: Select the rectangular milling position [front/back, the origin position (consistent with the selected reference point) and the depth.

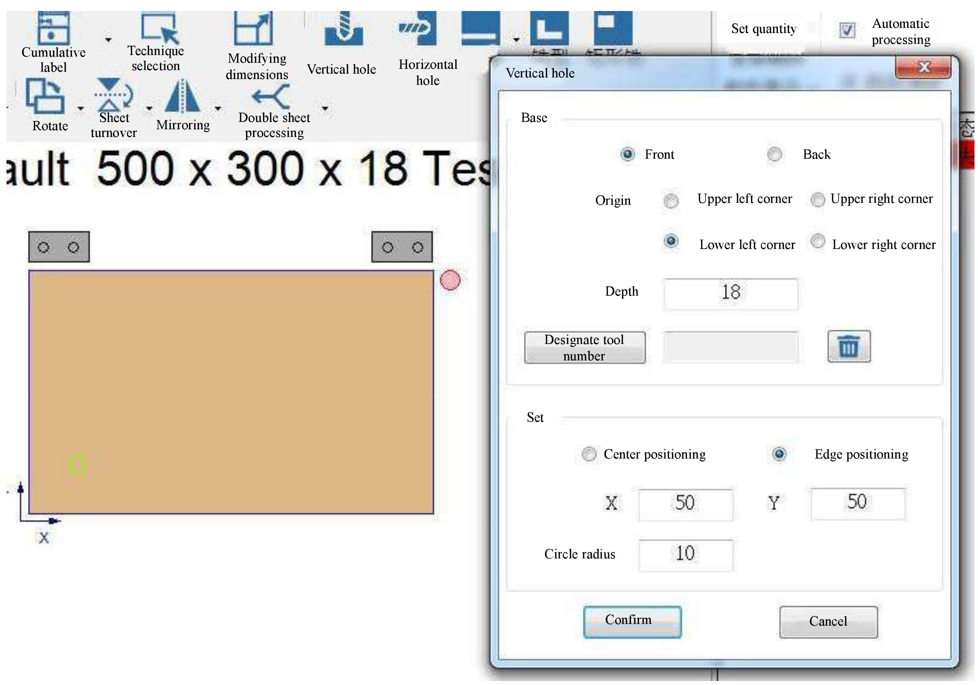
Settings: Center Positioning/Edge Positioning: Select the positioning points of the rectangle X, Y

Left/right/up/down: Left-right distance and up-down distance of rectangle.


Rectangular milling
Base
Front
Back
Origin
Depth
0
Set
Center positioning
Edge positioning
X
50
Y
50
Left/right
50
Up/down
50
Confirm
Cancel

9.4.12 Circular milling





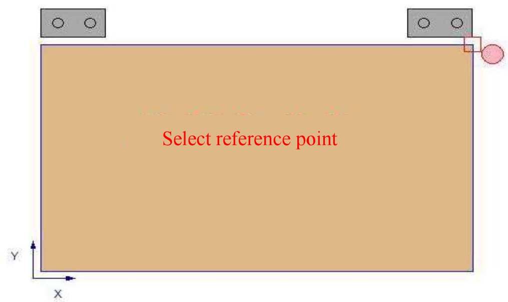
Steps: select the start point (four corner points) → select the end point → enter the milling type related information → after pressing confirm, a milling type has been established.

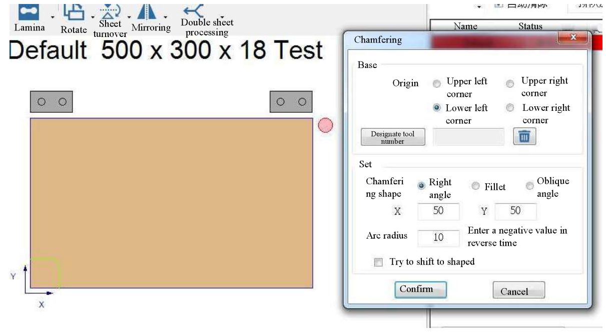
Circular milling related information:
Basics: Select the circular milling location (front, back), origin, and depth.

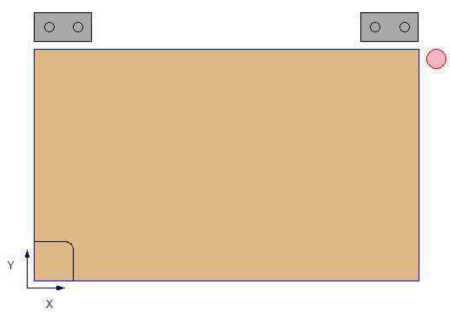
Setting: Center positioning/edge positioning: Select the center of the circle as the positioning or the edge of the circle as the positioning.

According to the positioning method, select the X and Y coordinates of the circle.
Circle radius

9.4.13 Chamfering





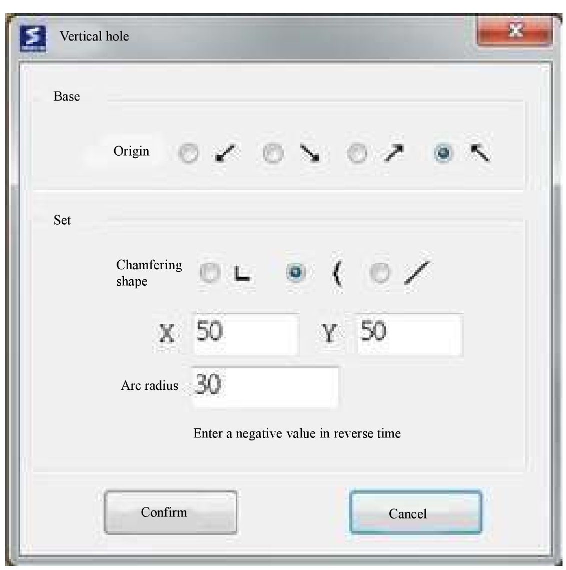


Step: Select the reference point (at the four corners of the sheet and plate) → enter the relevant information of cutting angle → after pressing confirm, a cutting angle has been established.

Basic: Origin position (consistent with the selected reference point)

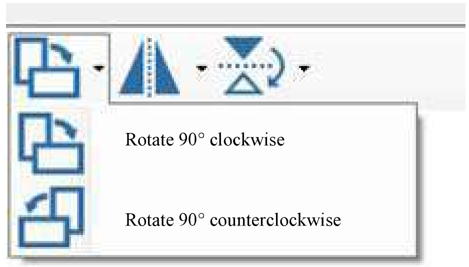
Setting: Chamfering shape/position: Select the chamfering shape and enter the X/ Y position

Arc radius: Enter the chamfering arc radius, if it is negative, the arc will be drawn counterclockwise



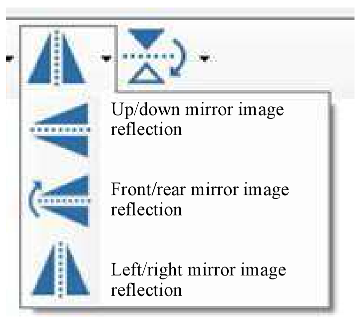
9.4.14 Rotation

Rotate the sheet in the selected direction



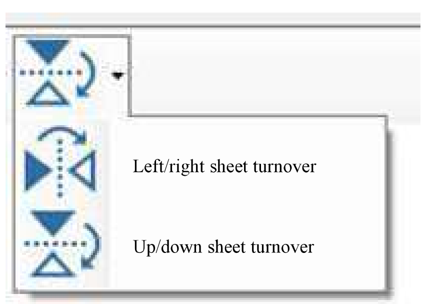
9.4.15 Mirroring

Mirror image reflection of the sheet in the selected direction



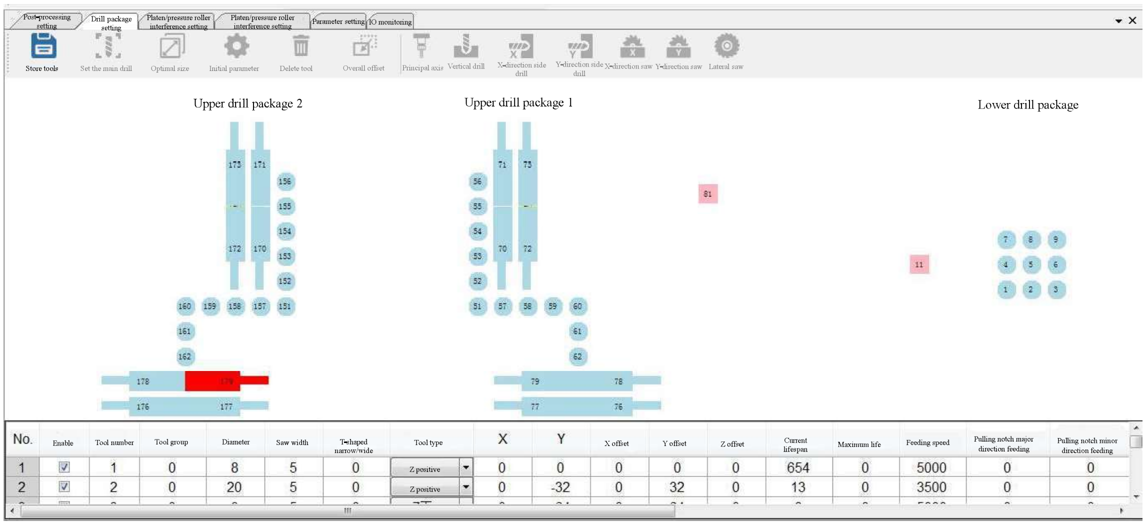
9.4.16 Sheet turnover

Sheet turnover in the selected direction



Drill package setting

Path: Parameter setting **-** drill package setting

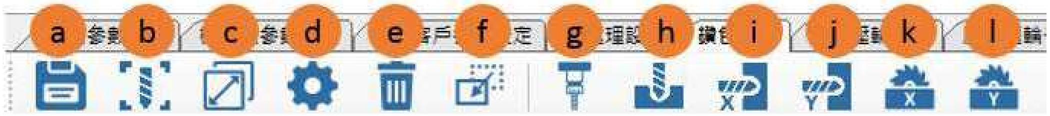


Tool number description:

1. The vertical drill tool number of lower drill package is T1 to T10, and the tool type is Z+ (representing tool nose upward); The tool number of the spindle for lower drill package is T11, and the tool type is spindle Z+ (representing tool nose upward).
2. The vertical drill tool number of the upper drill package 1 (Y1) is T51 to T66, the tool type: Z**-** (representing tool nose downward);
   The horizontal drill tool numbers are T70 to T79; the upper spindle tool numbers are T81 Tool types: spindle Z**-** (representing tool nose downward).
3. The vertical drill tool numbers of the upper drill package 2 (Y2) are T151 to T166; the horizontal drill tool numbers are T170 to T179.
4. Special tool number: T90, T96, T190;
   - T90: It is the simultaneously-moving tool of the upper drill package 1 (Y1), which is used to press the back hole with the movement of the lower drill package. The Z offset of T90 = (Z1 axis workpiece coordinate + safety height).
   - T190: It is the simultaneously-moving tool of the upper drill package 2 (Y2), which is used to press the back hole with the movement of the lower drill package. The Z offset of T190 = (Z2 axis workpiece coordinate + safety height).
   - T96: It is the simultaneously-moving tool when the lower spindle is grooving, which is used to ensure that the upper drill package can always follow the movement and the platen descends and floats on the surface of the sheet when the lower spindle is grooving; Z offset of T96 =(Z1 axis workpiece coordinate + safe height).

Notice: The tool radius of T90, T96 and T190 must be 0, and no value can be set, otherwise the effect will be invalid.

9.5.1 Operation setting

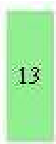


- **a. Save:** Save tool data;
- **b. Set the main drill:** Indicate a vertical drill as the reference drill;
- **c. Optimum size:** Each drill package displays the most appropriate size and position;
- **d. Preset parameter setting:** set the preset parameters when adding a new tool;
- **e.** Remove tool: delete the selected tool;
- **f. Overall offset:** According to the selected tool, make the overall offset of all the tools of the drill package;
- **g. Add spindle:** After clicking, press the left button in the drill package icon to add a new spindle;
- **h. Add vertical drill:** After clicking, press the left button in the drill package icon to add a vertical drill;
- **i. Add X-direction side drill:** After clicking, press the left button in the drill package icon to add X+ and X- side drills at the same time;
- **j. Add Y-direction side drill:** After clicking, press the left button in the drill package icon to add Y+ and Y- side drills at the same time;
- **k. Add X-direction saw blade:** After clicking, press the left button in the drill package icon to add an X-direction saw blade;
- **l. Add Y-direction saw blade:** After clicking, press the left button in the drill package icon to add a Y-direction saw blade.

9.5.2 Illustration of drill box

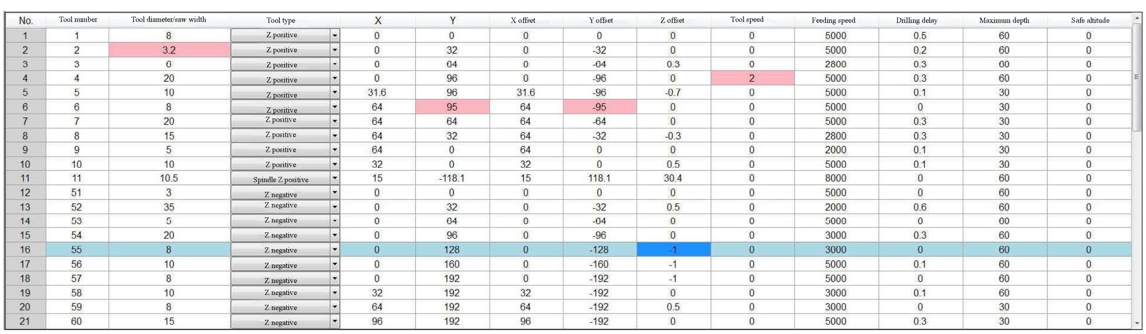
Displays a schematic diagram of each drill package (X+ up, Y+ left).

Illustration	Description
	Vertical Drill (Z+ or Z-)
	Principal axis
	Y-direction side drill (Y+)
	Y-direction side drill (Y+)

Illustration	**Description**
	X-direction side drill (X-)
	X-direction side drill (X-)
	Y-direction saw blade
	X-direction saw blade

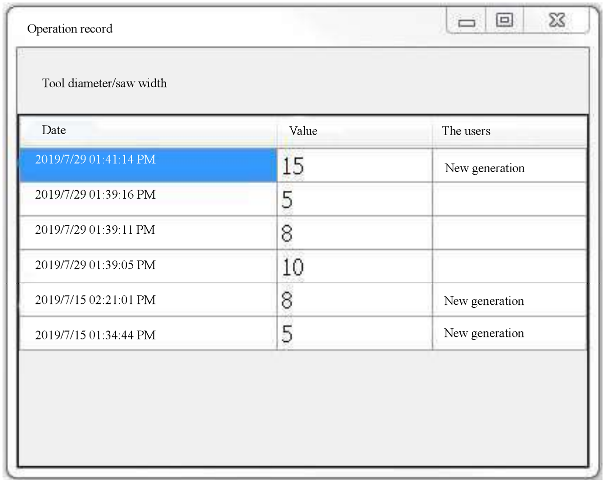
9.5.3 Tool data sheet

Set the parameters of each tool
After modifying the parameters, the background color of the grid will turn pink, prompting the user that this parameter has been modified. After pressing "Save", the background color will be removed.



No.	Tool number	Tool diameter/saw width	Tool type	X	Y	X offset	Y offset	Z offset	Tool speed	Feeding speed	Drilling delay	Maximum depth	Safe altitude
1	1	8	Z positive	0	0	0	0	0	0	5000	0.5	60	0
2	2	3.2	Z positive	0	32	0	-32	0	0	5000	0.2	60	0
3	3	6	Z positive	0	64	0	-64	0.3	0	2800	0.3	60	0
4	4	20	Z positive	0	96	0	-96	0	2	5000	0.3	60	0
5	5	10	Z positive	31.6	96	31.6	-96	-0.7	0	5000	0.1	30	0
6	6	8	Z positive	64	95	64	-95	0	0	5000	0	30	0
7	7	20	Z positive	64	64	64	-64	0	0	5000	0.3	30	0
8	8	15	Z positive	64	32	64	-32	-0.3	0	2800	0.3	30	0
9	9	5	Z positive	64	0	64	0	0	0	2000	0.1	30	0
10	10	10	Z positive	32	0	32	0	0.5	0	5000	0.1	30	0
11	11	10.5	Spindle Z positive	15	-118.1	15	118.1	30.4	0	8000	0	60	0
12	51	3	Z negative	0	0	0	0	0	0	5000	0	60	0
13	52	35	Z negative	0	32	0	-32	0.5	0	2000	0.6	60	0
14	53	5	Z negative	0	64	0	-64	0	0	5000	0	60	0
15	54	20	Z negative	0	96	0	-96	0	0	3000	0.3	60	0
16	55	8	Z negative	0	128	0	-128	-1	0	3000	0	60	0
17	56	10	Z negative	0	160	0	-160	-1	0	5000	0.1	60	0
18	57	8	Z negative	0	192	0	-192	-1	0	5000	0	60	0
19	58	10	Z negative	32	192	32	-192	0	0	3000	0.1	60	0
20	59	8	Z negative	64	192	64	-192	0.5	0	3000	0	30	0
21	60	15	Z negative	96	192	96	-192	0	0	5000	0.3	30	0

Right-click on any parameter of the tool to display the past operation records.

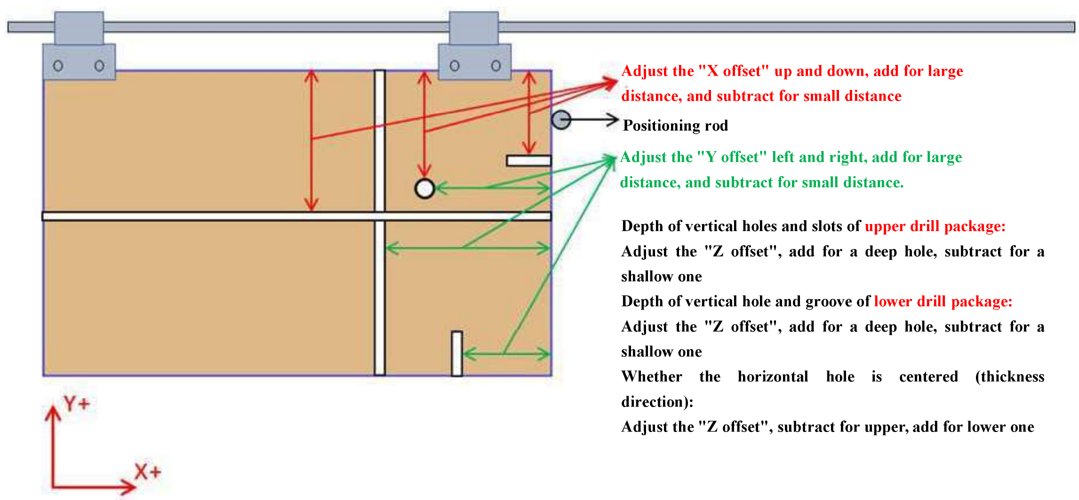

Operation record
Tool diameter/saw width
Date
Value
The users
2019/7/29 01:41:14 PM
15
New generation
2019/7/29 01:39:16 PM
5
2019/7/29 01:39:11 PM
8
2019/7/29 01:39:05 PM
10
2019/7/15 02:21:01 PM
8
New generation
2019/7/15 01:34:44 PM
5
New generation

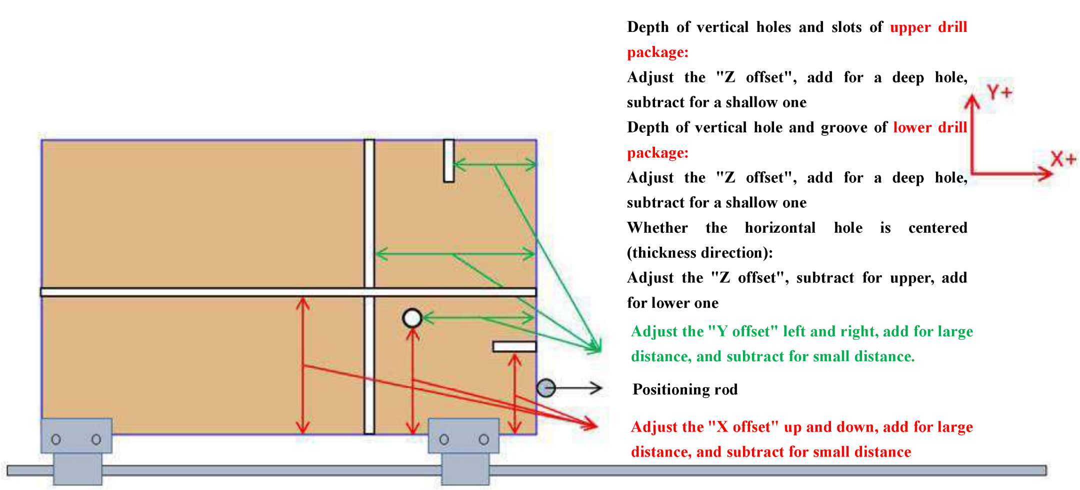
Machining accuracy confirmation

1. To adjust the accuracy, first adjust the accuracy of the **reference tool**, and then adjust the accuracy of the horizontal tool and the spindle;
2. **Accuracy overall offset**, adjust G54 coordinates and parameters 4711/4714/4720;

- Parameter 4711: reference point X of lower drill box 1;
- Parameter 4714, reference point X of the upper drill box 1.
- Parameter 4720, reference point X of the upper drill box 2.

3. If a single tool is not allowed, adjust the tool data sheet.





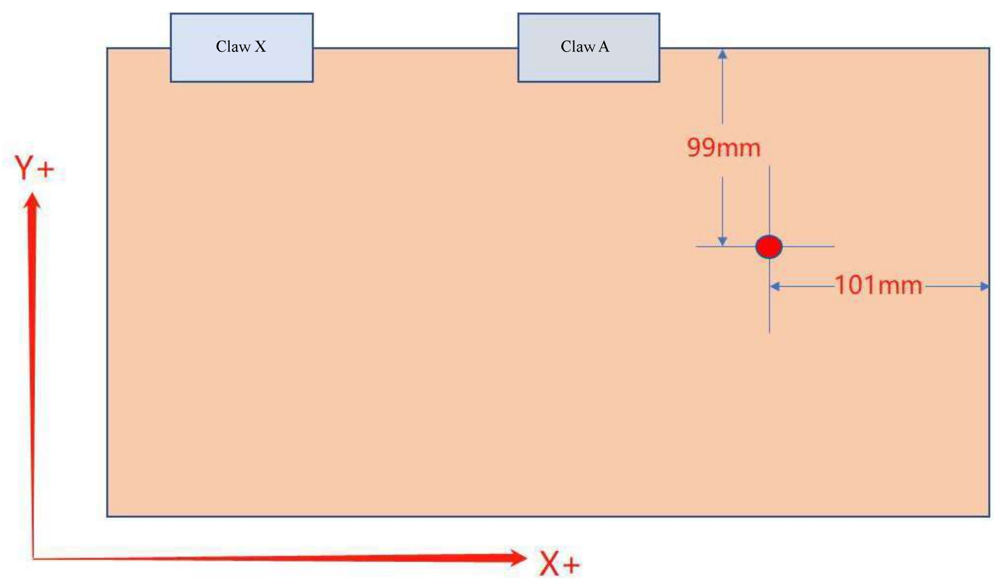
9.6.1 Vertical hole X axis, Y axis position adjustment

If the positional accuracy of the X axis and Y axis of the vertical hole is not accurate, it is only necessary to adjust the accuracy of the reference tool (T1/T51/T151) of each drill box; other vertical tools do not need to be adjusted. Because the spacing between other vertical tools and the reference tool is an integral multiple of 32 mm, which has been set in the tool list, so the drilling precision of the reference tool is accurate, and the drilling precision of other vertical tools is accurate;

Note: If the precision of the reference tool is accurate, and the precision of other vertical tools is inaccurate at this time, you need to modify the X offset and Y offset of the tool data list according to the actual situation.

1. It is expected to drill a horizontal hole with a depth of 100mm at a distance of 100mm away from the gripper side and 100mm from the sheet head;

2. After the actual processing, it is found that the accuracy is not accurate, the X-direction is 1mm more, and the Y direction is 1mm less, as shown below:



3. Y direction: Modify the external coordinate offset in the Y direction from 0 to -1, as shown in the figure below:

4. X direction: Modify parameter 4714, change 166900 (representing 166.9 mm) to 165900 (representing 165.9 mm).

No.	Parameter number	Parameter description	Value
		Cutter	
1	4526	Distance from reference tool to front edge of front platen	0
2	4711	Lower drill package 1 reference point X	294900
3	4714	Upper drill package 1 reference point X	166900
4	4720	Upper drill package 2 reference point X	166900
5	4900	Feeding speed	0
6	4906	Saw blade cutting speed (mm/min)m	0
7	4907	Spindle cutting rate (mm/min)	0
8	4958	The "through" technique is changed to up/down parts (bit control hole +1 slot +2 milling type +4).	0
9	4959	Overlapping amount (length: positive, percentage: negative) when "through technique" is changed to "up/down technique"	0
10	4963	Combine "through" and "vertical" techniques (No =0, drilling +1, slotting +2, milling type +4, turning front +128, turning back +256)	0
11	4964	Merging through the vertical process, sheet and plate will not be merged if the thickness reaches this upper limit	0
12	4965	Merging through the vertical process, add additional depth	0
13	4966	"Through vertical" technique face changing (no =0, drilling +1, slotting +2, milling type +4, turning front +128, turning back +256)	0
		Claw	
1	4520	Claw X interference distance	10000
2	4560	Claw Y interference distance	15000
3	4521	Claw X length	145000
4	4522	Claw Y length	33000
5	4523	Claw Z thickness	30000
6	4524	Minimum clamping distance in the X-direction of the claw	30000
7	4561	Minimum clamping distance in the Y-direction of the claw	0
8	4529	Claw is at the G54 positive limit position	2330000
9	4530	Claw is in the negative limit position of G54	-1800000
10	4556	Allow a single claw to feed plate (No=0, Yes=1)	0

9.6.2 Vertical hole Z axis depth adjustment

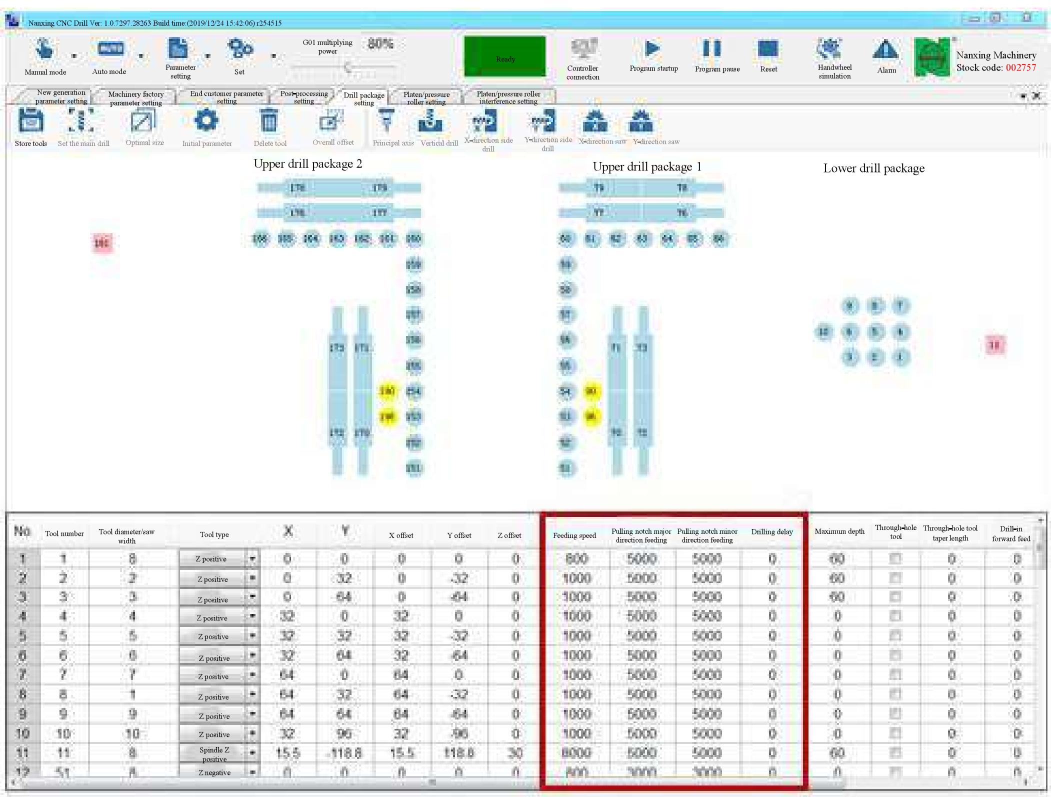
Take the upper drill box 1 as an example, and its reference tool is T51 with a diameter of 8 mm.

Use the reference tool T51 to drill a 10 mm deep hole, the actual measurement is only 9 mm, which is 1 mm shallower; then modify the external coordinate offset, modify the external coordinate offset of Z from 0 to -1, as shown in the following figure:

Operation path: Machine setting - workpiece coordinate system

Although the tool noses of all vertical tools (including T51) are on the same horizontal plane, the tool diameters are different. Drill at the same speed. If the air pressure is not particularly sufficient, a tool with a larger tool diameter (such as a 20 mm knife) will drill shallower than a tool with a smaller tool diameter (such as a 5 mm knife).

Therefore, the feeding speed shall be modified according to the actual situation, or the hole bottom delay shall be set.



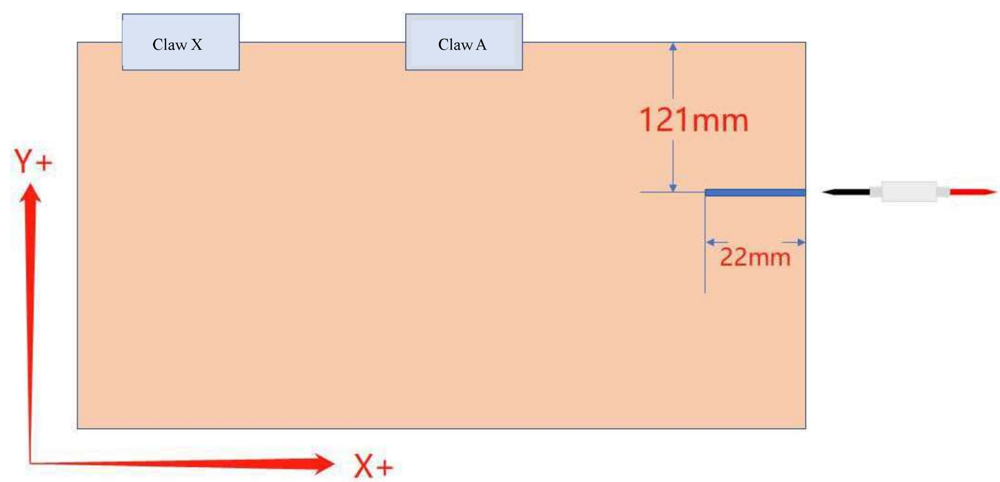
Notice: It is also possible that the tool noses of other tools and the reference tool are not on the same level. In this case, the Z offset of the tool data sheet needs to be modified according to the actual situation;

9.6.3 Side hole accuracy adjustment

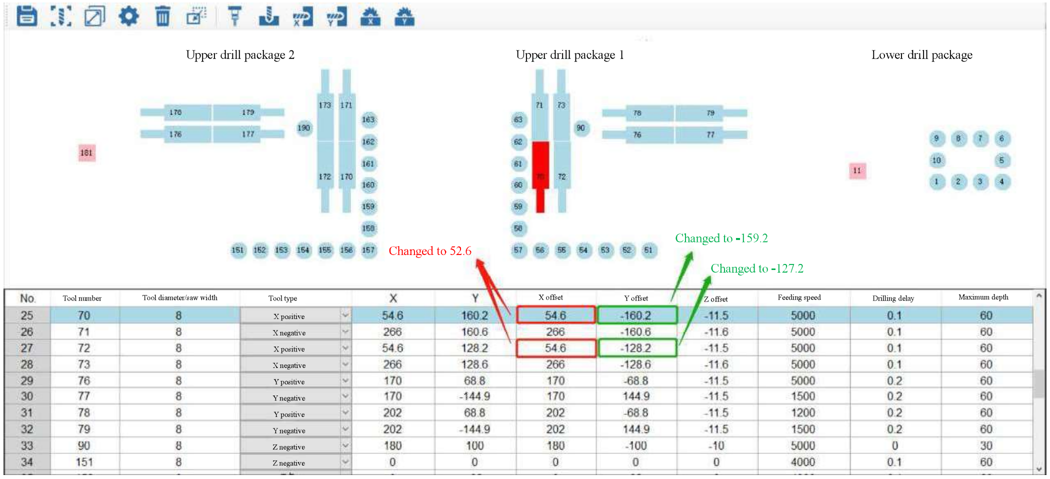
The horizontal tool diameter is calculated in 8mm here;

Accuracy adjustment of X-direction side hole

1. It is expected to drill a horizontal hole with a depth of 20mm at a distance of 120mm from the gripper side;
2. After the actual processing, it is found that the accuracy is not accurate, the X-direction is 2mm deep, and the Y-direction is 1mm more, as shown below:



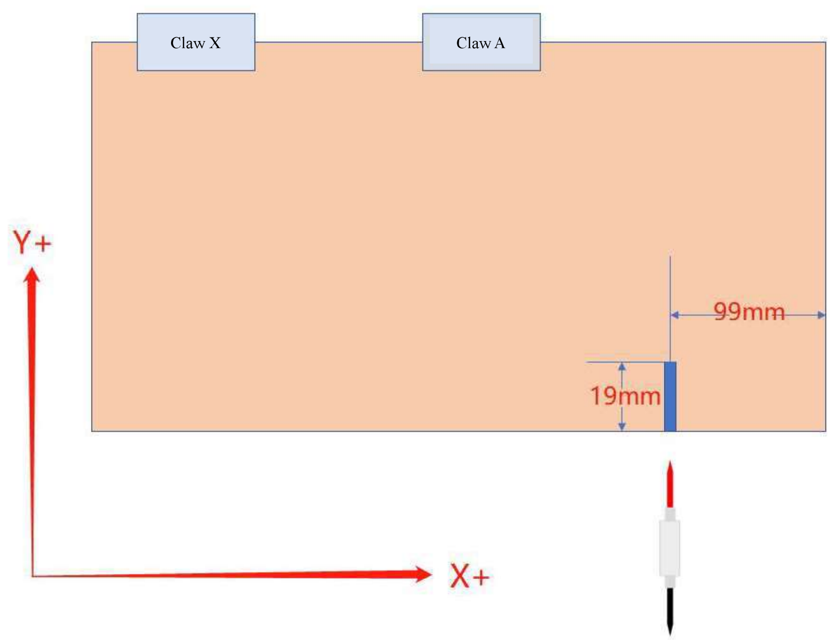
3. Enter the tool data list and find the corresponding tool (this example shows the horizontal hole in the positive X direction, as shown below):
4. At this time, we hope that the corresponding tool moves 2mm in the X+ direction and 1mm in the Y+ direction, so that the hole position is correct;
5. Modify the X offset and Y offset of the tool list, subtract 2 from the X offset, and add 1 to the Y offset;
6. Both T70 and T72 are horizontal tools in the positive X-direction, but the difference in the Y-direction is 32mm, which can be modified together.



No.	Tool number	Tool diameter/saw width	Tool type	X	Y	X offset	Y offset	Z offset	Feeding speed	Drilling delay	Maximum depth
25	70	8	X positive	54.6	160.2	54.6	-160.2	-11.5	5000	0.1	60
26	71	8	X negative	266	160.6	266	-160.6	-11.6	5000	0.1	60
27	72	8	X positive	54.6	128.2	54.6	-128.2	-11.5	5000	0.1	60
28	73	8	X negative	266	128.6	266	-128.6	-11.6	5000	0.1	60
29	76	8	Y positive	170	68.8	170	-68.8	-11.5	5000	0.2	60
30	77	8	Y negative	170	-144.9	170	144.9	-11.5	1500	0.2	60
31	78	8	Y positive	202	68.8	202	-68.8	-11.5	1200	0.2	60
32	79	8	Y negative	202	-144.9	202	144.9	-11.5	1500	0.2	60
33	90	8	Z negative	180	100	180	-100	-10	5000	0	30
34	151	8	Z negative	0	0	0	0	0	4000	0.1	60

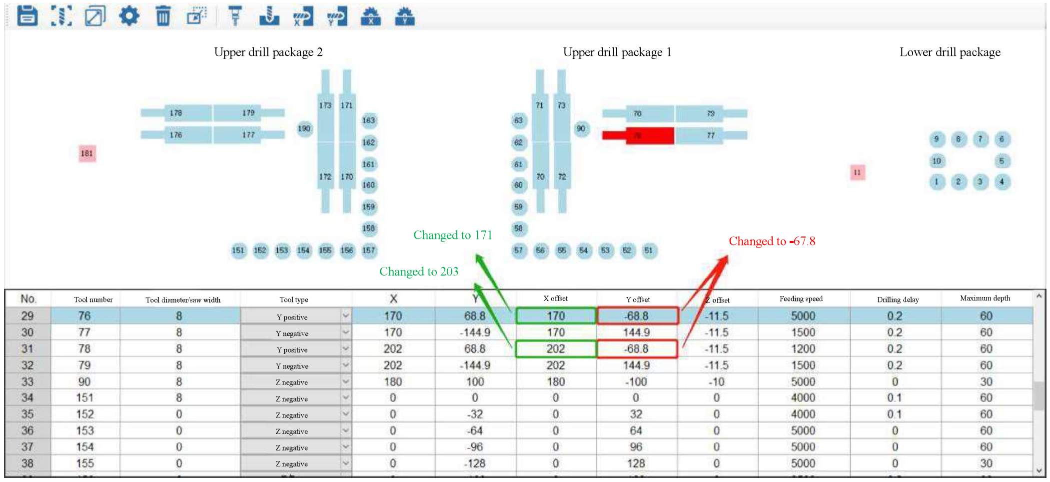
Accuracy adjustment of Y-direction side hole

1. It is expected to drill a horizontal hole with a depth of 20mm at a distance of 100mm from the sheet head;
2. After the actual processing, it is found that the accuracy is not accurate, and the X and Y directions are both less than 1mm, as shown in the following figure:



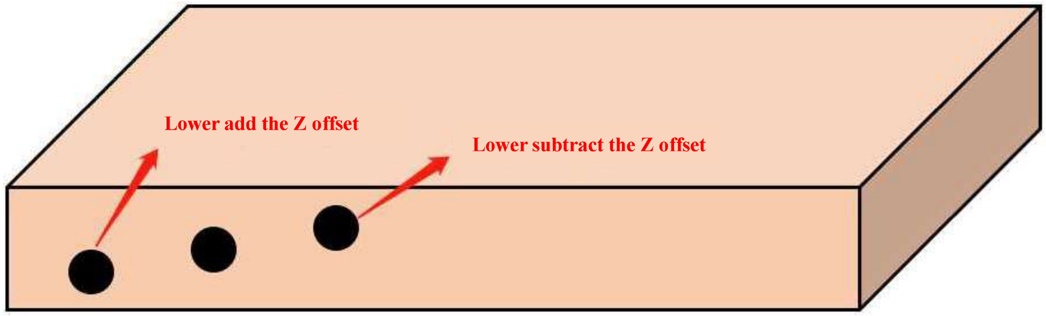
3. Enter the tool data list and find the corresponding tool (this example shows the horizontal hole in the positive Y direction, as shown below):
4. At this time, we hope that the tool moves 1mm in the X- direction and 1mm in the Y+ direction, so that the hole position is correct;
5. Modify the X offset and Y offset of the tool list, subtract 2 from the X offset, and add 1 to the Y offset;

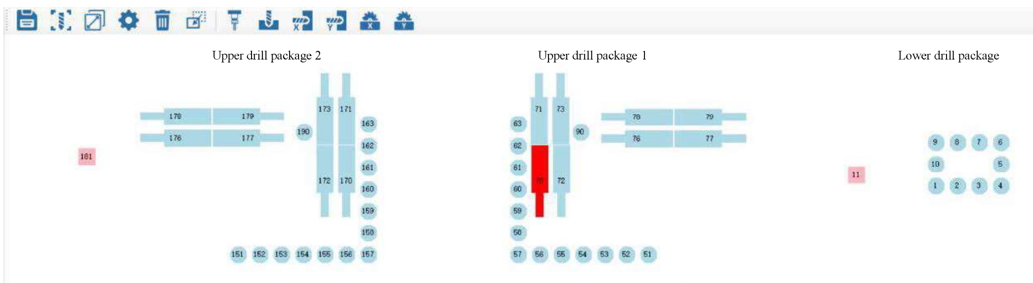
6. Both T76 and T78 are horizontal tools in the positive Y-direction, but the difference in the X-direction is 32mm, which can be modified together;



Accuracy adjustment of Z-direction side hole

If the side hole is not in the middle of the sheet, adjust the Z offset of the horizontal tool (as shown in the figure below);





No.	Tool number	Tool diameter/saw width	Tool type	X	Y	X offset	Y offset	Z offset	Feeding speed	Drilling delay	Maximum depth
25	70	8	X positive	54.6	160.2	54.6	-160.2	-11.5	5000	0.1	60
26	71	8	X negative	266	160.6	266	-160.6	-11.6	5000	0.1	60
27	72	8	X positive	54.6	128.2	54.6	-128.2	-11.5	5000	0.1	60
28	73	8	X negative	266	128.6	266	-128.6	-11.6	5000	0.1	60
29	76	8	Y positive	170	68.8	170	-68.8	-11.5	5000	0.2	60
30	77	8	Y negative	170	-144.9	170	144.9	-11.5	1500	0.2	60
31	78	8	Y positive	202	68.8	202	-68.8	-11.5	1200	0.2	60
32	79	8	Y negative	202	-144.9	202	144.9	-11.5	1500	0.2	60
33	90	8	Z negative	180	100	180	-100	-10	5000	0	30
34	151	8	Z negative	0	0	0	0	0	4000	0.1	60

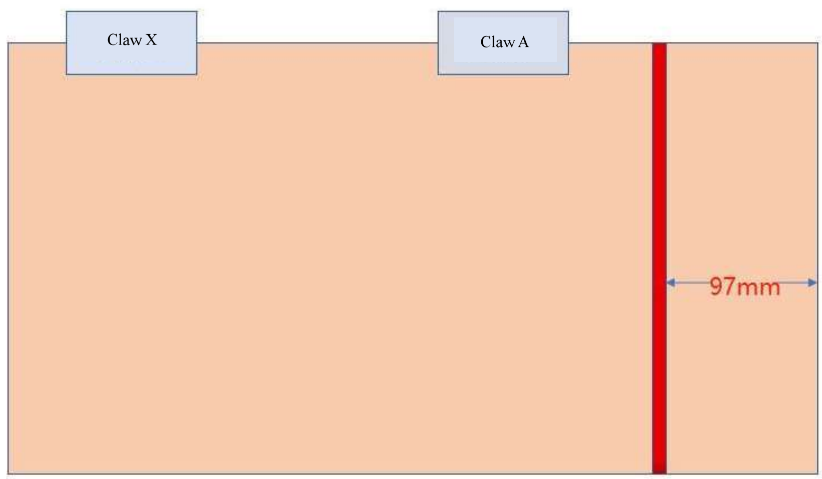
9.6.4 Spindle accuracy adjustment

Take the spindle tool diameter of 8mm as an example here;

Accuracy adjustment of X-direction

1. As shown in the figure, a slot is expected to be drawn at a distance of 100mm from the sheet head; in the case of accurate accuracy, the X-direction measured by the caliper should be 96mm (100mm-tool radius 4mm);

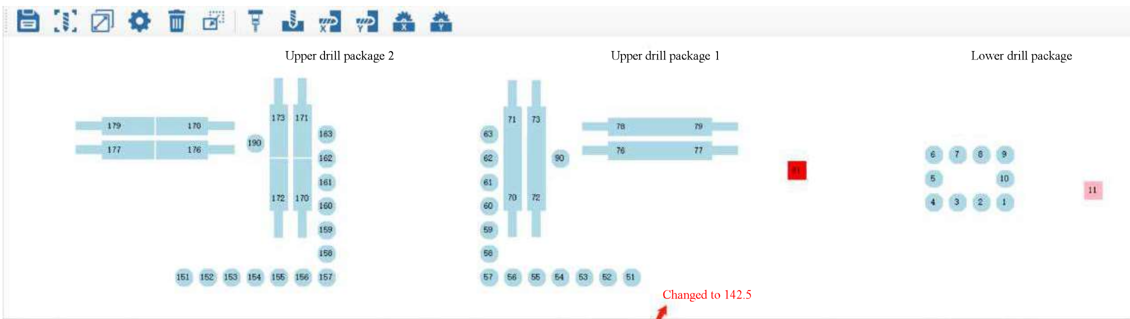
2. After the actual processing, it is found that the accuracy is not accurate, and the X-direction is more than 1mm;



3. Enter the tool data list to find the corresponding spindle (this example shows the spindle Z- as below):

4. At this time, we hope that the corresponding spindle moves 1mm in the X+ direction, so that the slot position is correct;

5. Modify the X offset of the tool sheet and subtract the X offset by 1.

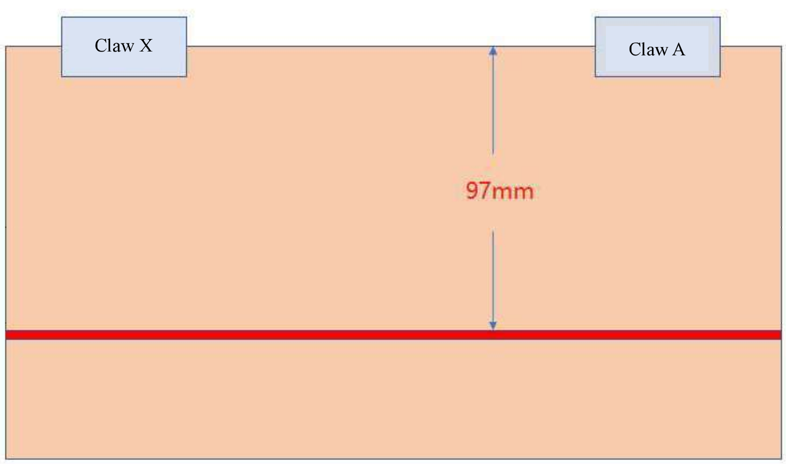


No.	Tool number	Tool diameter/saw width	Tool type	X	Y	X offset	Y offset	Z offset	Feeding speed	Drilling delay	Maximum depth
33	81	8	Spindle Z negative	143.5	-222.4	143.5	222.4	40.5	5000	0	0
34	90	0	Z negative	160	96.2	160	-96.2	-10	5000	0	30
35	96	0	Z negative	143.5	-222.4	143.5	222.4	35.167	5000	0	30
36	151	8	Z negative	0	0	0	0	0	5000	0	60
37	152	0	Z negative	0	-32	0	32	0	5000	0	60
38	153	0	Z negative	0	-64	0	64	0	5000	0	60
39	154	0	Z negative	0	-96	0	96	0	5000	0	60
40	155	0	Z negative	0	-128	0	128	0	5000	0	30
41	156	0	Z negative	0	-160	0	160	0	5000	0	30
42	157	0	Z negative	0	-192	0	192	0	5000	0.2	30

Accuracy adjustment of Y-direction

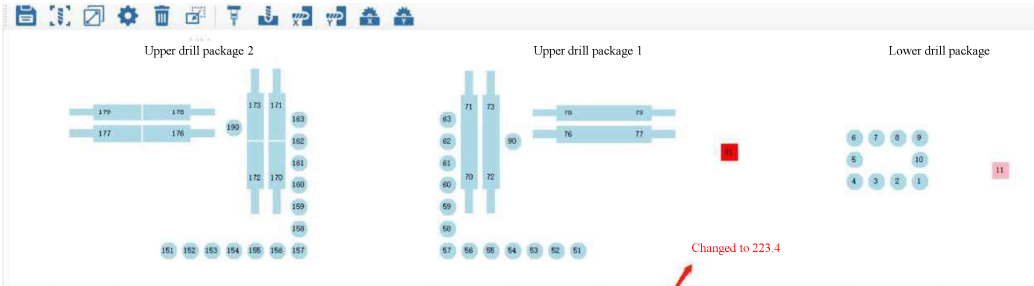
1. As shown in the figure, it is expected to draw a groove at a position 100mm away from the jaw side; in the case of accurate accuracy, the caliper should measure 96mm in the Y-direction (100mm-tool radius 4mm);

2. After the actual processing, it is found that the accuracy is not accurate, and the Y-direction is more than 1mm;



3. Enter the tool data list to find the corresponding spindle (this example shows the spindle Z- as below):

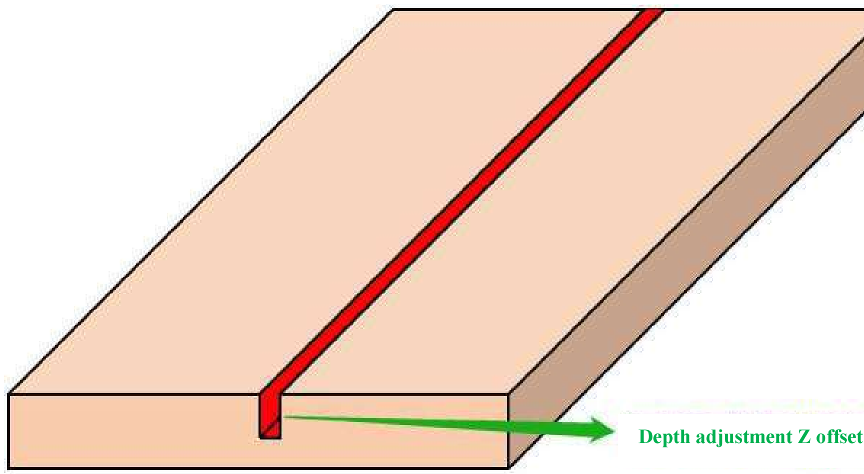
4. At this time, we hope that the corresponding spindle moves 1mm in the Y+ direction, so that the slot position is correct;
   Modify the Y offset of the tool sheet and subtract the Y offset by 1.



No.	Tool number	Tool diameter/saw width	Tool type	X	Y	X offset	Y offset	Z offset	Feeding speed	Drilling delay	Maximum depth
33	81	8	Spindle Z negative	143.5	-222.4	143.5	222.4	40.5	5000	0	0
34	90	0	Z negative	160	96.2	160	-96.2	-10	5000	0	30
35	96	0	Z negative	143.5	-222.4	143.5	222.4	35.167	5000	0	30
36	151	8	Z negative	0	0	0	0	0	5000	0	60
37	152	0	Z negative	0	-32	0	32	0	5000	0	60
38	153	0	Z negative	0	-64	0	64	0	5000	0	60
39	154	0	Z negative	0	-96	0	96	0	5000	0	60
40	155	0	Z negative	0	-128	0	128	0	5000	0	30
41	156	0	Z negative	0	-160	0	160	0	5000	0	30
42	157	0	Z negative	0	-192	0	192	0	5000	0.2	30

Accuracy adjustment of Z-direction

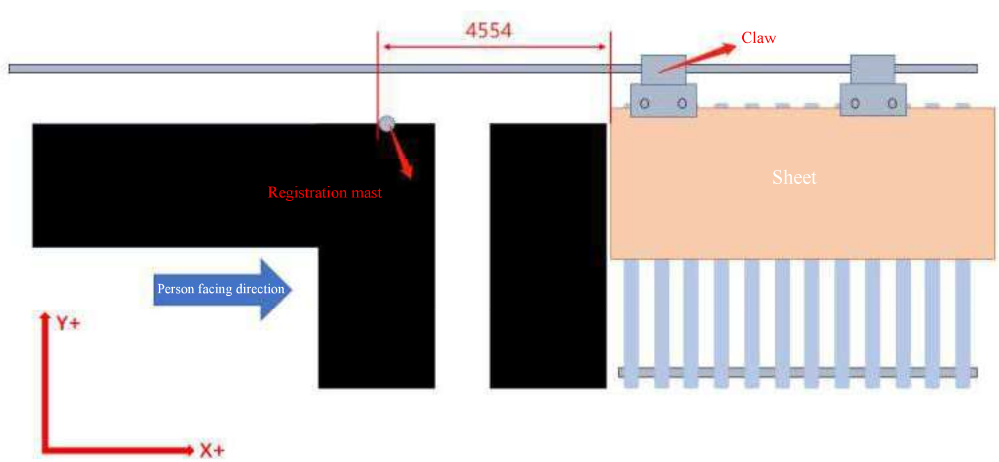
To adjust the depth of the notch, adjust the Z offset of the spindle; add when it's deeper and subtract if it's shallower;



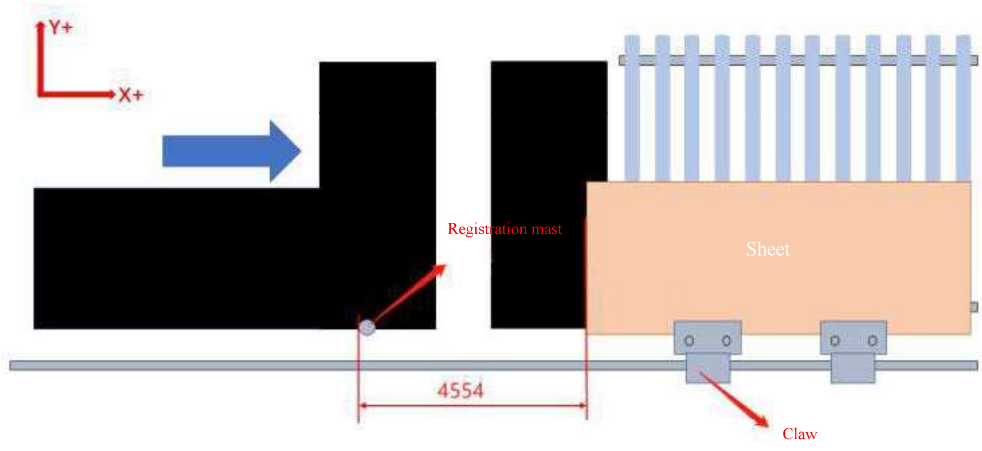
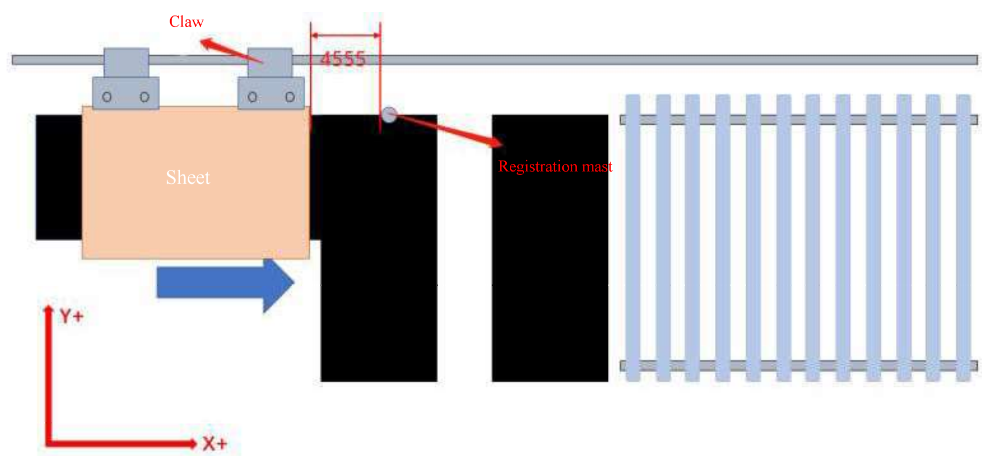
Common parameter settings

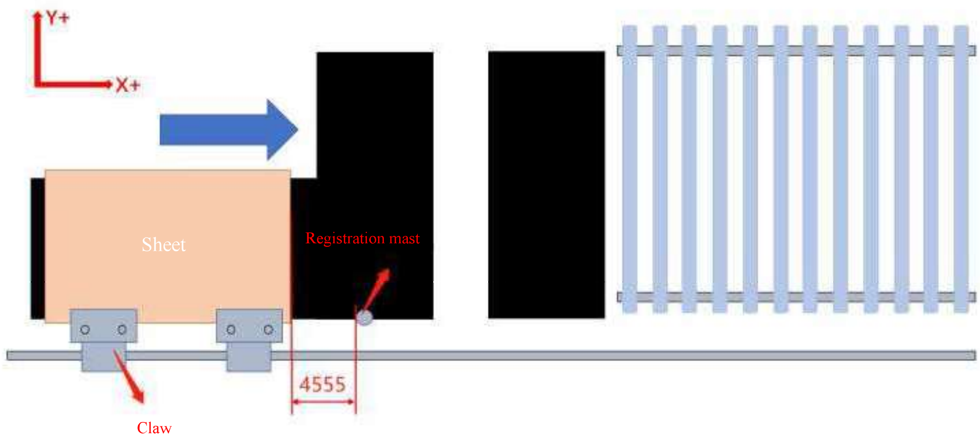
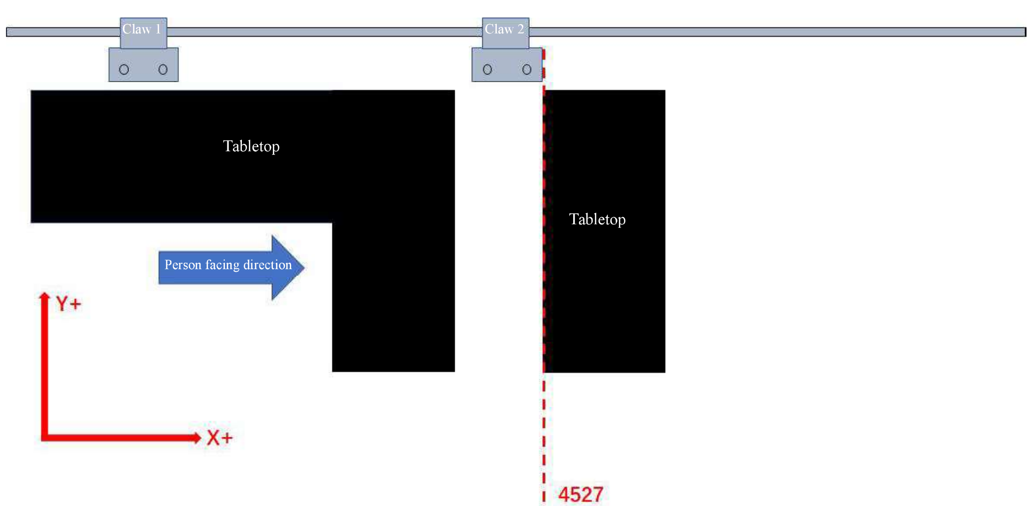
After restoring the backup and setting the machinery origin and G54 origin, move each axis to the specified position and fine tune the following parameters.

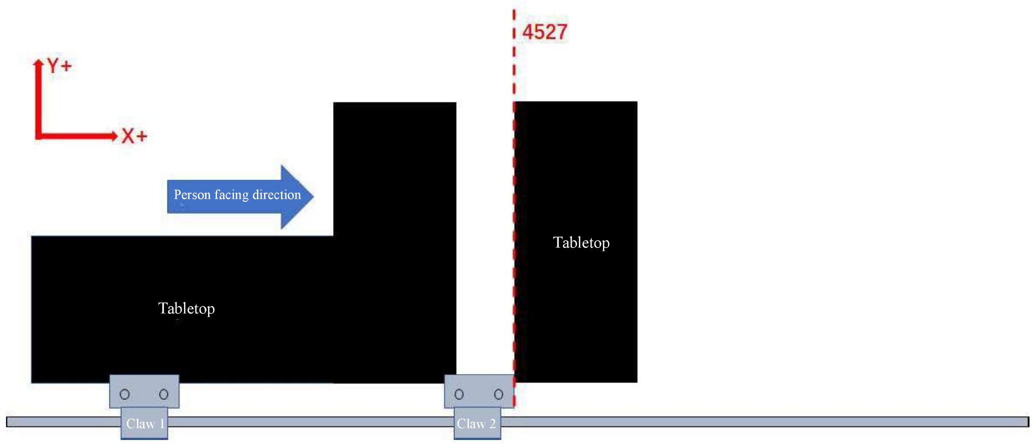
Note: Due to different models, the numbers of some parameters may be slightly different.

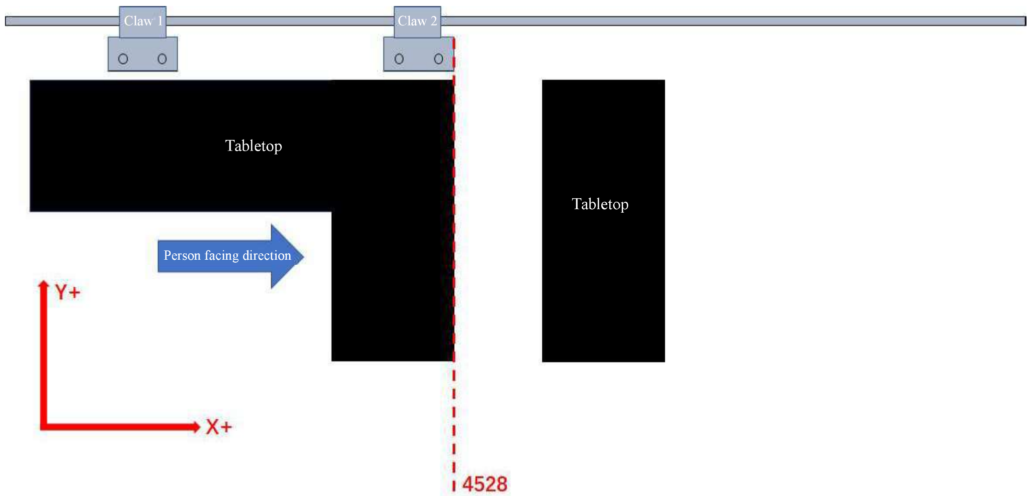
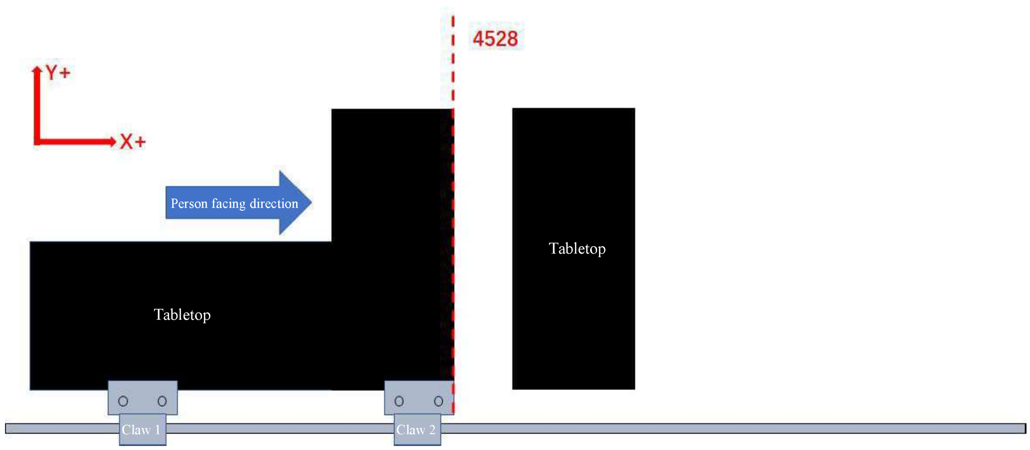
<table>
<tr><th>Serial number</th><th>Parameter No.</th><th>Parameter name</th><th>Setting method</th></tr>
<tr><td rowspan="2">1</td><td>4554</td><td>The distance between the end face of the exit plate and the positioning rod (rear exit)</td><td>The plate goes backward, and the distance between the plate head and the positioning cylinder is measured with a ruler, and check what is the distance from the positioning post to the board feeding position (board tail), and fill in this parameter.</td></tr>
<tr><td colspan="3">The claws are on the left side of the sheet:


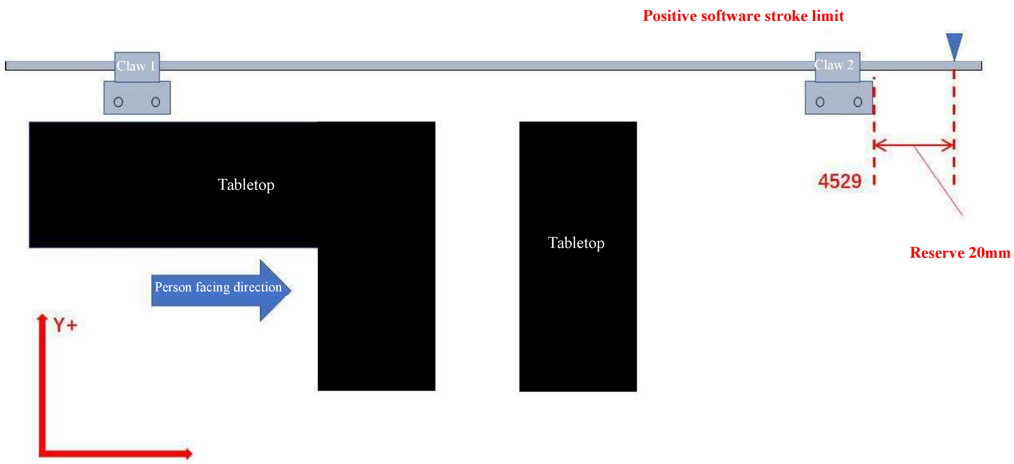
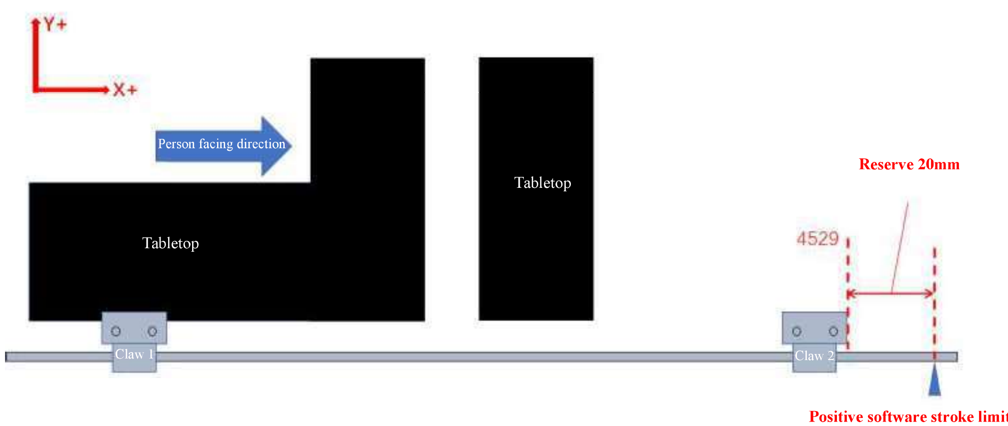
The claws are on the right side of the sheet:</td></tr>
</table>

Serial number	Parameter No.	Parameter name	Setting method
			
2	4555	The distance between the end face of the plate and the positioning rod (front out)	The plate goes forward, and the distance between the plate head and the positioning cylinder is measured with a ruler, and check what is the distance from the positioning post to the board feeding position (board tail), and fill in this parameter.
	The claws are on the left side of the sheet:  **The claws are on the right side of the sheet:**		

<table>
<tr><th>Serial number</th><th>Parameter No.</th><th>Parameter name</th><th>Setting method</th></tr>
<tr><td></td><td colspan="3">
</td></tr>
<tr><td rowspan="2">3</td><td>4527</td><td>Work area positive boundary program coordinates</td><td>After the claw moves to the position shown in the figure, write the program coordinates of the claw into this parameter.</td></tr>
<tr><td colspan="3">The claws are on the left side of the sheet:

</td></tr>
</table>

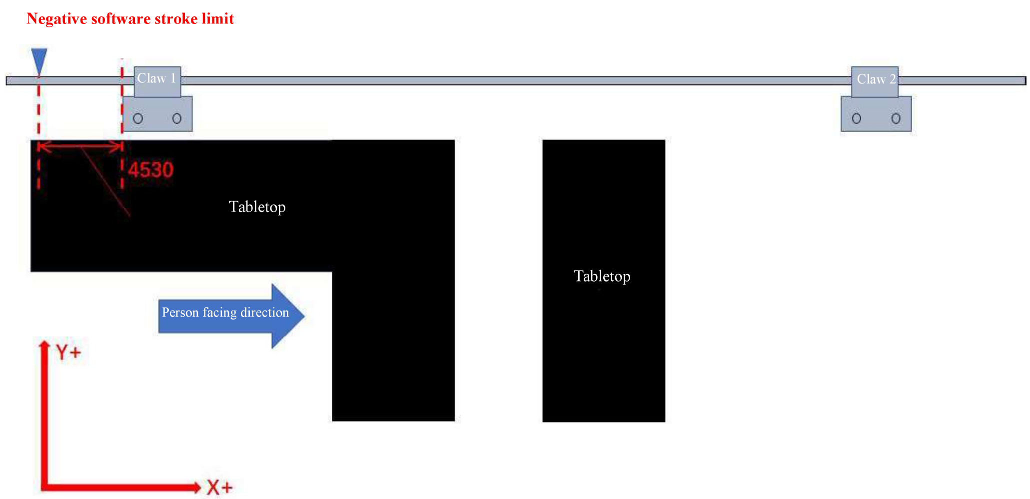
Serial number	Parameter No.	Parameter name	Setting method
	The claws are on the right side of the sheet: 		
4	4528	Work area negative boundary program coordinates	After the claw moves to the position shown in the figure, write the program coordinates of the claw into this parameter.
	The claws are on the left side of the sheet:		

Serial number	Parameter No.	Parameter name	Setting method
	 **The claws are on the right side of the sheet:**  		
5	4529	Claw is at the G54 positive limit position	After the claw moves to the position shown in the figure, write the program coordinates of the claw into this parameter.

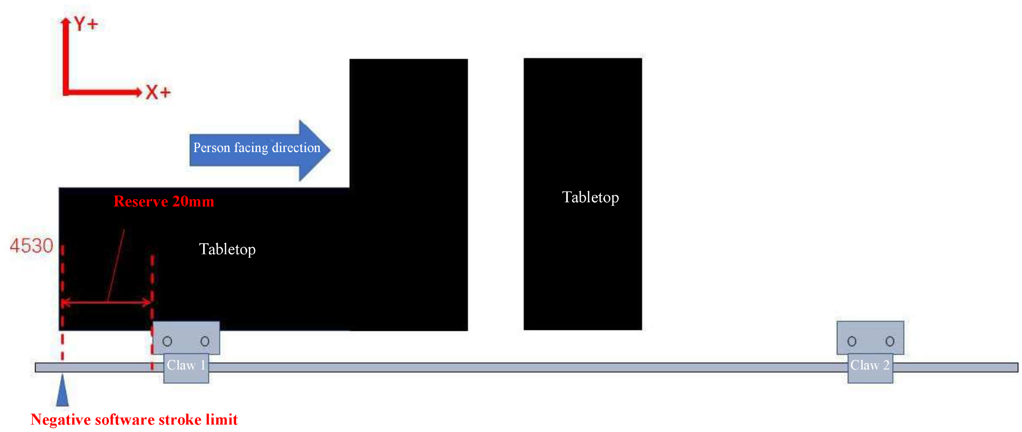
Serial number	Parameter No.	Parameter name	Setting method
	The claws are on the left side of the sheet:  The claws are on the right side of the sheet: 		

<table>
<tr><th>Serial number</th><th>Parameter No.</th><th>Parameter name</th><th>Setting method</th></tr>
<tr><td rowspan="2">6</td><td>4530</td><td>Claw is in the negative limit position of G54</td><td>After the claw moves to the position shown in the figure, write the program coordinates of the claw into this parameter.</td></tr>
<tr><td colspan="3">

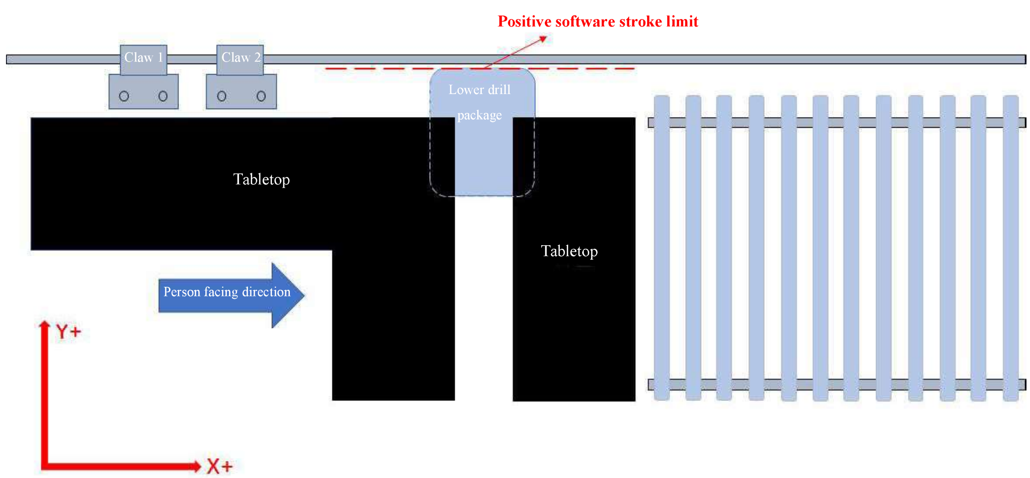
The claws are on the left side of the sheet:

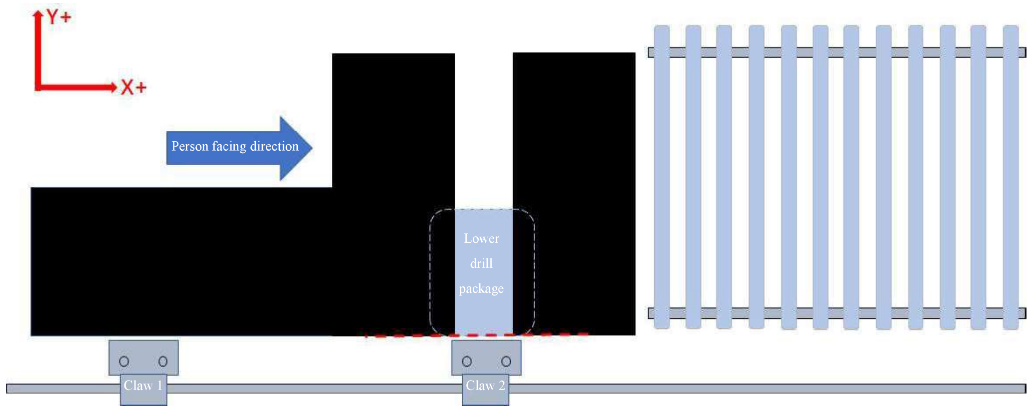


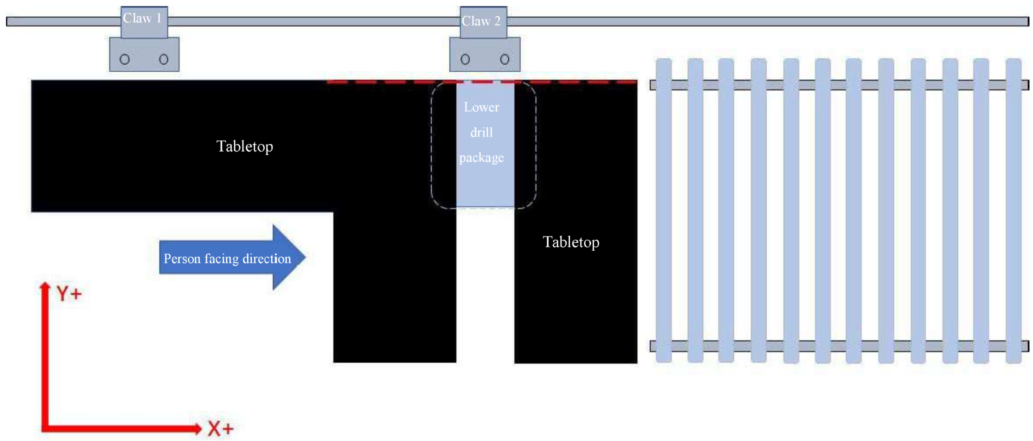
The claws are on the right side of the sheet:



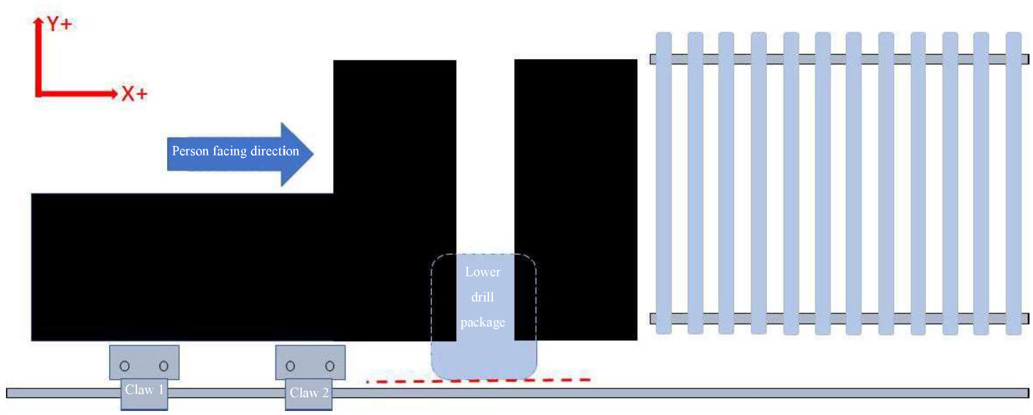
</td></tr>
</table>

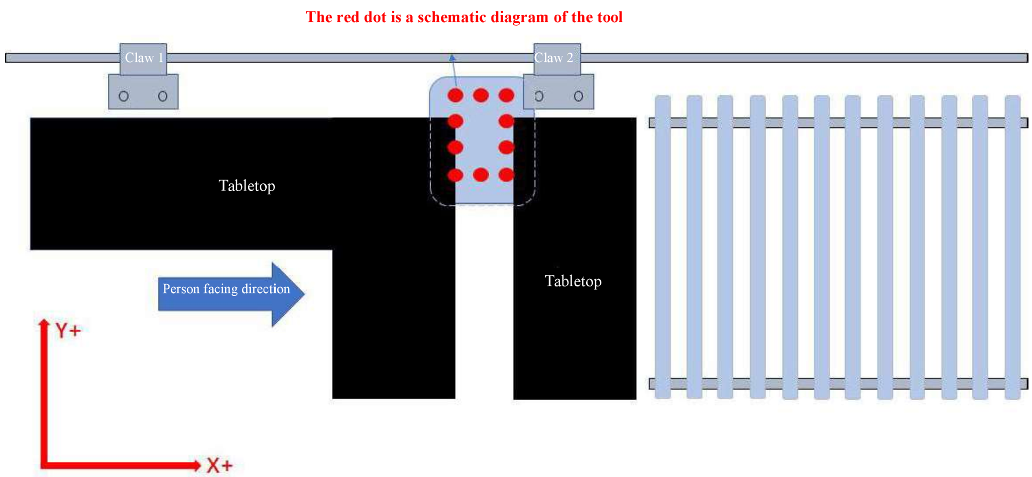
Serial number	Parameter No.	Parameter name	Setting method
7	4585	Claw avoidance lower drill box condition Lower drill box Y range upper limit 1	The claws are on the left side of the sheet: After the lower drill box reports over the positive stroke limit, the alarm is cleared by retreating 1~2 mm, and the program coordinates of the lower drill box shall be included in this parameter. The claws are on the right side of the sheet: Move the lower drill box in the Y-direction of the handwheel mode to ensure that when the tools of lower drill box are all ejected, they will not hit the maximum coordinate of the claw. Reserve about 20mm, and write the program coordinates of lower drill box at this time into this parameter
	The claws are on the left side of the sheet:  **The claws are on the right side of the sheet:**		

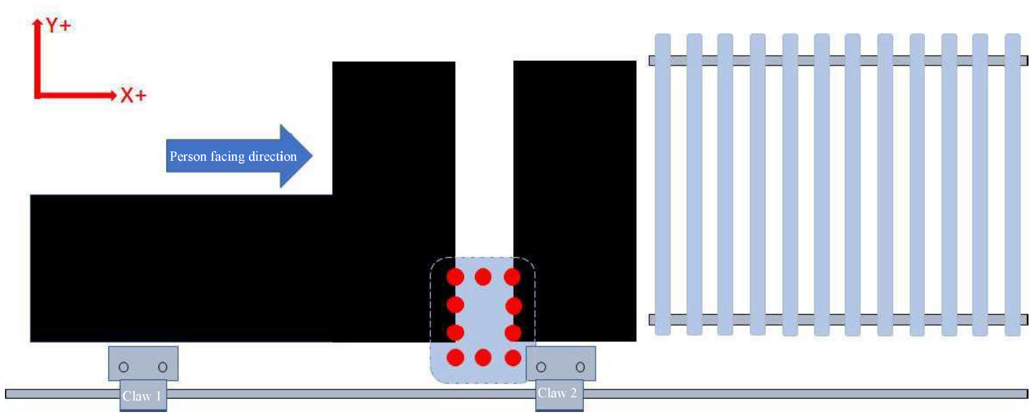
Serial number	Parameter No.	Parameter name	Setting method
			
8	4586	Claw avoidance lower drill box condition lower drill box Y range upper limit 1	The claws are on the left side of the sheet: Move the lower drill box in the Y-direction of the handwheel mode to ensure that when the tools of lower drill box are all ejected, they will not hit the maximum coordinate of the claw. Reserve about 20mm, and write the program coordinates of lower drill box at this time into this parameter The claws are on the right side of the sheet: After the lower drill box reports over the negative stroke limit, the alarm is cleared by retreating 1~2 mm, and the program coordinates of the lower drill box shall be included in this parameter.
	The claws are on the left side of the sheet:		

<table>
<tr><th>Serial number</th><th>Parameter No.</th><th>Parameter name</th><th>Setting method</th></tr>
<tr><td></td><td colspan="3">

The claws are on the right side of the sheet:

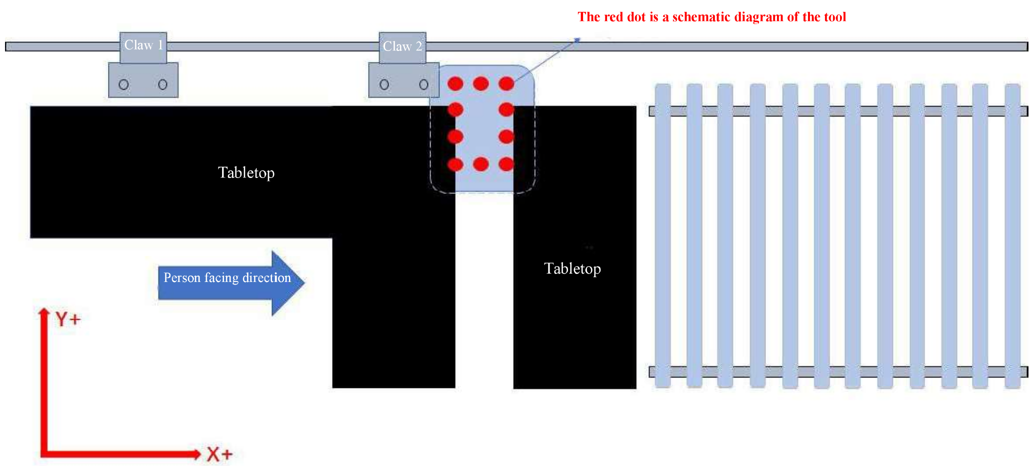

</td></tr>
<tr><td>9</td><td>4587</td><td>Claw avoidance lower drill box condition claw avoidance X range upper limit 1</td><td>Move the claw in the X- direction in the handwheel mode, ensure that the maximum coordinate will not collide when all the tools in lower drill box pop up, and reserve about 20mm. Subtract the length of the claw in the X-direction from the program coordinate of the claw at this time, and then write this parameter.
Example:
Parameter 4521, the length of the claw in the X-direction is 145000 (145mm)
At this time, the coordinate where the claw drill</td></tr>
</table>

<table>
<tr><th>Serial number</th><th>Parameter No.</th><th>Parameter name</th><th>Setting method</th></tr>
<tr><td rowspan="2"></td><td></td><td></td><td>box will not collide is 538.152,
then the parameter is 393152
538.152-145.0=393.152mm</td></tr>
<tr><td colspan="3">The claws are on the left side of the sheet:


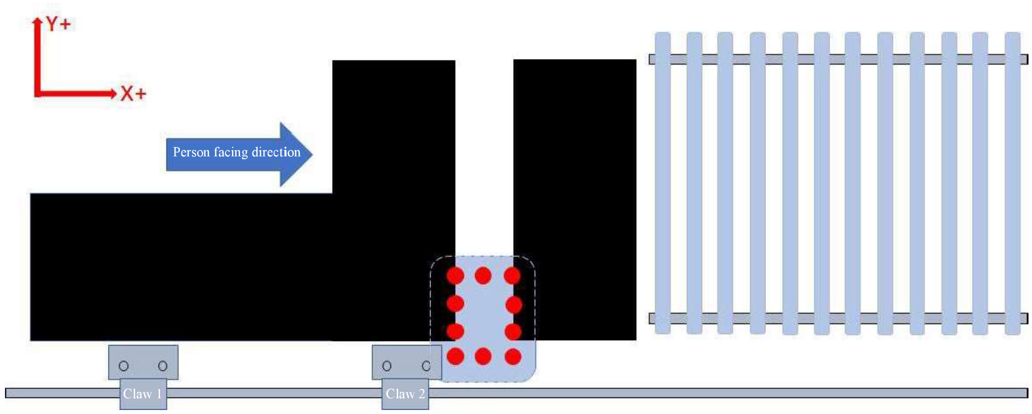
The claws are on the right side of the sheet:

</td></tr>
<tr><td>10</td><td>4588</td><td>Conditions for claw to avoid lower drill box Lower limit 1 for claw to avoid X range</td><td>Move the claw in the X+ direction in the handwheel mode, and reserve the maximum coordinate of about 20mm, so as to ensure that all the tools in lower drill box will not collide when they pop out. Subtract the length of the</td></tr>
</table>

Serial number	Parameter No.	Parameter name	Setting method
			claw in the X-direction from the program coordinate of the claw at this time, and then write this parameter.

The claws are on the left side of the sheet:

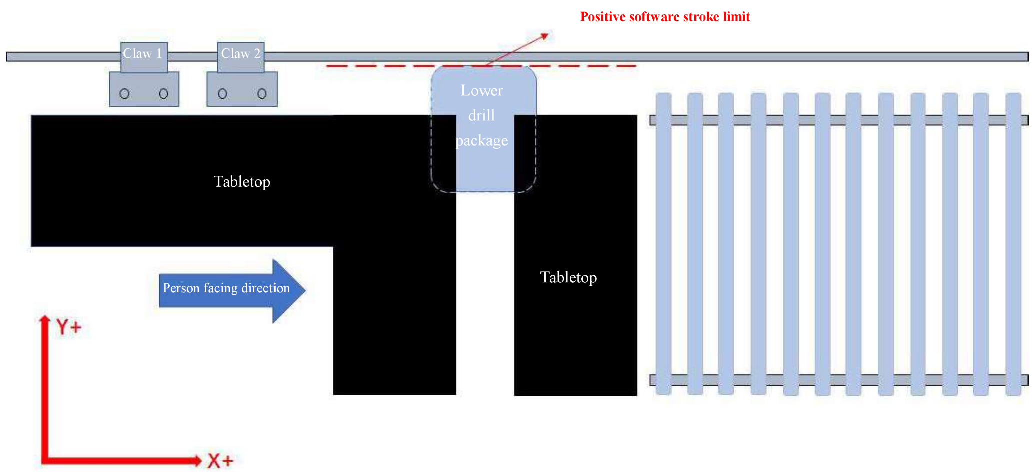


The claws are on the right side of the sheet:

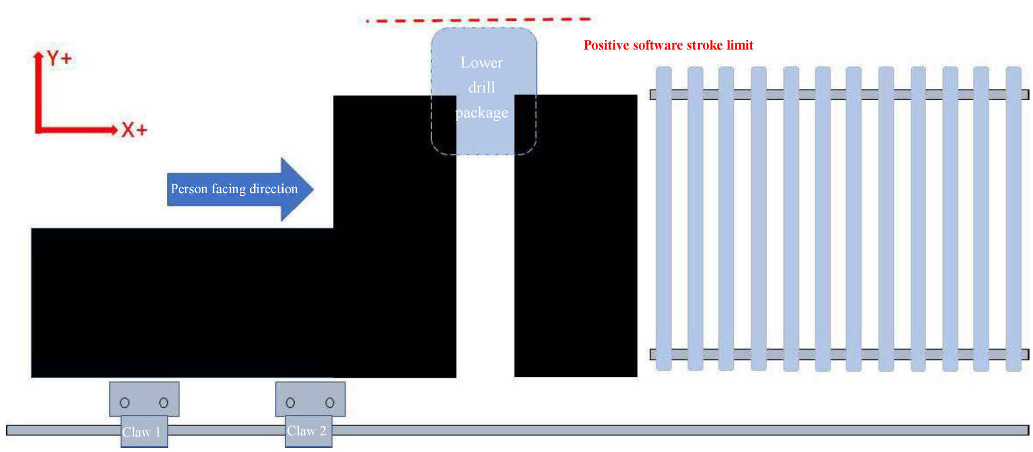


Serial number	Parameter No.	Parameter name	Setting method
11	4893	Lower drill box G54 Y-direction positive limit	After the lower drill box reports over the positive stroke limit, the alarm is cleared by retreating 1~2 mm, and the program coordinates of the lower drill box shall be included in this parameter.

The claw is on the left side of the sheet:

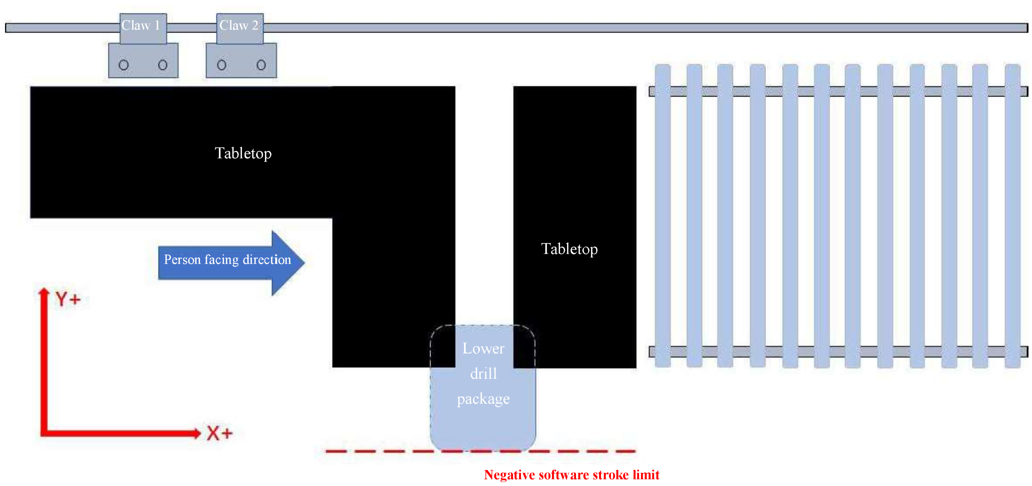


The claws are on the right side of the sheet:



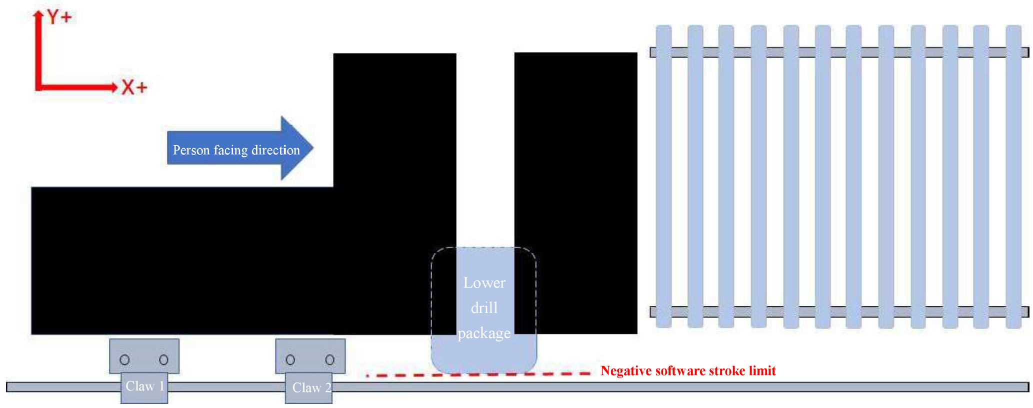
<table>
<tr><th>Serial number</th><th>Parameter No.</th><th>Parameter name</th><th>Setting method</th></tr>
<tr><td></td><td colspan="3"></td></tr>
<tr><td rowspan="2">12</td><td>4894</td><td>Lower drill box G54 Y-direction negative limit</td><td>After the lower drill box reports over the negative stroke limit, the alarm is cleared by retreating 1~2 mm, and the program coordinates of the lower drill box shall be included in this parameter.</td></tr>
<tr><td colspan="3">

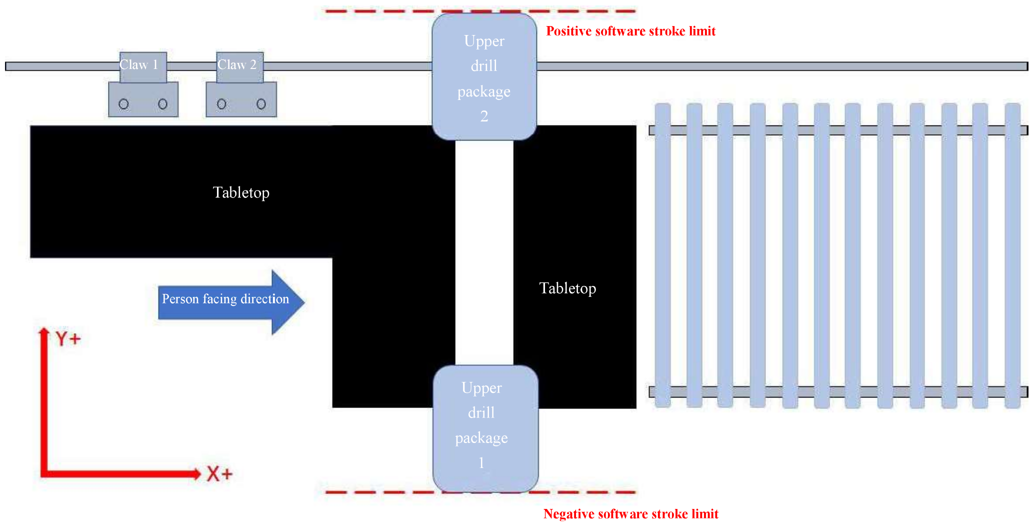
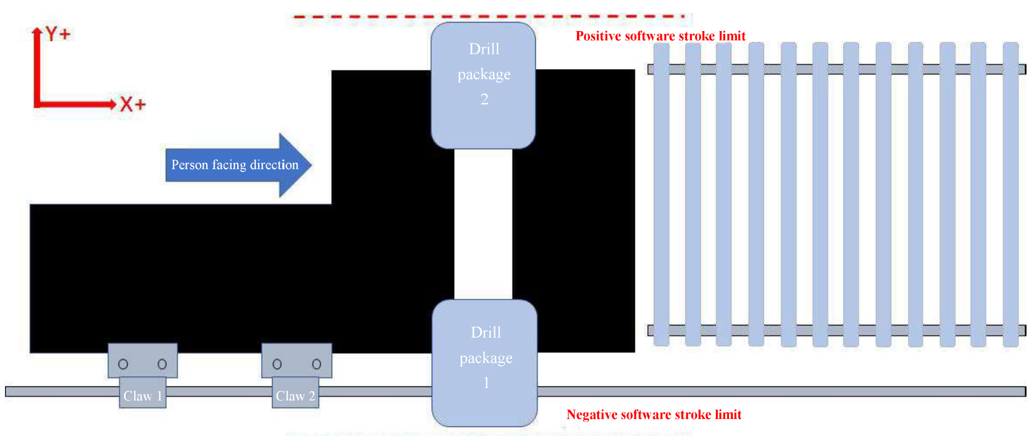
The claw is on the left side of the sheet:



The claws are on the right side of the sheet:

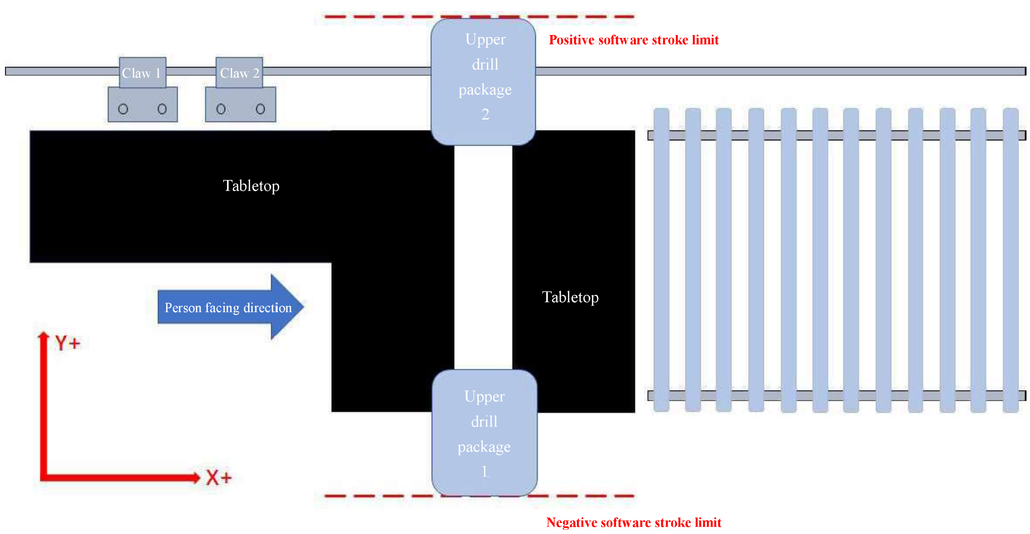
</td></tr>
</table>

Serial number	Parameter No.	Parameter name	Setting method
			
13	4897	Upper drill package G54 Y-direction positive limit	Single drill box machine: After the upper drill box reports that it exceeds the positive travel limit, back 1~2mm to make the alarm disappear, and write the program coordinates of the upper drill box into this parameter at this time. Double drill box machine: After the upper drill box 2 reports that it exceeds the positive travel limit, back 1~2mm to make the alarm disappear, and write the program coordinates of the upper drill box 2 into this parameter at this time.
	The claws are on the left side of the sheet:		

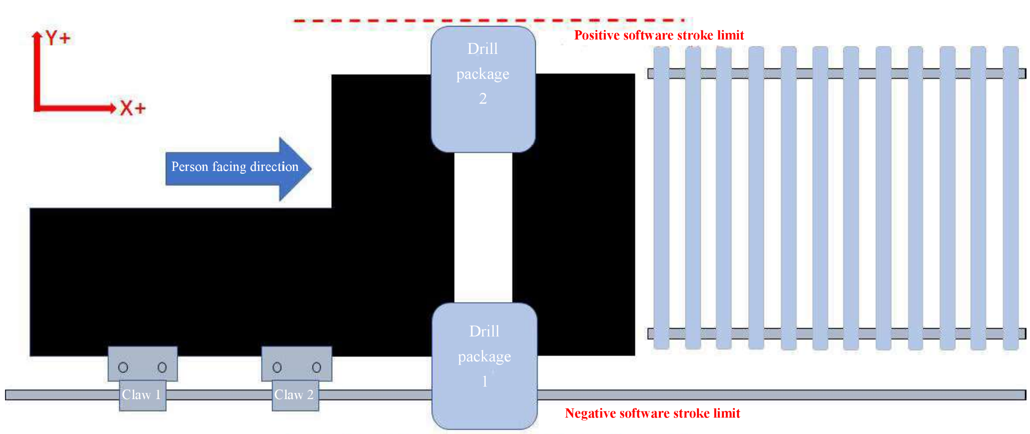
Serial number	Parameter No.	Parameter name	Setting method
	 **The claws are on the right side of the sheet:** 		
14	4898	Upper drill box G54 Y-direction negative limit	Single drill box machine: After the upper drill box reports over the negative stroke limit, the alarm is cleared by retreating 1~2 mm, and the program coordinates of the upper drill box shall be included in this parameter.

<table>
<tr><th>Serial number</th><th>Parameter No.</th><th>Parameter name</th><th>Setting method</th></tr>
<tr><td rowspan="2"></td><td></td><td></td><td>Double drill box machine:
After the upper drill box 1 reports over the negative stroke limit, the alarm is cleared by retreating 1~2 mm, and the program coordinates of the upper drill box 1 shall be included in this parameter.</td></tr>
<tr><td colspan="3">

The claws are on the left side of the sheet:



The claws are on the right side of the sheet:

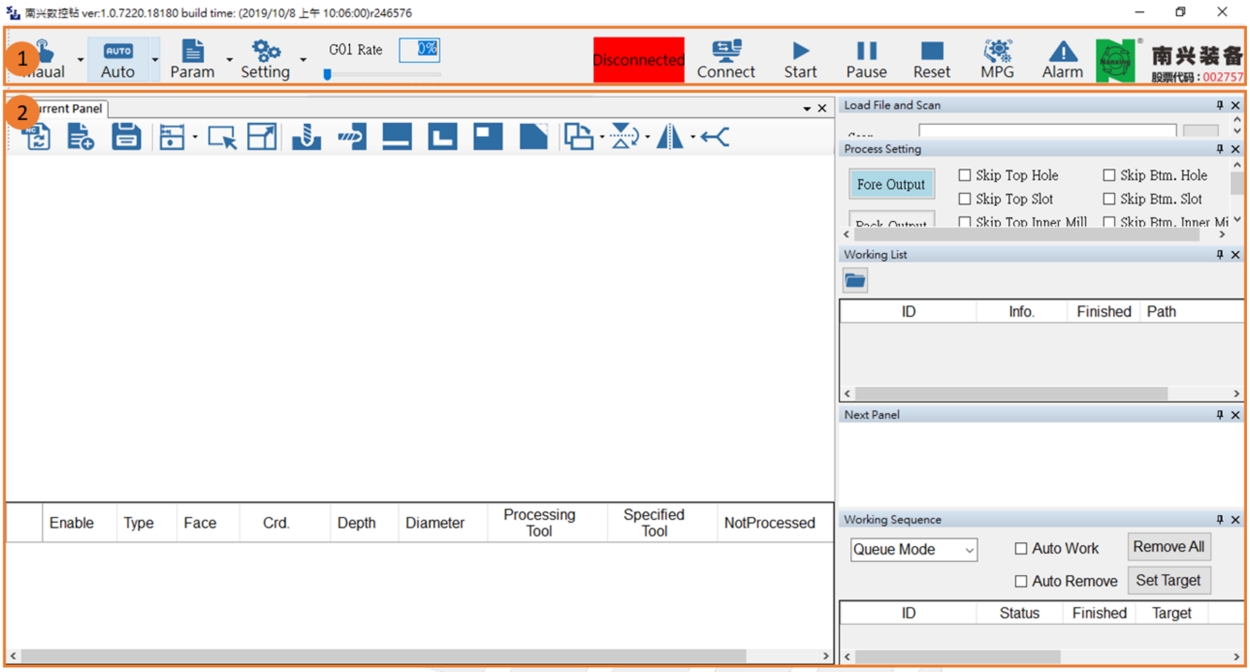


</td></tr>
</table>

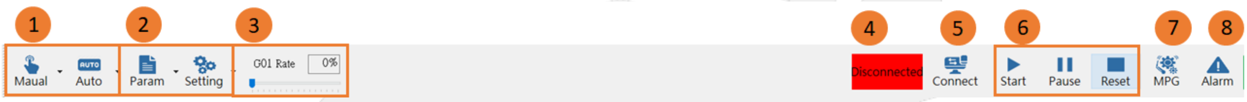
PC CAM Drill Machine Software Operatio Manual

匯出日期：2019-12-24
修改日期：2019-11-27

1 Main screen



1.1 **Tool Strip**



1.1.1 1. Switch Mode

- Click button or focus on application, switch CNC mode to current mode of application.
- Manual：Switch to manual mode
  - If CNC state is "Cycle Start", "Feed Hold", "Stop" and "Execute", button of maunal mode is not enable.
- Auto：Switch to auto mode

1.1.2 2. Application settings

- Param：Parameter setting for processing
- Setting：Basic application setting
  - Connect Setting：Setting the IP of controller and time out
  - Authority Setting：Setting the authority of user
  - Save Layout：Save the current layout
  - Backup and Recover：Backup the current environment and recover

1.1.3 3. Working Rate

- Slider：Setting the CNC feed rate(G01)
  a. Tick：10%
  b. Span：0%~200%
- Display：Display the CNC feed rate(G01) as a percentage

1.1.4 4. CNC Status

- Display the current state of controller
  - Disconnected
  - Not Ready
  - Ready
  - Cycle Start
  - Feed Hold
  - Stop
  - Execute
- If CNC state is "Cycle Start", "Feed Hold", "Stop" and "Execute", button of maunal mode is not enable.
- If controller is connected, the button of "connect to controller" is not enable

1.1.5 5. Connect to Controller

- Connect：Connect to Controller
- If controller is connected, the button of "connect to controller" is not enable

1.1.6 6. Cycle Start/Feed Hold/Reset

- Start：Cycle start
- Pause：Feed hold
- Reset：Reset the CNC state

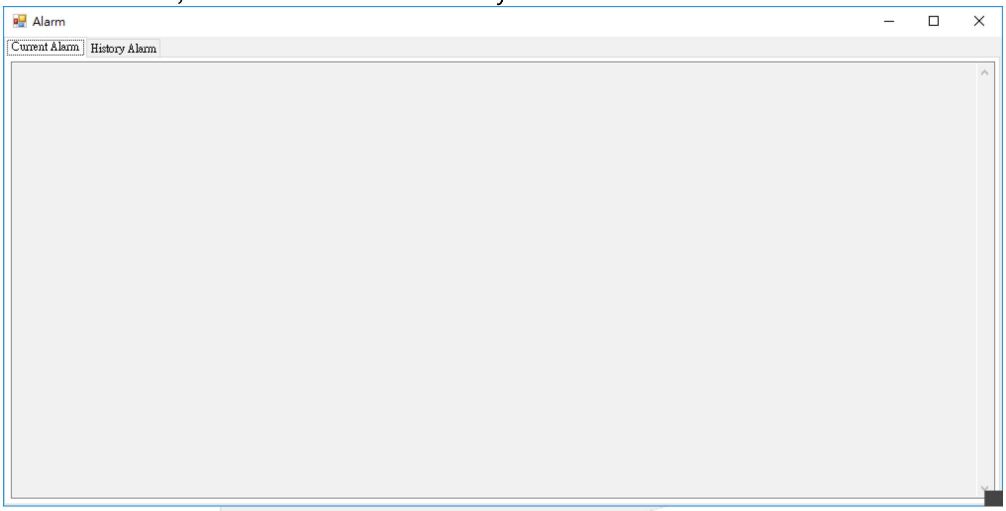
1.1.7 7. MPG

- MPG：MPG Simulate

1.1.8 8. Alarm

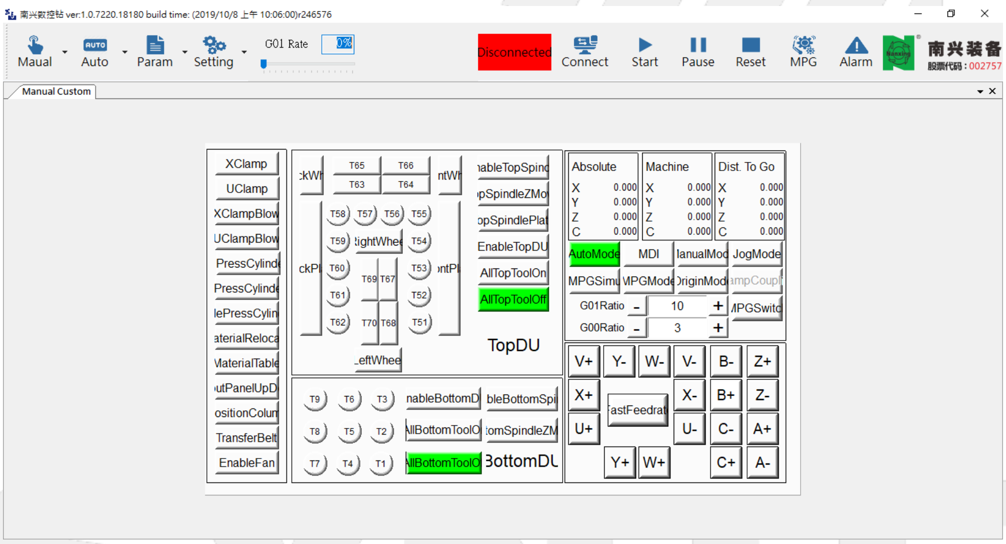
- Alarm：If there are alarms, display the alarm count and flashing red-white alternately

- Click the button, there are current and history alarm can be check

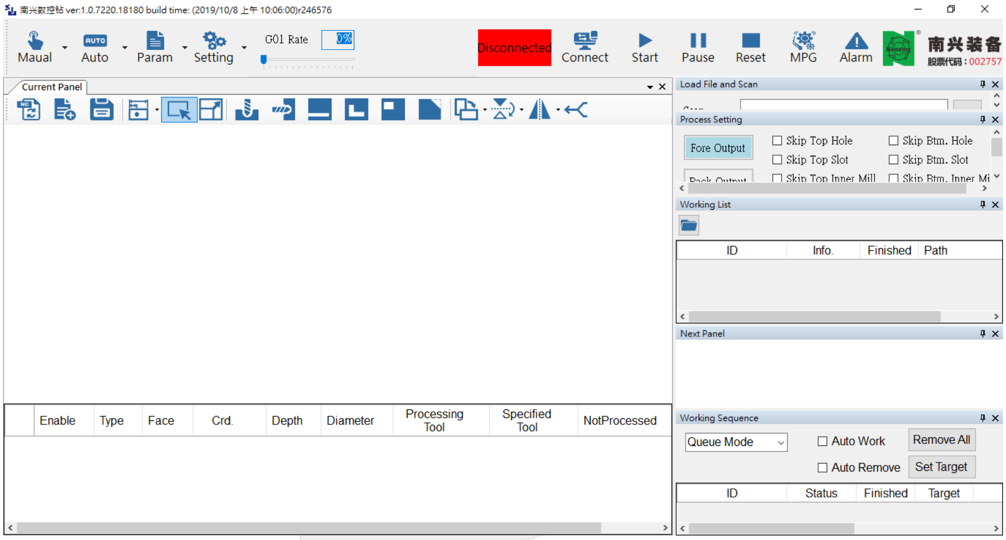


1.2 **Display Area**

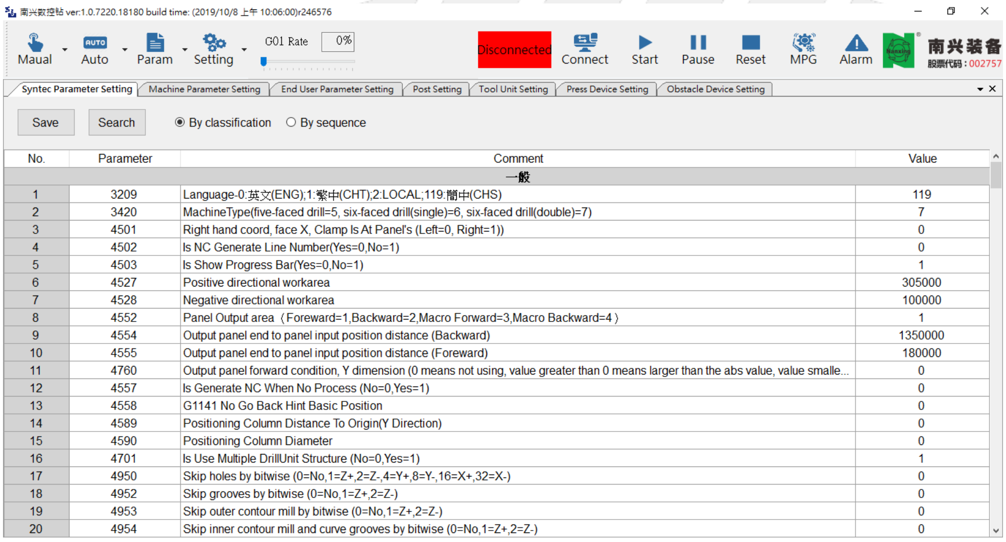
1.2.1 1. Manual Mode



1.2.2 2. Auto Mode

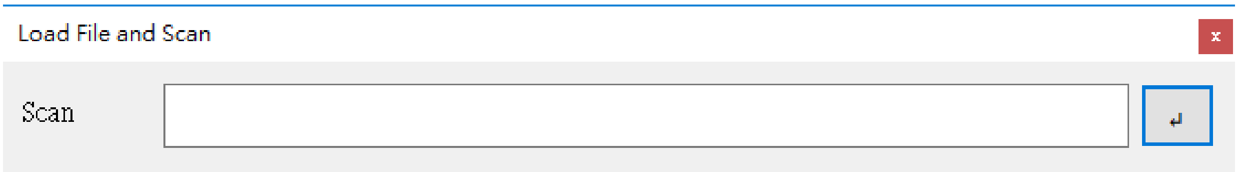


1.2.3 3. Parameter Setting



2 Auto Mode

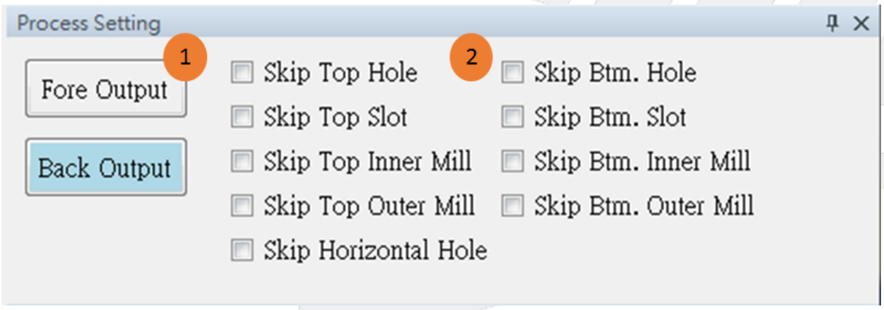
2.1 Scan and Load File



2.1.1 1. Scan Bar code

- Scanning bar code is available after import files to working list.
- Bar code scanner：After scan the bar code on the panel, the NC file of processing would be generated and added to list of queue.
- Input Box：User can enter the panel ID by yourself, the NC file of processing would be generated and added to list of queue.
- Enter the "Error" bar code：If bar code can not be found in the working list, the string of bar code would turn red and show "XXX FileNotFound"

2.2 Process



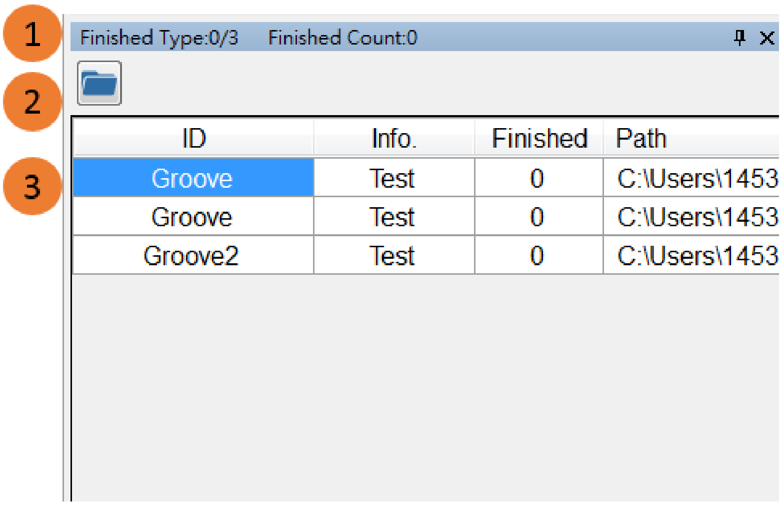
2.2.1 1.Mode

- Fore output: After processing, the panel is output from the front.
- Back output: After processing, the panel is output from the back.

2.2.2 2.Process

- Skip top hole: When processing the front panel, ignore the front hole.
- Skip bottom hole: When processing the back panel, ignore the back hole.
- Skip horizontal hole: When processing the side panels, ignore the side holes.
- Skip top groove: When processing the front panel, ignore the front groove.
- Skip bottom groove: When processing the back panel, ignore the back groove.
- Skip top inner mill: When processing the front panel, ignore the front inner mill.
- Skip bottom inner mill: When processing the back panel, ignore the back inner mill.
- Skip top outer mill: When processing the front panel, ignore the front outer mill.
- Skip bottom outer mill: When processing the back panel, ignore the back outer mill.

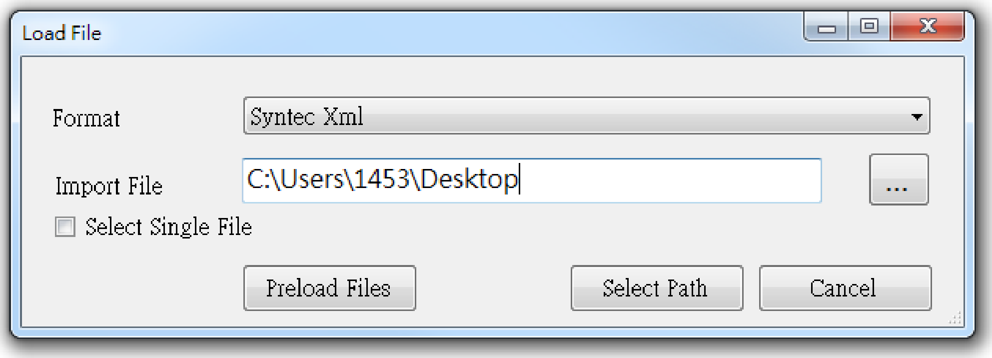
2.3 Working List



2.3.1 1. Title

- Finished Type：Show how many types of panel has been processed.
- Finished Count：Show the amount of processed panel.

2.3.2 2. Import File

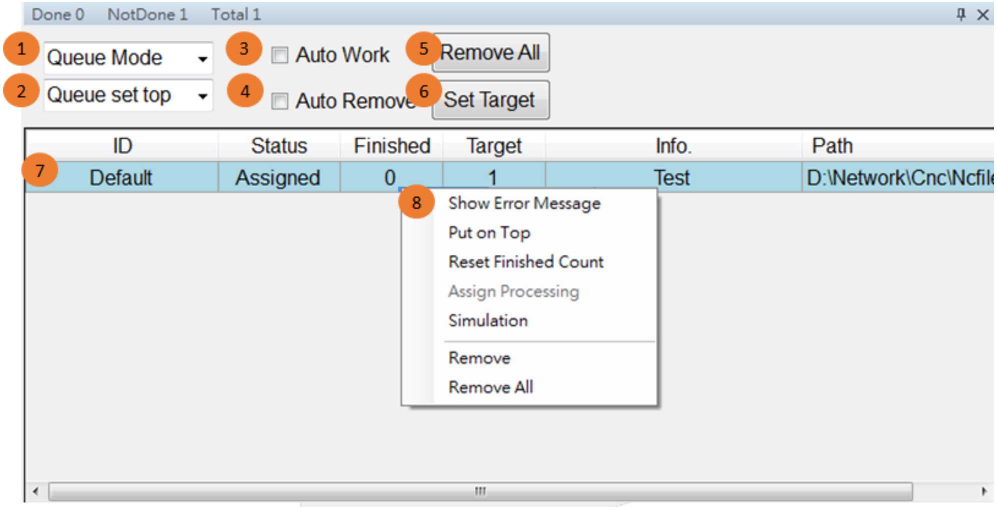


- Step：Choose file format → Select the folder → Click "Preload Files" or "Select Path".
- Supported file format：Anderson、 Biesse (Bpp)、 EWood (Xml)、 FinChina (Xml)、 MPR Ban、 Pytha (DXF)、 Syntec (Xml)、 Syntec (DXF)、 Wanshi (PDX)、 SCM (XXL)

2.3.3 3. Working List

- Infomation：Panel ID、 Info.、 Number of finished panel、 File path
- Add to Sequence List：Double-click the item to add to the sequence list.

2.4 Sequence List



2.4.1 1. Operation Mode

- Queue mode：Only process next type of panel when the finished number of current panel reaches to the target number.
- Cycle mode：Process next type of panel when finishing one current type of panel.

2.4.2 2. Queue Mode

- Queue set top: Let the new item set to top. (In the process, it cannot add an item)
- Queue set bottom: Let the new item set to bottom.

2.4.3 3. Auto Work

- Start to process the next panel automatically when a panel is finished.

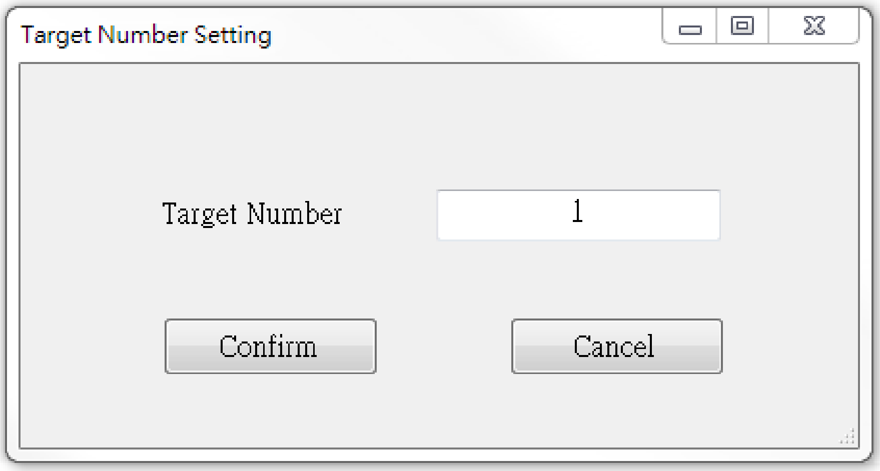
2.4.4 4. Auto Remove

- Remove the item automatically when its finished number reaches its target number.

2.4.5 5. Remove All

- Remove all the items.

2.4.6 6. Set Target



- Set target number to all the items.

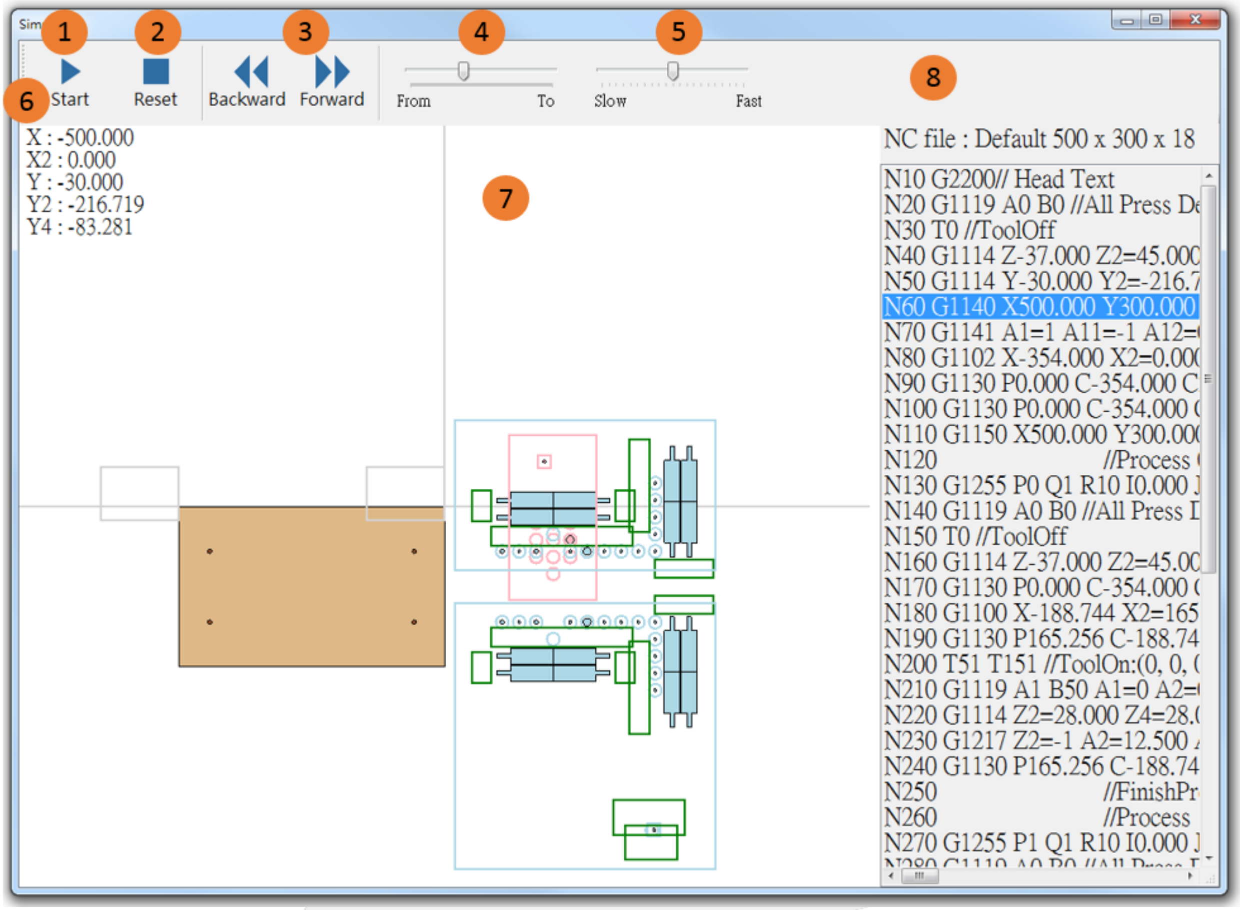
2.4.7 7. Sequence List

- Display the information such as Panel ID, Status, Finished Number, Target Number, Panel Information and File path.
- Status：
    - Done：This item is processed.
    - Generated：The NC of this item is generated.
    - Assigned：This item is assigned to the main program.
    - Running：This item is processing.
    - Failed：Fail to convert to NC. Reasons can be found out by right-clicking the item and choosing "Show Error Message".

2.4.8 8. Right-Click Function

- Select an item and right-click it.
- Function：
    - Show Error Message：Find out reasons that failing to convert to NC.
    - Put on Top：put the selected item to the top.
    - Reset Finished Count：Set the finished count of the selected item to zero.
    - Assign Processing：Assign the selected item to process.
    - Simulation：Simulate the selected item.
    - Remove：Remove the selected item.
    - Remove All：Remove all the items.

2.5 Simulation



- Entrance：Select an item in working list → Right click it → Simulation

2.5.1 1. Start/Pause

- Start：Start simulation. This button would become "Pause" when the simulation is started.
- Pause：Pause simulation. This button would become "Start" when the simulation is paused.

2.5.2 2. Reset

- Reset the simulation.

2.5.3 3. Forward/Backword

- Move forward/backward with a time unit.

2.5.4 4. Progress

- Adjust the progress of the simulation.

2.5.5 5. Speed

- Adjust the speed of the simulation.

2.5.6 6. Axis Position

- Display every position of the axis.

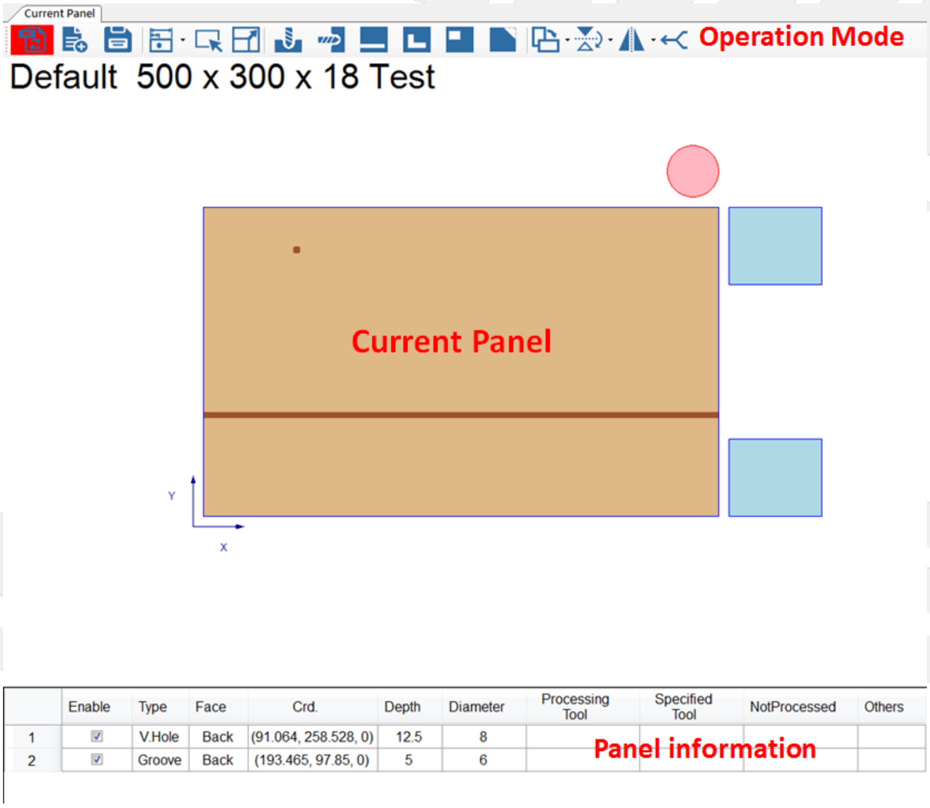
2.5.7 7. Display

- Display simulation.
- Press then move the mouse to translate.
- Scroll the mouse wheel to zoom in or out.

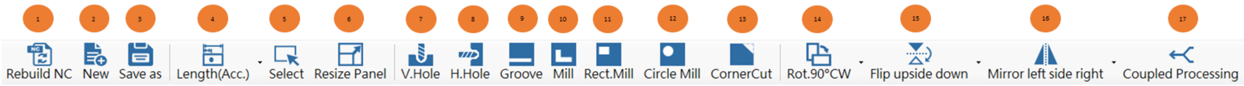
2.5.8 8. NC

- Display information of NC.
- Move to the specific moment when clicking a row.

2.6 Current Panel



2.6.1 **Operation Mode**

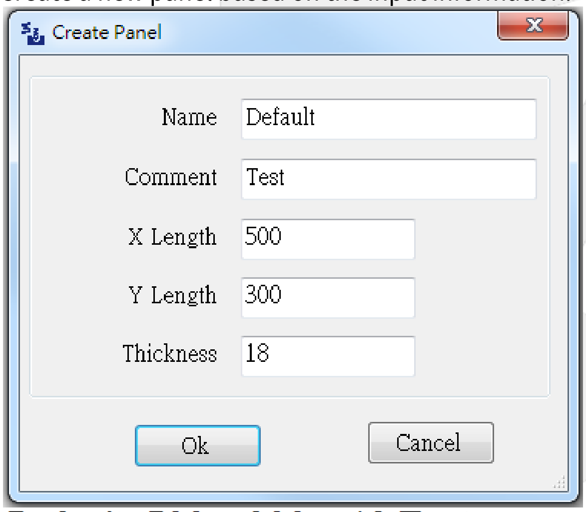


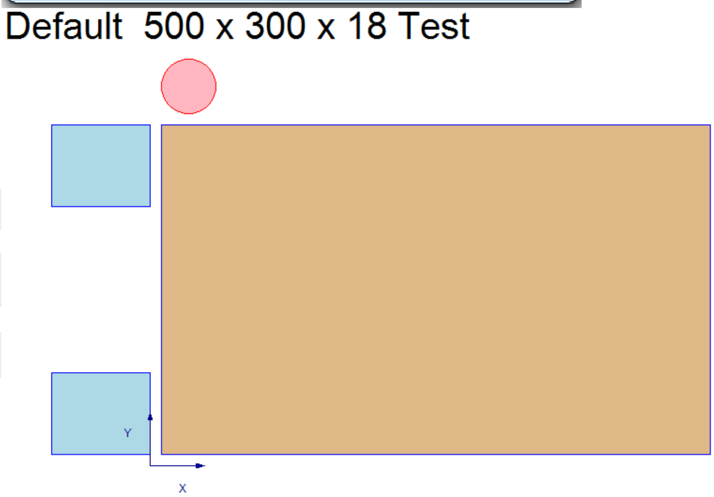
1. Rebuild NC 

- Rebuild the NC program.

2. Create a panel 

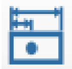
- Create a new panel based on the input information.

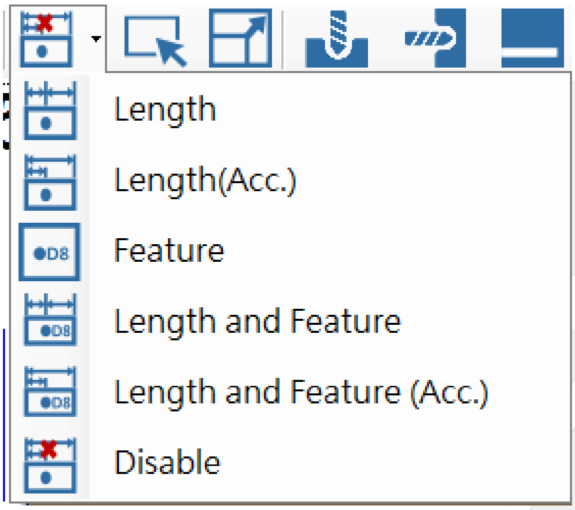




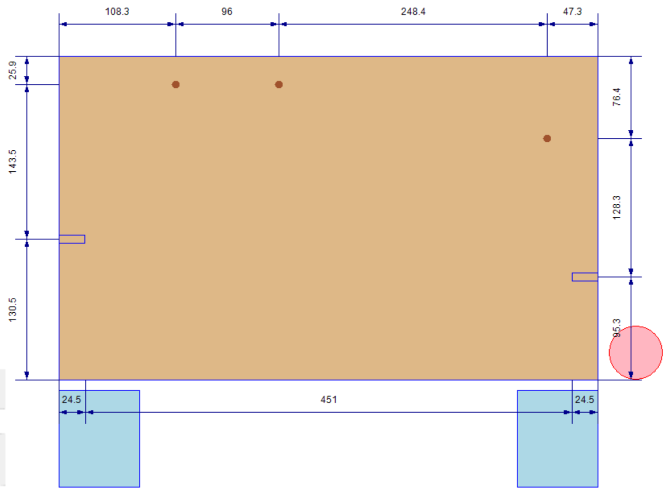
3. Save As... 

- The Save As... dialog asks the user to provide a name for the file to be created, to select a folder in which to place the new file, and to select a file format type for the file.

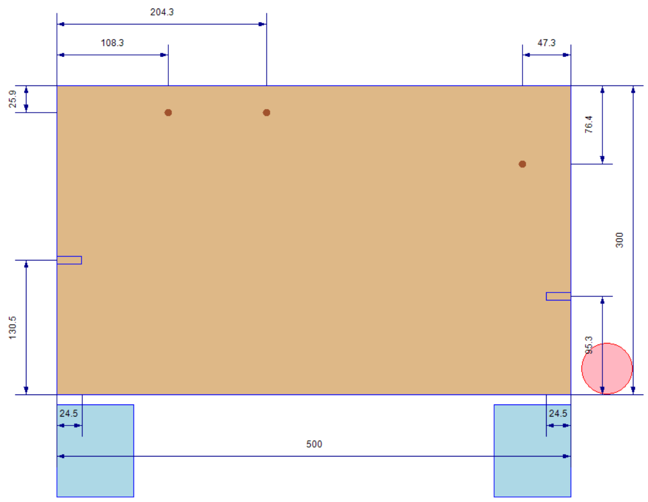
4. Dimension 



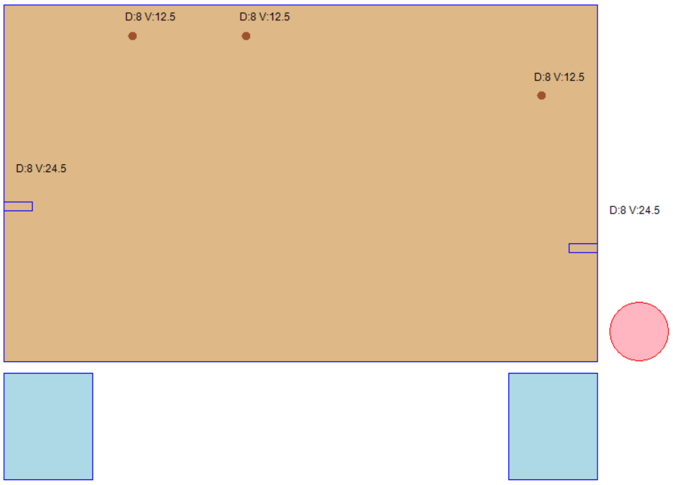
1. Length: Display the relative distance between the feature and the feature or the feature and the edge of the panel.



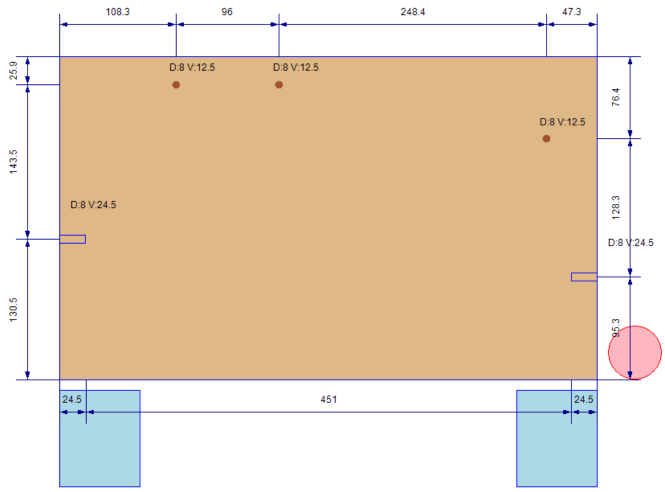
2. Length (Accumulate): Display the relative distance between the feature and the edge of the panel.



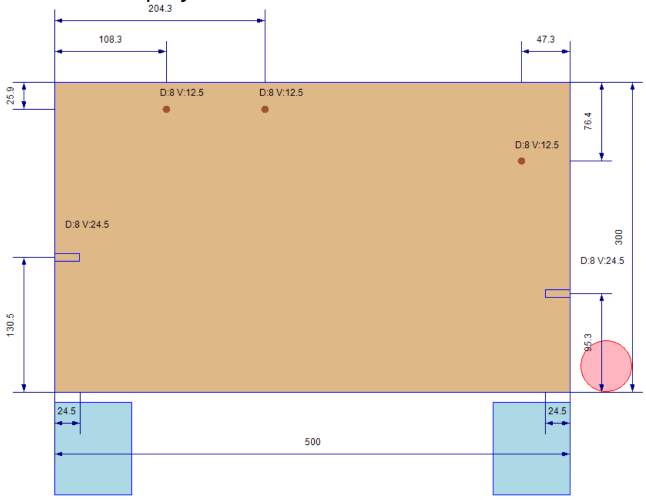
3. Feature: Display features data. (D: Diameter、V: Hole depth)



4. Length and Feature: Display the relative distance between the feature and the feature or the feature and the edge of the panel.
   And display features data. (D: Diameter、V: Hole depth)



5. Length and Feature (Accumulate): Display the relative distance between the feature and the edge of the panel. And display features data.



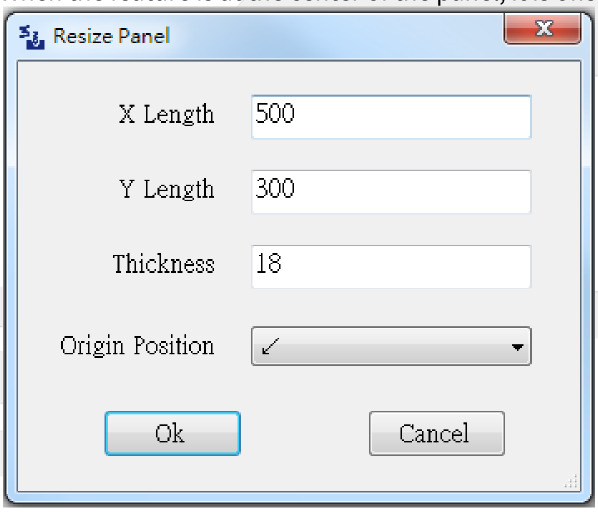
6. Disable

5. Select 

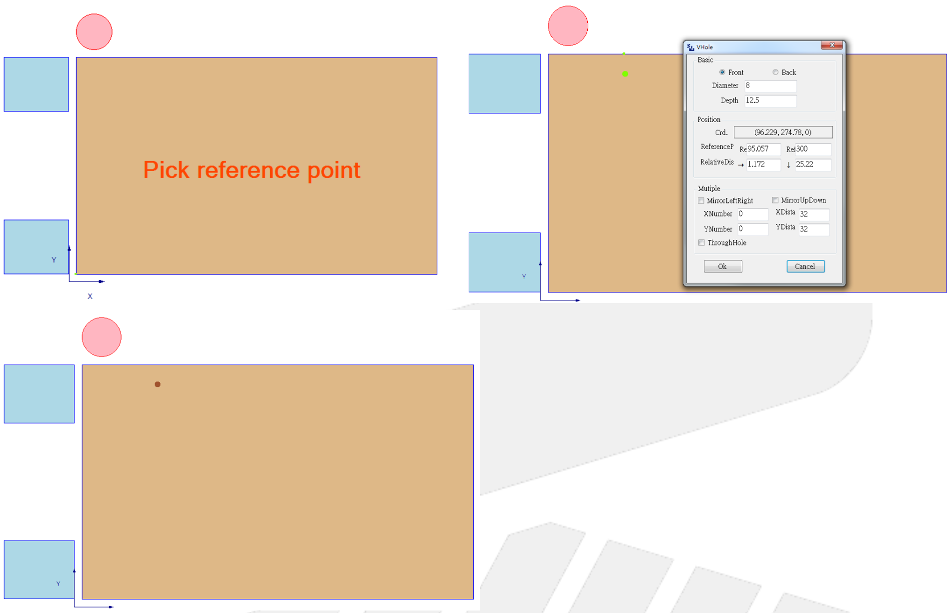
- Re-select any mode.

6. Resize panel 

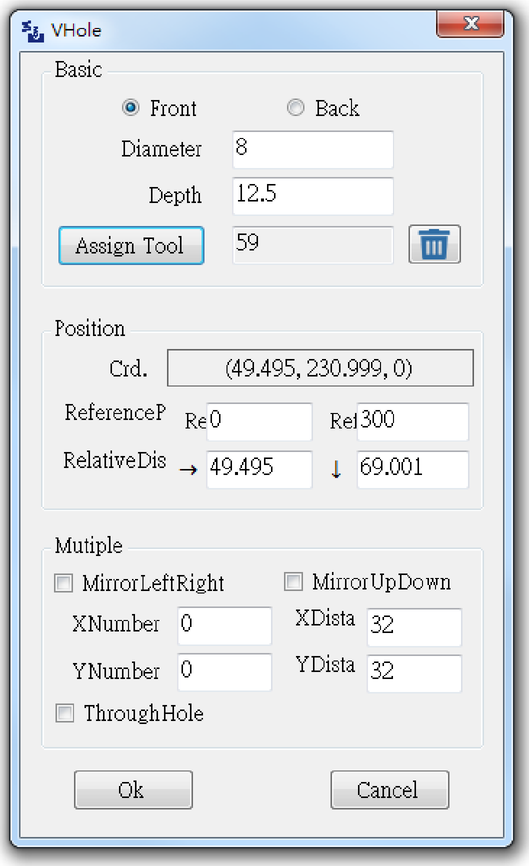
- Adjust the feature position according to the modified size.
- When the feature is at the center of the panel, it is offset according to the origin position.



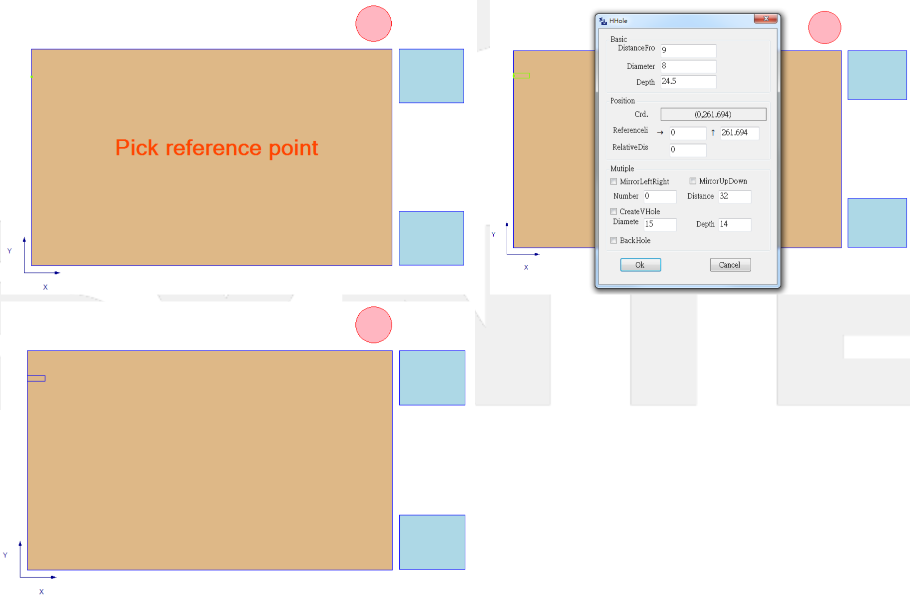
7. Vertical hole



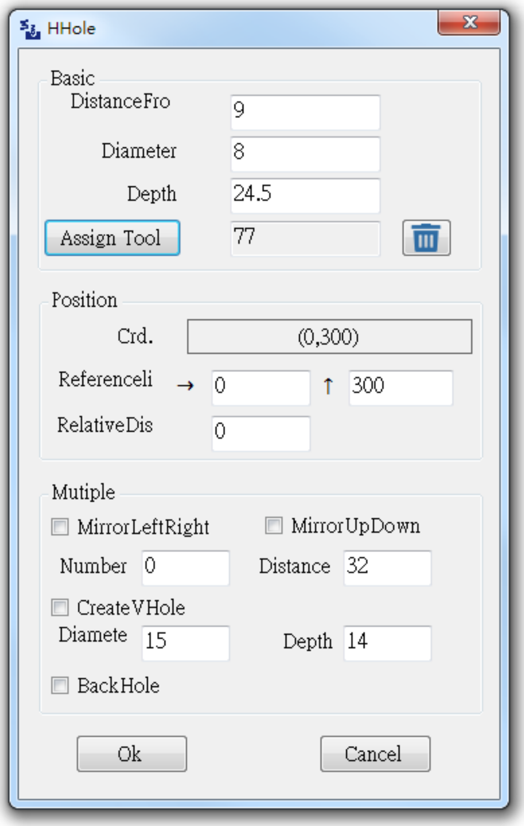
a. Step: Pick reference point on the panel → Pick feature point → Input the data → Pressed OK → Create a vertical hole。
b. Vertical hole information:
   i. Basic: Choose the hole position. [Front / Back]、 Input the diameter and depth of the hole、 Assign the tool if needed.
   ii. Position:
      - Coordinate: Display coordinate relative to the origin point of the panel.
      - Reference point: Reference point position.
      - Relative distance: Relative distance between feature position and reference point.
   iii. Multiple:
      - Mirror reverse left and right: Copy the same features to another side.
      - Mirror upside and down: Copy the same features to another side.
      - Number/Distance: Copy the same hole based on the input data.
      - Through Hole



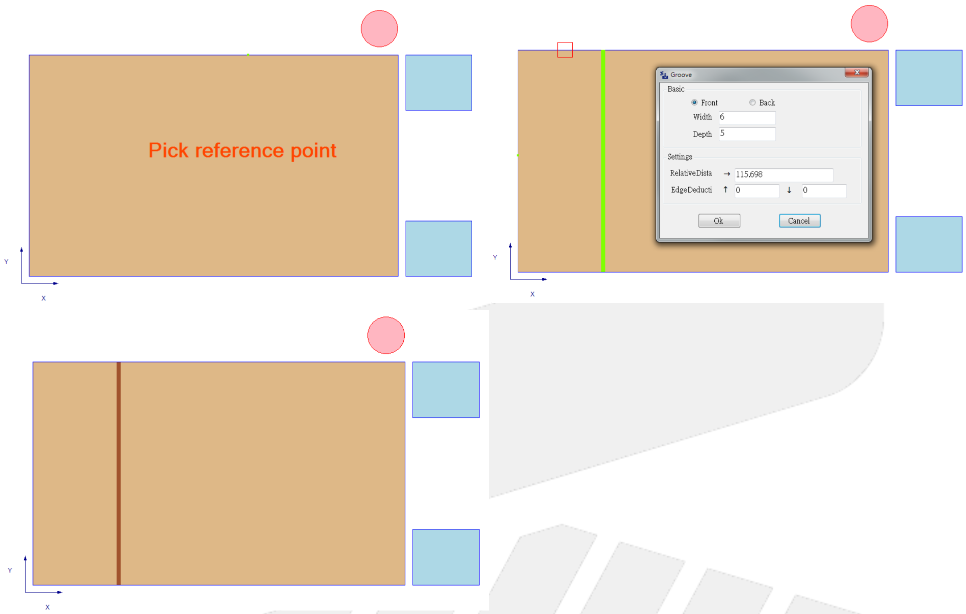
8. Horizontal hole



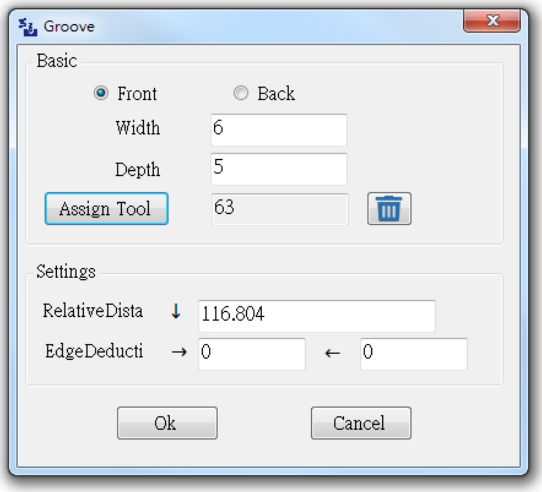
a. Step: Pick reference point on the four sides of the panel → Pick feature point → Input the data → Pressed OK → Create a horizontal hole
b. Horizontal hole information:
    i. Basic: Input the diameter and depth of the hole. And input the distance from the surface(Need to rebuild NC)、Assign the tool if needed.
    ii. Position:
        - Coordinate: Display coordinate relative to the origin point of the panel.
        - Reference line: Display relative distance to the reference line.
        - Reference distance: Display relative distance to the reference point.
    iii. Multiple:
        - Mirror reverse left and right: Copy the same features to another side.
        - Mirror upside and down: Copy the same features to another side.
        - Number/Distance: Copy the same hole based on the input data.
        - Create a vertical hole: Input the hole data.
        - Back hole: Create a back vertical hole



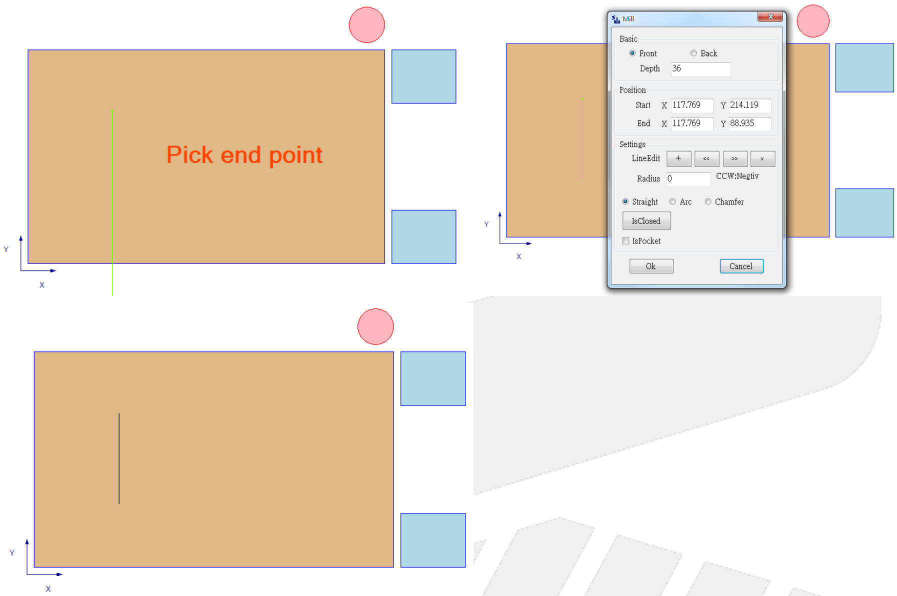
9. Groove



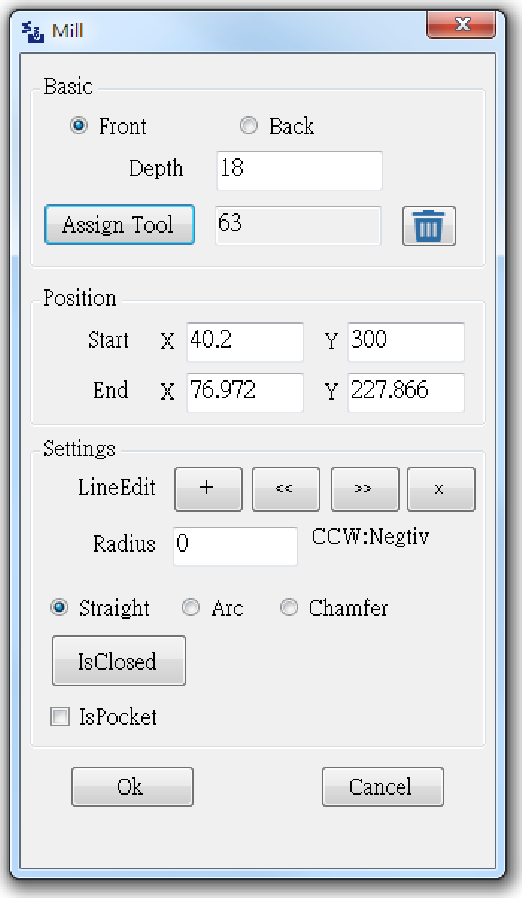
a. Step: Pick reference point on the four sides of the panel → Pick feature point → Input the data → Pressed OK → Create a groove.
b. Groove information:
    i. Basic: Choose the groove position. [Front / Back]、Input the width and depth of the groove、Assign the tool if needed.
    ii. Settings
        - Relative distance: Display relative distance to the reference line.
        - Edge deduction: Input the deduction distance on both sides.



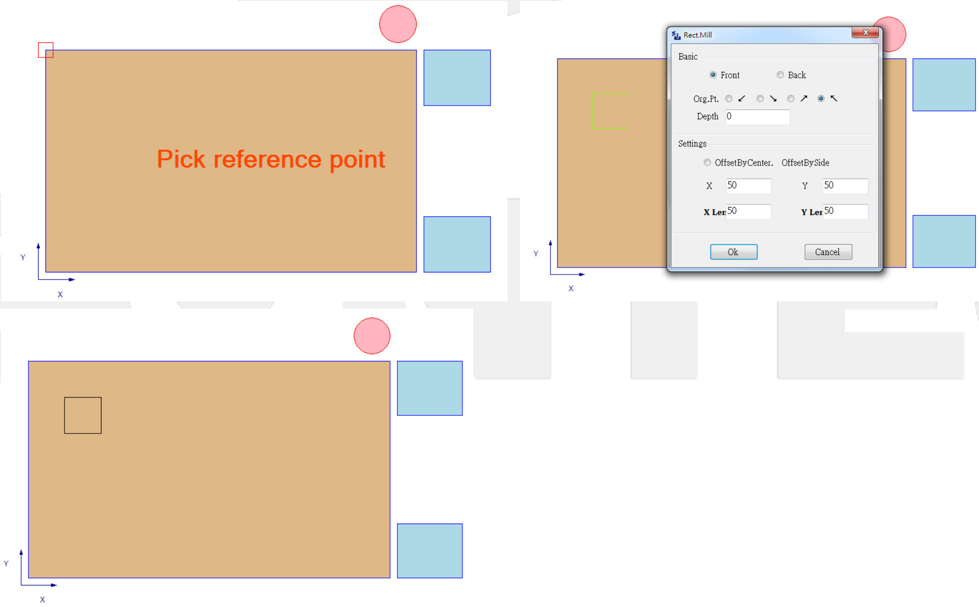
10. Mill 



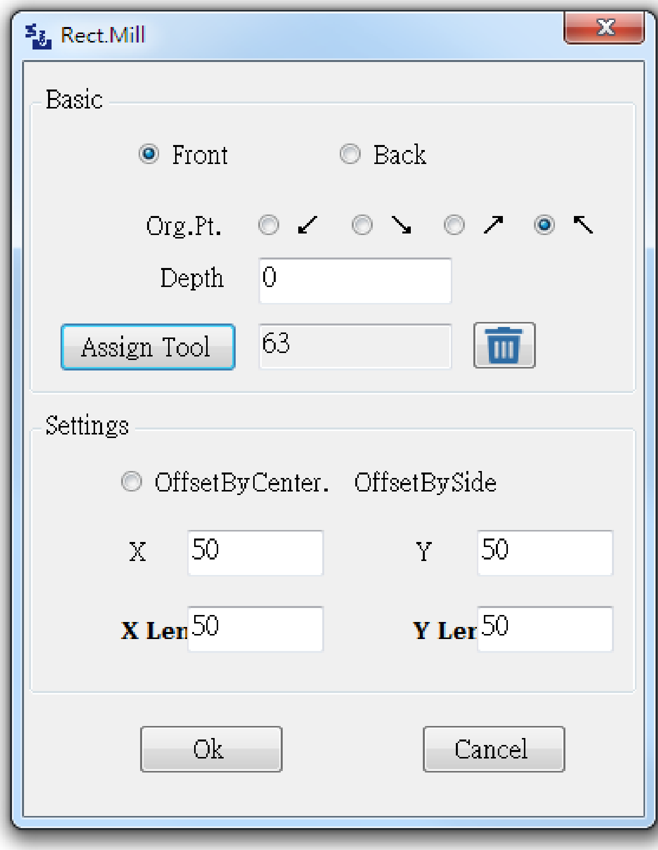
a. Step: Pick start point→ Pick endpoint → Input the data →Pressed OK → Create a mill.
b. Mill information:
    i. Basic: Choose the mill position. [Front / Back]、Input the depth of the mill、Assign the tool if needed.
    ii. Position:
        - Start/End : Input X,Y position.
    iii. Settings:
        - Line edit : + Add line segment、 << / >> Choose the last or the neat line、 x Delete line segment
        - Radius: Only use for Arc. Adjust the radius of arc (If the input value is less than the minimum arc radius, it will be changed to the minimum arc radius.)
        - Straight/Arc/Chamfer: Line type.
        - Is closed: Close the two segments.
        - Is pocket: only use for the closed mill. Set the enclosed area to the milling range.



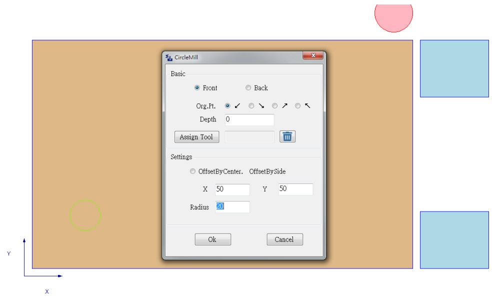
11. Rectangle mill



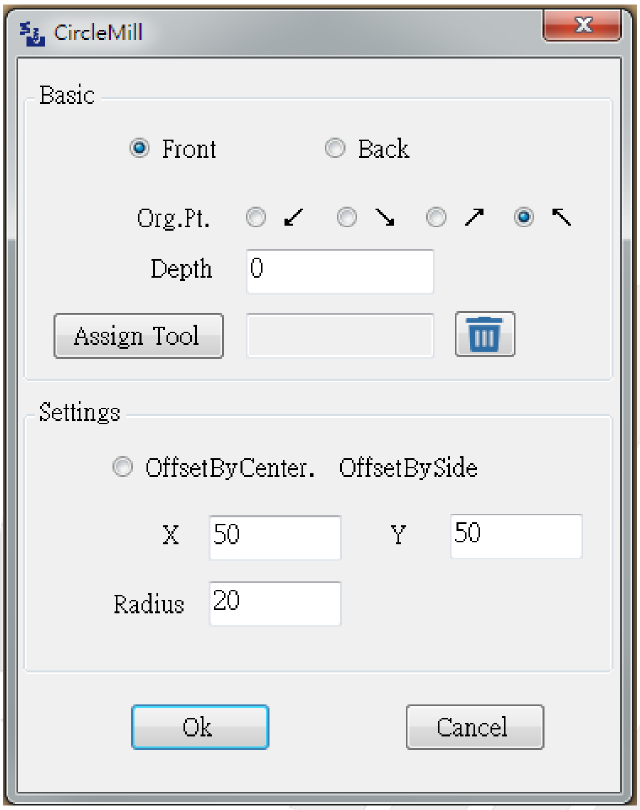
a. Step: Pick reference point on the four corners of the panel → Input the data → Pressed OK → Create a rectangle mill.
b. Rectangle mill information:
   i. Basic: Choose the rectangle mill position. [Front / Back]、Choose the origin point and input the depth of the mill、Assign the tool if needed.
   ii. Settings:
      - Offset by center or side: Choose the anchor point of the X, Y of the rectangle.
      - X Length / Y Length



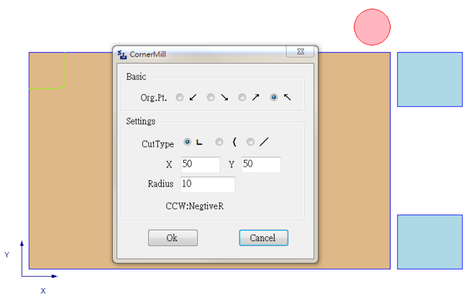
12. Circle mill

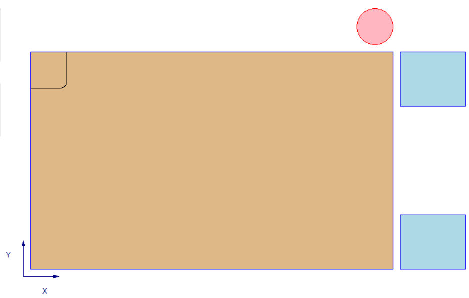


a. Step: Pick reference point on the four corners of the panel → Input the data → Pressed OK → Create a circle mill.
b. Circle mill information:
   i. Basic: Choose the rectangle mill position. [Front / Back]、 Choose the origin point and input the depth of the mill、 Assign the tool if needed.
   ii. Settings:
      - Offset by center or side: Choose the anchor point of the X, Y of the circle.
      - Radius

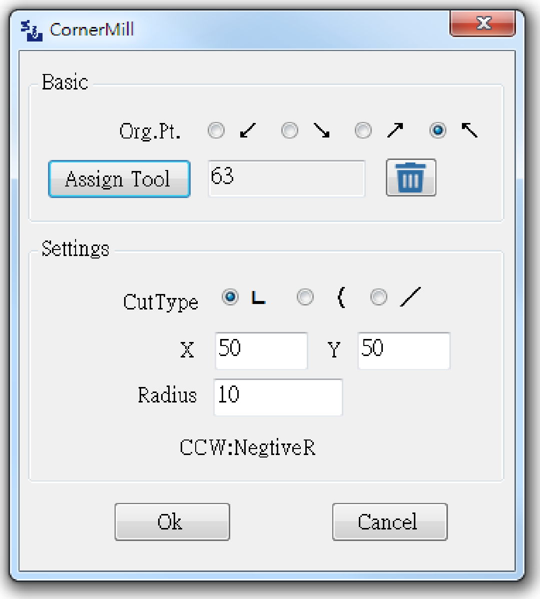


13. Corner mill 



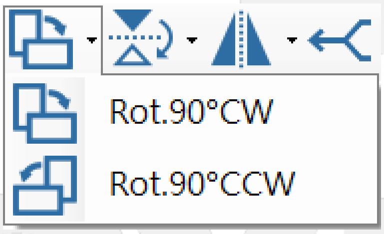


a. Step: Pick reference point on the four corners of the panel → Input the data → Pressed OK → Create a corner mill.
b. Corner mill information:
    i. Basic: Choose the origin point、 Assign the tool if needed..
    ii. Settings:
        • Cut type/Position
        • Radius: Input the arc radius.



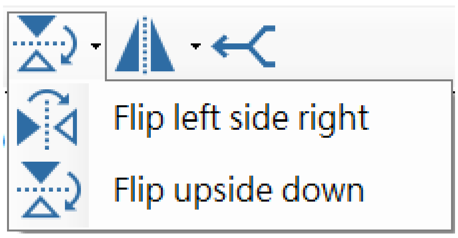
14. Rotate 

• Rotate the panel according to the selected direction.



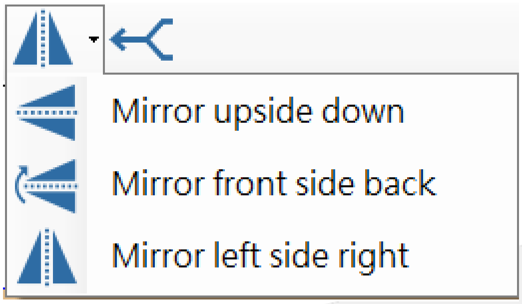
15. Flip the panel 

• Flip the panel according to the selected direction.



16. Mirror 

- Mirror the panel according to the selected direction.

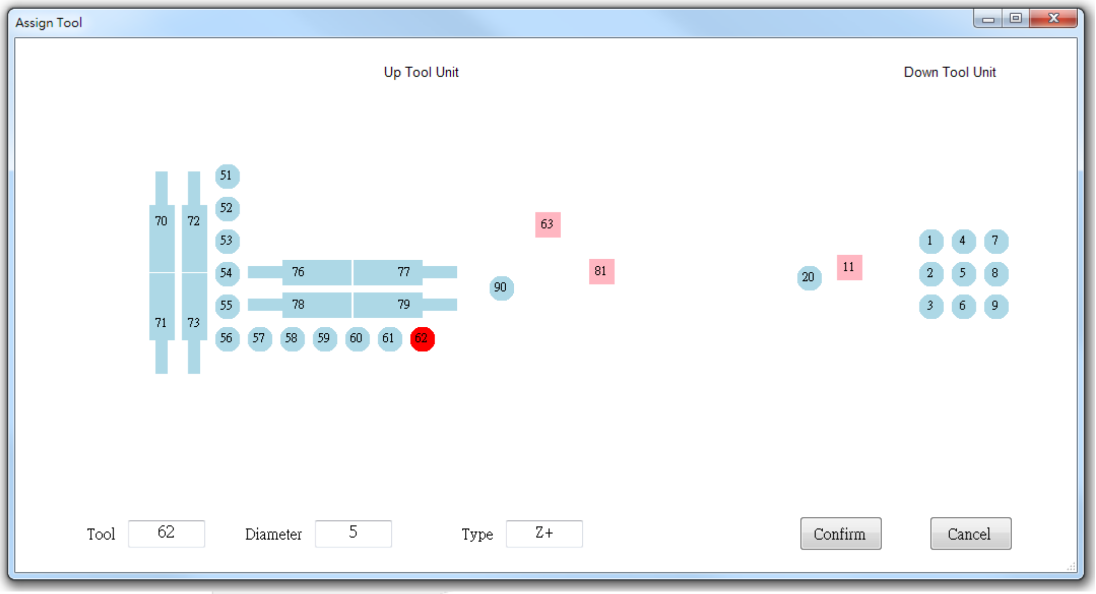


17. Coupled processing 

- Step
    a. Load the Cad file.
    b. Press button of coupled processing.
        i. The panel can not be couple processing for an empty panel or loading failure.
        ii. If the panel has features on both the top face and bottom face, it can not be couple processing so you need to skip the feature one of them.
        iii. Coupled processing does not process milling.
    c. Save as a new file to the sequence list and the filename is original panel ID adds a suffix _Coupled Processing.
- Rules
    a. Generate a panel that it has twice the thickness of the original panel and mirror the process feature upside down.
    b. Coupled processing does not process milling.
    c. Coupled processing does not process the ignored features.

Assign Tool

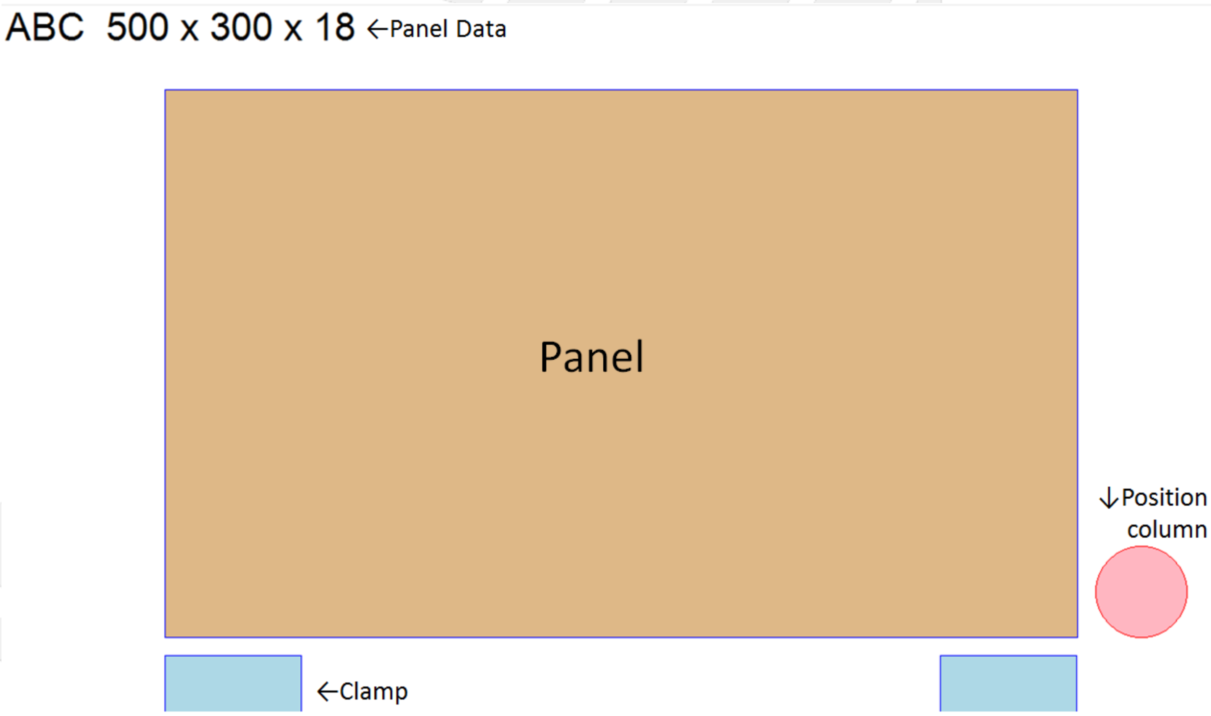
- Step : Click "Assign Tool" when editing the process → choose a tool (click it) → click "Confirm"



•

2.6.2 Current panel display area

Display the current panel.

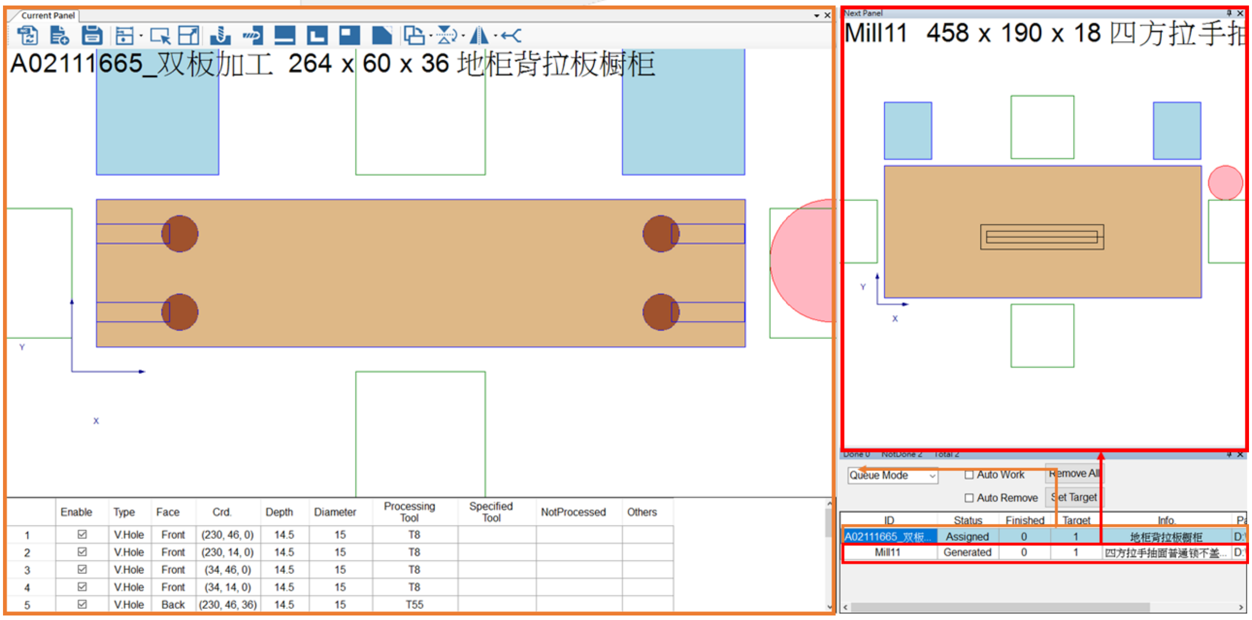


2.6.3 Panel information display area

	Enable	Type	Face	Crd.	Depth	Diameter	Processing Tool	Specified Tool	NotProcessed	Others
1	☑	V.Hole	Back	(50, 477, 18)	12.5	8	T155			
2	☑	Groove	Back	(208.679, 0, 18)	5	6	T81			

- Enable: Enable this feature if it is not skipped.
- Type: The feature type.， ex: Hole、Groove、Mill...
- Face: The face where the feature is located， ex: Front、Back
- Coordinate: Feature coordinates.
- Depth: Feature depth.
- Diameter: Feature diameter.
- Processing Tool: After it converts to NC, the processing tool used.
- Specified Tool: Feature specified tool.
- Not Processed: Unprocessed reason.

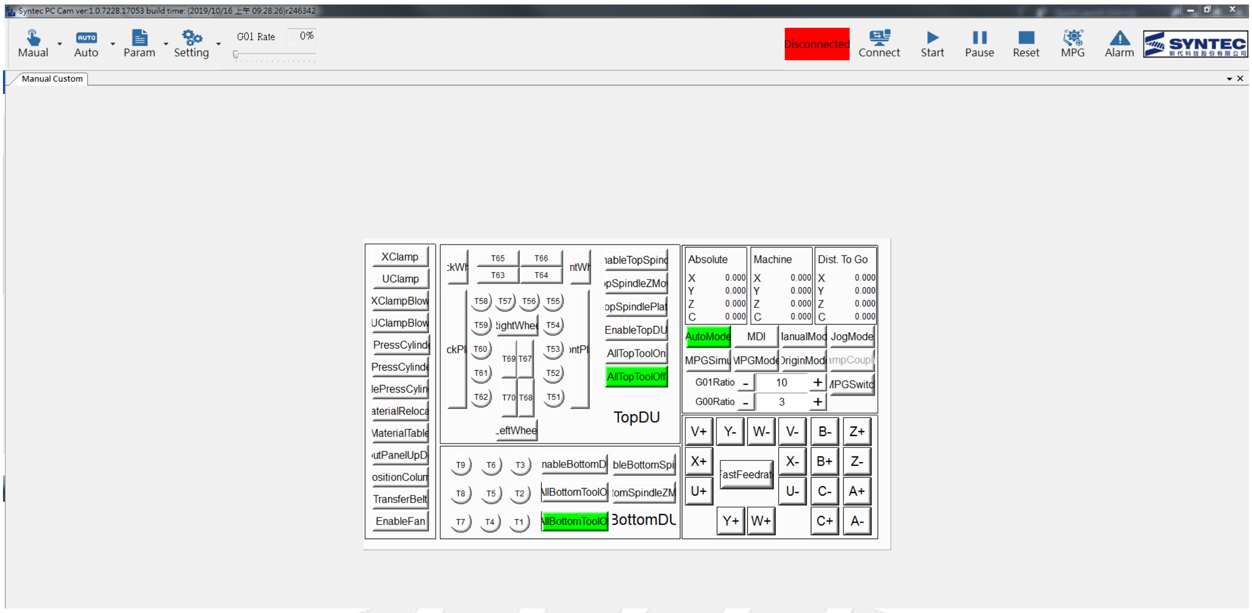
2.7 Next Panel



2.7.1 1. Next Panel

- Preview the next processing panel in the queue.
- The panel can be zoom in&out or move but not edited in the preview window.
- If there is no next processing panel exits, the window is empty.

3 Manual Mode



3.1 Customize

1. Path: OCRes\Common\Appdata, Add "ManualCustom.xml" file.The screen will appear on the manual custom screen of PC CAM.
2. The green text framed by the red line is the size of the customized screen on the PC CAM.

ManualCustom.xml

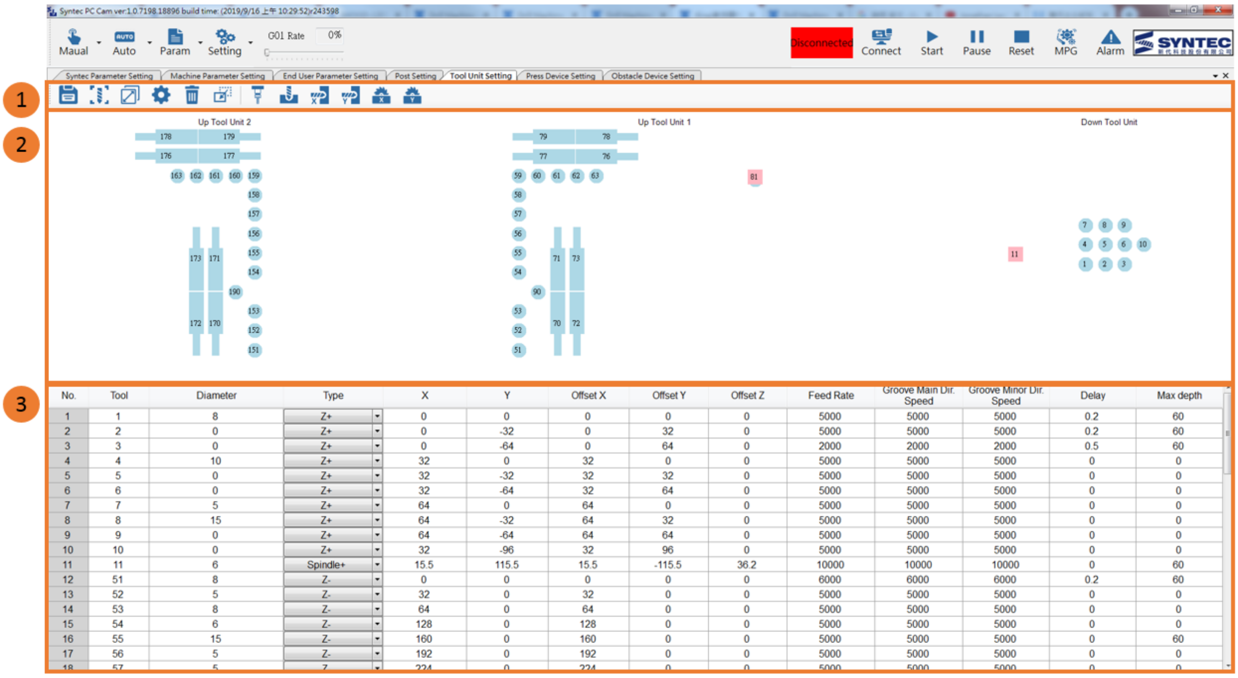
```
<?xml version="1.0" encoding="utf-8"?>
<!--BROWSER SIZE|H:768|W:1024-->
<Screen Type="DotNetExtension">
    <Name>Manualcustom</Name>
    <Panel>
        <Name>Panel62113</Name>
        <Left>-19</Left>
```

3. Currently, only some custom actions (Action) are provided, as follows

Action	說明	範例
CUSTOMDEV	Modify the variable value Variable: R、P、@、#、L	CUSTOMDEV_R253=100 CUSTOMDEV_R253++ CUSTOMDEV_R253-- CUSOMDEV_ON R100.1 CUSOMDEV_OFF R100.1 CUSOMDEV_INV R100.1
CUSTOMXML	Switch screen	CUSTOMXML_XXXX.xml

4 Parameter Setting

4.1 Drill Setting



4.1.1 1. Settings

- a. Save：Save tool data.
- b. Set main tool：Select a vertical drill to be the main tool.
- c. Zoom to Fit：Zoom the figure to fit size and location.
- d. Default parameter：Set the default parameter for adding new tools.
- e. Delete tool：Remove the selected tool.
- f. Offset location：Offset all the tool in the drill unit based on the selected tool.
- g. Spindle：Add a spindle by left-clicking a location in the desired drill unit figure.
- h. Drill Z：Add a vertical drill by left-clicking a location in the desired drill unit figure.
- i. Drill X：Add a pair of horizontal drills (X Dir.) by left-clicking a location in the desired drill unit figure.
- j.Drill Y：Add a pair of horizontal drills (Y Dir.) by left-clicking a location in the desired drill unit figure.
- k. Saw X：Add a saw (X Dir.) by left-clicking a location in the desired drill unit figure.
- l. Saw Y：Add a saw (Y Dir.) by left-clicking a location in the desired drill unit figure.

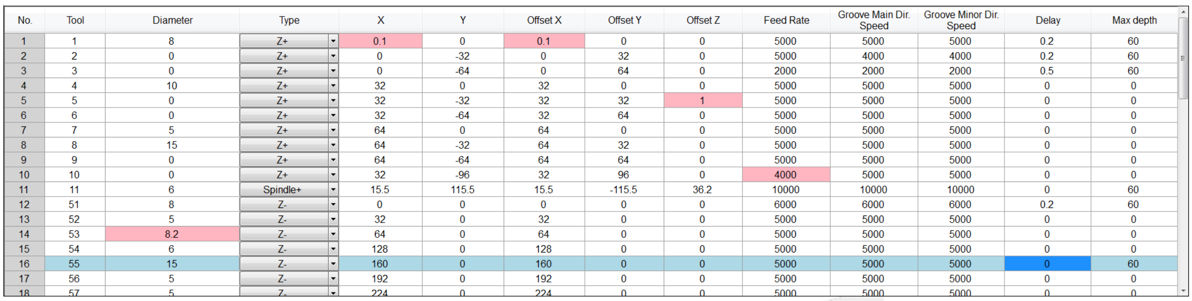
4.1.2 2. Figures

- Display different kinds of tools (Up → X+, Left → Y+)

圖示	說明
	Vertical drill (Z+ or Z-)
	Spindle
	Horizontal drill (Y+)
	Horizontal drill (Y-)
	Horizontal drill (X-)
	Horizontal drill (X+)
	Saw (Y)
	Saw (X)

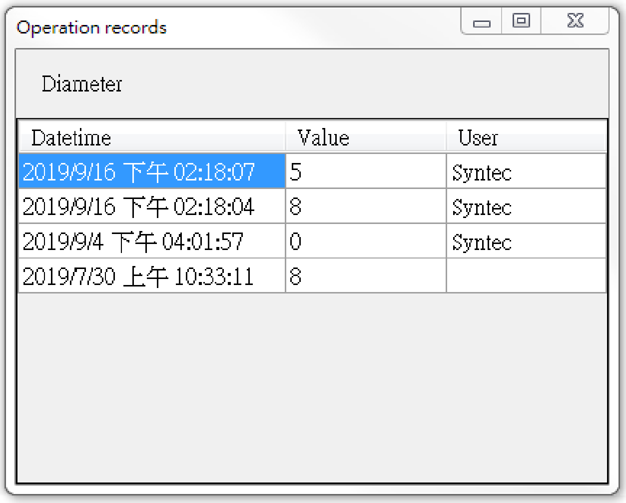
4.1.3 3. Tool Data Table

- Setting parameters for every tool.
- The back color of the cell would become pink after modifying it, giving hints that the parameter has been modified. The back color would be removed once saving tool data.



No.	Tool	Diameter	Type	X	Y	Offset X	Offset Y	Offset Z	Feed Rate	Groove Main Dir. Speed	Groove Minor Dir. Speed	Delay	Max depth
1	1	8	Z+	0.1	0	0.1	0	0	5000	5000	5000	0.2	60
2	2	0	Z+	0	-32	0	32	0	5000	4000	4000	0.2	60
3	3	0	Z+	0	-64	0	64	0	2000	2000	2000	0.5	60
4	4	10	Z+	32	0	32	0	0	5000	5000	5000	0	0
5	5	0	Z+	32	-32	32	32	1	5000	5000	5000	0	0
6	6	0	Z+	32	-64	32	64	0	5000	5000	5000	0	0
7	7	5	Z+	64	0	64	0	0	5000	5000	5000	0	0
8	8	15	Z+	64	-32	64	32	0	5000	5000	5000	0	0
9	9	0	Z+	64	-64	64	64	0	5000	5000	5000	0	0
10	10	0	Z+	32	-96	32	96	0	4000	5000	5000	0	0
11	11	6	Spindle+	15.5	115.5	15.5	-115.5	36.2	10000	10000	10000	0	60
12	51	8	Z-	0	0	0	0	0	6000	6000	6000	0.2	60
13	52	5	Z-	32	0	32	0	0	5000	5000	5000	0	0
14	53	8.2	Z-	64	0	64	0	0	5000	5000	5000	0	0
15	54	6	Z-	128	0	128	0	0	5000	5000	5000	0	0
16	55	15	Z-	160	0	160	0	0	5000	5000	5000	0	60
17	56	5	Z-	192	0	192	0	0	5000	5000	5000	0	0
18	57	5	Z-	224	0	224	0	0	5000	5000	5000	0	0

- Right-click any parameter can show its record.



Operation records

Diameter

Datetime	Value	User
2019/9/16 下午 02:18:07	5	Syntec
2019/9/16 下午 02:18:04	8	Syntec
2019/9/4 下午 04:01:57	0	Syntec
2019/7/30 上午 10:33:11	8	

- Operation Manual • Version History

Syntec Wood Panel CAD DXF Format

木工產品組

Exported on 12/02/2020

Table of Contents

1 Introduction

- Users can use CAD software such as AutoCAD to create wood panels with holes, grooves, and mills, then output it by DXF file.
- When drawing, some specific rules must be followed so that the information on the panel can be recognized correctly.
- One DXF file can describe many panels, but it is recommended that one DXF file for one panel.

2 Step to Use Syntec DXF

1. Set config file (\DiskC\SysData\CadConfig\SyntecDxf.config) to define every LAYER name and set the definition of Height.
2. Use CAD software such as AutoCAD to build LAYER. (The names of all LAYER must match the config file)
3. Following the rules to draw panel and create DXF files.
4. Import to PCCAM.

3 CAD Environment

- DXF support version：AutoCad2000, AutoCad2004, AutoCad2007, AutoCad2010, AutoCad2013 和 AutoCad2018

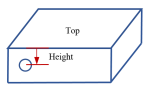
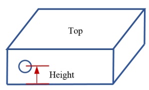
4 Config Setting

- Config file：SyntecDxf.config[1] (please put it in \DiskC\SysData\CadConfig\ )

- LAYER name
  - Define the following Layer Name so that the information on the panel can be recognized correctly.
  - Default LAYER name would be assigned if not set.

	Code	Comment	Default
1	PanelInfo	Panel Information, Outline, Edgebanding	PanelInfo
2	DrillXY1	Top hole	Drill_XY1
3	DrillXY0	Bottom hole	Drill_XY0
4	DrillSide	Side hole	Drill_Side
5	GrooveXY1	Top groove	Groove_XY1
6	GrooveXY0	Bottom groove	Groove_XY0
7	MillXY1	Top mill	Mill_XY1
8	MillXY0	Bottom mill	Mill_XY0

- Height definition of the side hole
  - Set the height definition of the side hole.
  - The default definition would be set if not set.

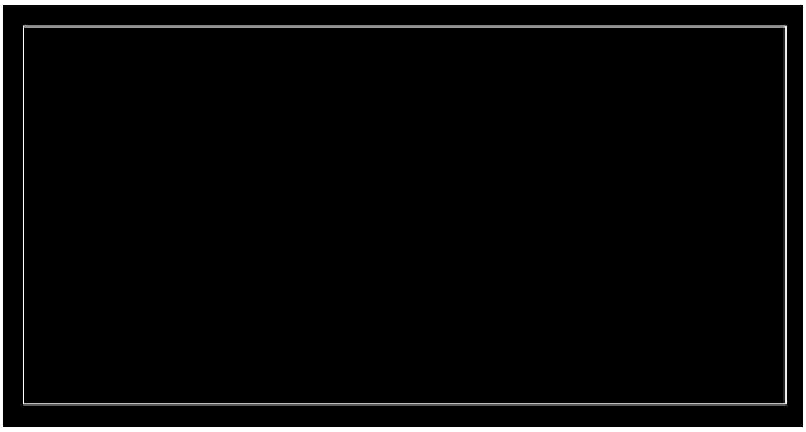
	Code	Value	Figure	Comment
1	SideHoleHeightDefinition	BaseXY1		From top to bottom
2		BaseXY0		From bottom to top

1 https://confluence.syntecclub.com.tw/download/attachments/584925127/SyntecDxf.config?api=v2&modificationDate=1584954692640&version=1

5 Panel Setting

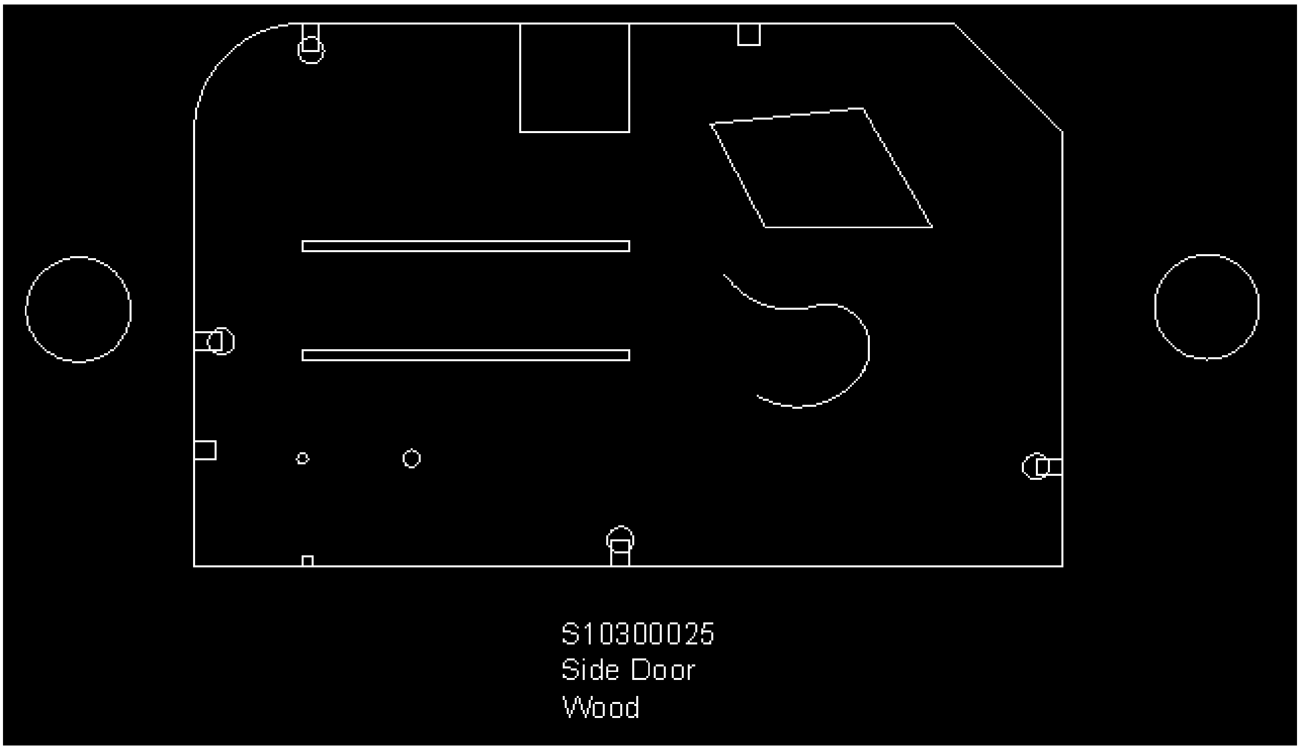
5.1 Panel Outline

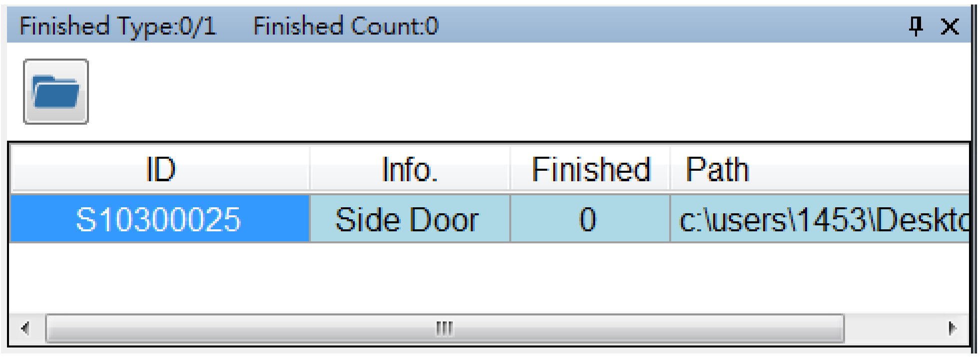
- **LAYER：PanelInfo (Default)**
- Use DXF LwPolyline to describe a panel outline. It must be closed.
- **The thickness of the panel**：The thickness of the LwPolyline.



5.2 Panel Info.

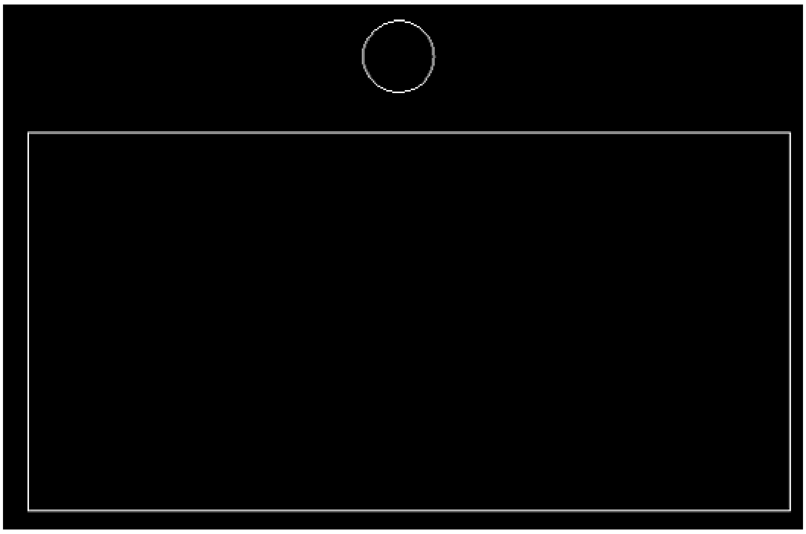
- **LAYER：PanelInfo (Default)**
- Use a DXF MText to describe panel info.
- **Panel ID**：First line of the text. (Must exist, and cannot be the same as the others.)
- **Panel Name**：Second line of the text. (Not required necessarily)
- **Panel Material**：Third line of the text. (Not required necessarily)
- The DXF MText must be placed right under the panel.

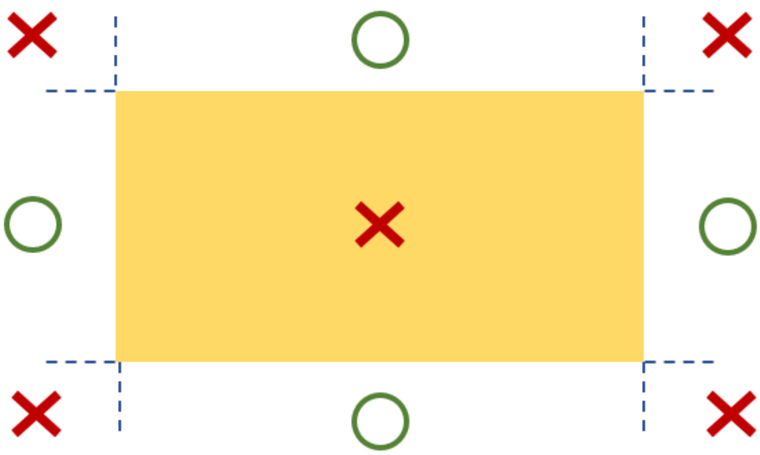




5.3 Edgebanding

- **LAYER：PanelInfo (Default)**
- Use a DXF Circle near the desired edge to describe edgebanding.
- The center of the circle must be placed in the valid regions.
- **The thickness of the edgebanding**：The thickness of the circle.





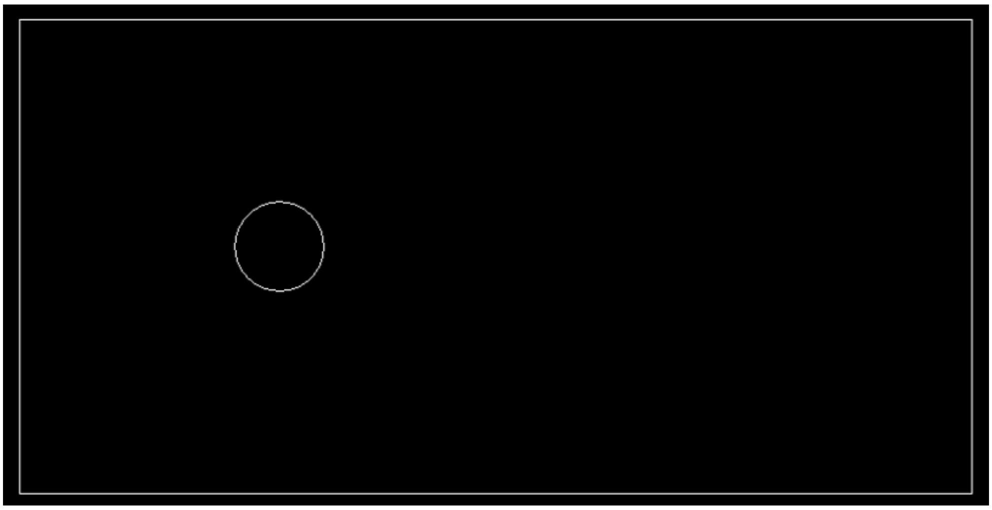
6 Process Setting

- **BLOCK can be used, but changing length or width ratio is prohibited.**

6.1 Hole

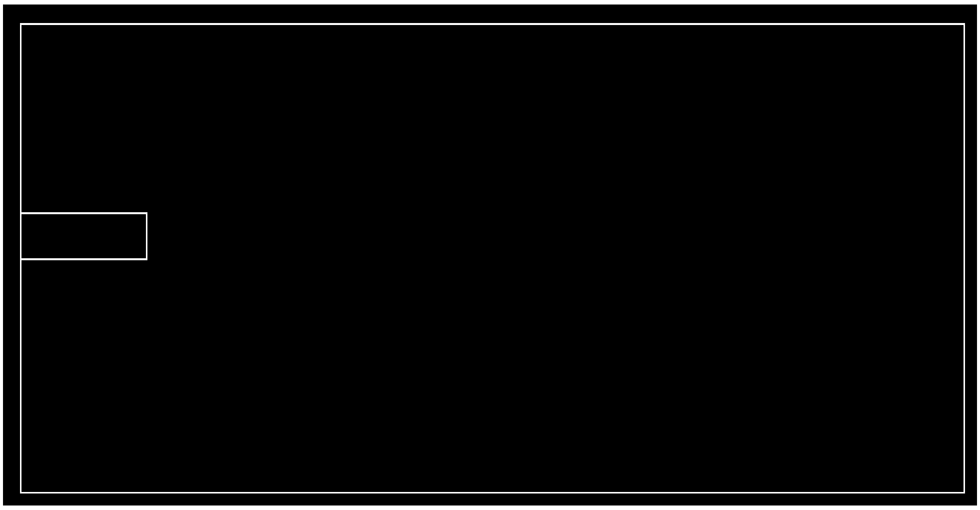
6.1.1 Top hole、Bottom hole

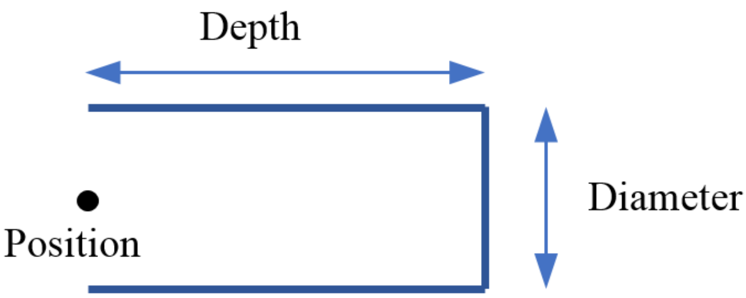
- **LAYER：Drill_XY1 (default) for top hole、Drill_XY0 (default) for bottom hole**
- Use a DXF Circle to describe a vertical hole
- **Hole diameter**：diameter of the circle
- **Hole depth**：thickness of the circle



6.1.2 Side hole

- **LAYER：Drill_Side (default)**
- Use a DXF LwPolyline with three lines to describe a side hole. The notch is at the edge of the outline.
- The first line must be parallel to the third line, and the first line must be perpendicular to the second line.
- **Hole X Y position**：The middle of the first point and the fourth point.
- **Hole Z position**：The thickness of the LwPolyline. Also based on the height definition of the config.
- **Hole depth**：The length of the first line
- **Hole diameter**：The length of the second line





6.2 **Groove**

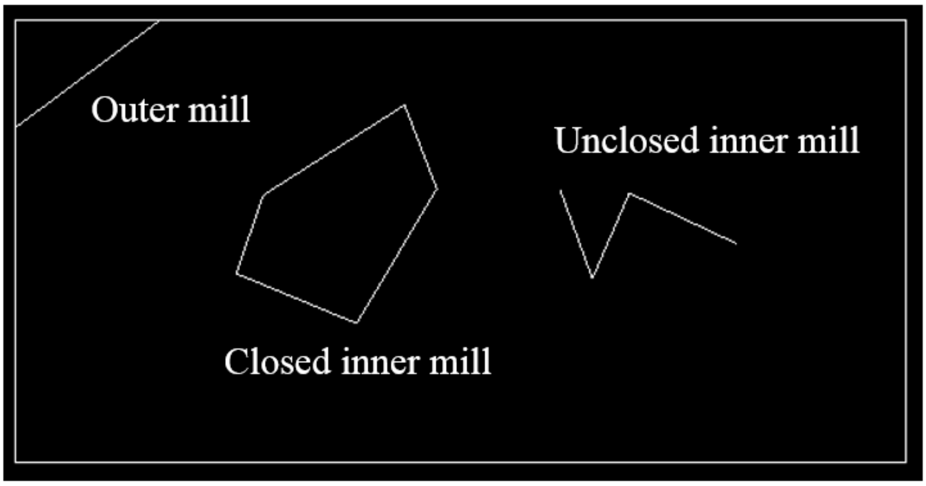
6.2.1 Top groove、Bottom groove

- **LAYER：Groove_XY1 (default) for top groove、Groove_XY0 (default) for bottom groove**
- Use a rectangular DXF LwPolyline to describe the groove.
- The groove must be in the X or Y direction.
- **Groove depth**：The thickness of the LwPolyline

6.3 Mill

6.3.1 Top mill、Bottom mill

- **LAYER：Mill_XY1(default) for top mill、Mill_XY0 (default) for bottom mill**
- Use a DXF LwPolyline to describe the mill.
- **Mill depth**：The thickness of the LwPolyline.
- Inner mill
  - Closed：The closed region is the removed region.
  - Unclosed：The tool would work on the path. (tool center)
- Outer mill
  - The start point and the end point must be on the panel outline. The tool would work on the path. (tool edge)



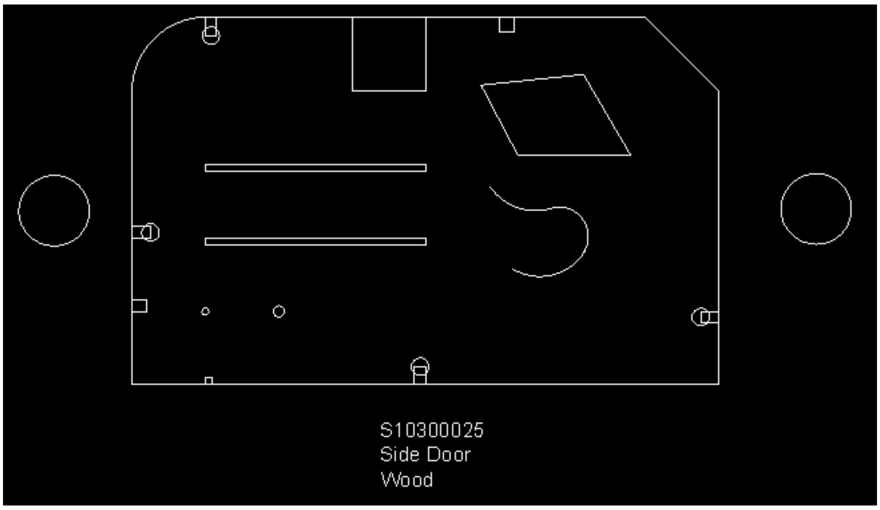
7 Drawing SOP

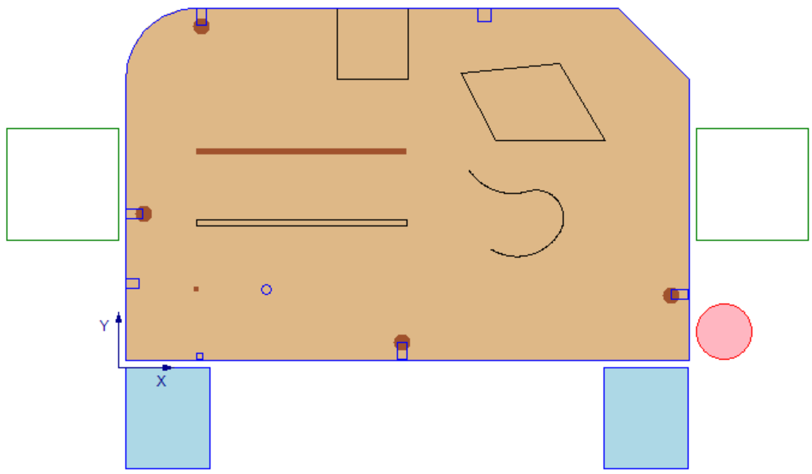
	Step	Introduction
1	Panel outline	1. Use a LwPolyline to draw a closed outline. 2. Assign the LAYER. 3. Set the thickness of the LwPolyline to present panel thickness.
2	Panel info	1. Use an MText right under the panel outline to describe panel info. 2. The first line is the panel ID, and the second line is the panel name. 3. Assign the LAYER.
3	Edgebanding	1. Use a Circle outside the panel outline to present the edgebanding. 2. The diameter of the circle can be random. 3. Assign the LAYER. 4. Set the thickness of the circle to present the thickness of the edgebanding.
4	Top/bottom hole	1. Use a Circle inside the panel outline to present the top/bottom hole. 2. Assign the LAYER. 3. Set the diameter of the circle to present the hole diameter. 4. Set the thickness of the circle to present the hole depth.
5	Side hole	1. Use a three-line LwPolyline to present a side hole. 2. For detailed settings, please refer to the process setting → side hole. 3. Assign the LAYER. 4. Set the thickness of the LwPolyline to present the hole height.
6	Top/bottom groove	1. Use a rectangular LwPolyline to describe a groove. 2. Assign the LAYER. 3. Set the thickness of the LwPolyline to present the groove depth.

	Step	Introduction
7	Top/bottom mill	1. Use a LwPolyline to describe a mill. 2. Assign the LAYER. 3. Set the thickness of the LwPolyline to present the mill depth.

8 Example

File：DxfTest.dxf[2] (AutoCAD 2004)





2 https://confluence.syntecclub.com.tw/download/attachments/584925127/DxfTest.dxf?api=v2&modificationDate=1584954692620&version=1

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